

Accelerators for medical and industrial applications

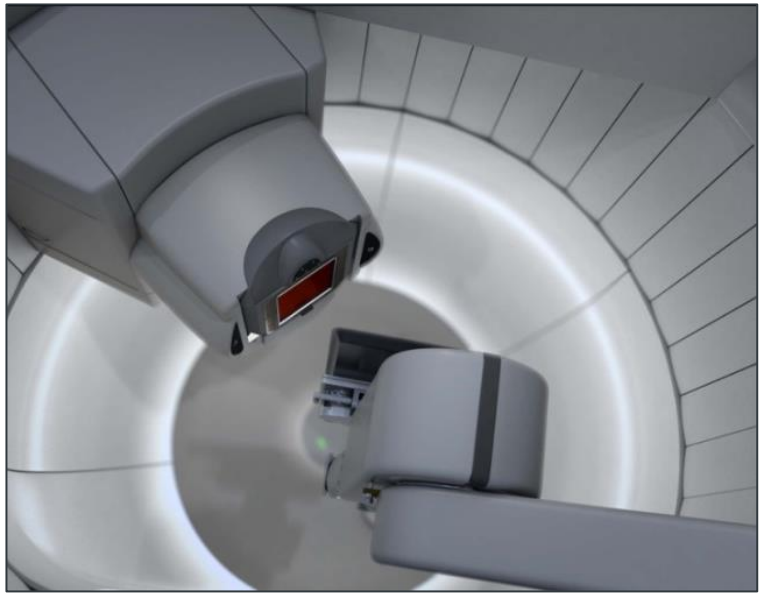
JUAS Archamps, March 9th, 2022

Erik van der Kraaij

Accelerator Physicist

Erik.vanderkraaij@iba-group.com

IBA – Ion Beam Applications



MEDICAL TECHNOLOGY

- RadioPharma Solutions
- Proton Therapy (PT)
- Industrial Applications
- Dosimetry

NOWADAYS

- **R&D:**
  - 12% of turnover
  - 13% of workforce

- Over **500 accelerator** systems installed worldwide

- ~**270 M€ sales** per year
- ~1600 employees worldwide, 40 nationalities

➢ If interested in job openings, see http://www.iba-careers.com/?page=career

Organization of the lecture

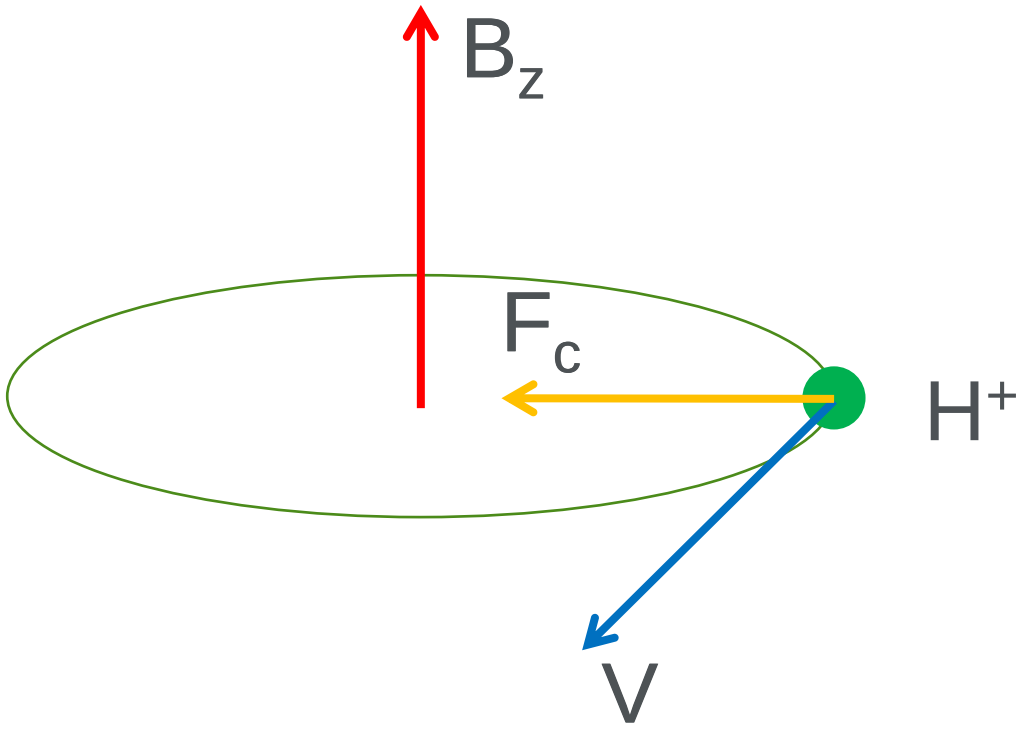
- Part 1: Cyclotron design and beam dynamics
- Part 2: Particle therapy systems for cancer treatment
- Part 3: Accelerators for industrial applications

- Addendum: Radioisotopes for medical applications

Part 1: Cyclotron design and beam dynamics

- Introduction
- Acceleration, focusing and isochronism
- Injection
- Extraction

Introduction: most basic equation of cyclotron

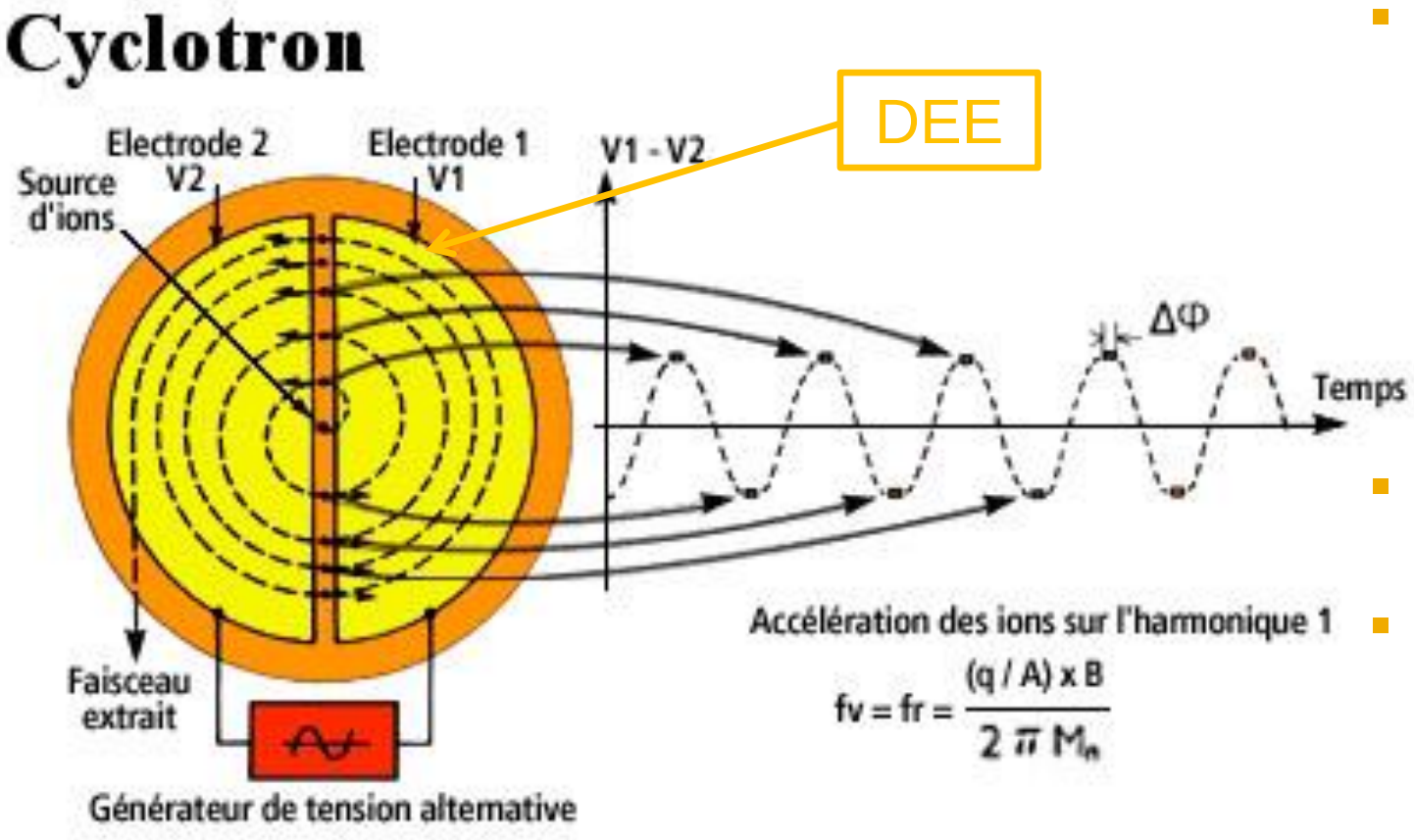


- A charged particle in a uniform magnetic field moves on a circle
- The centripetal force is provided by the Lorentz force acting on the particle

$$\frac{mv^2}{r} = qvB \Rightarrow \omega = \frac{v}{r} = \frac{qB}{m}$$

- **Thus the rotation frequency of the particle is constant => independent on radius ,velocity , energy or time** (in non-relativistic limit)

Consequences of constant cyclotron frequency

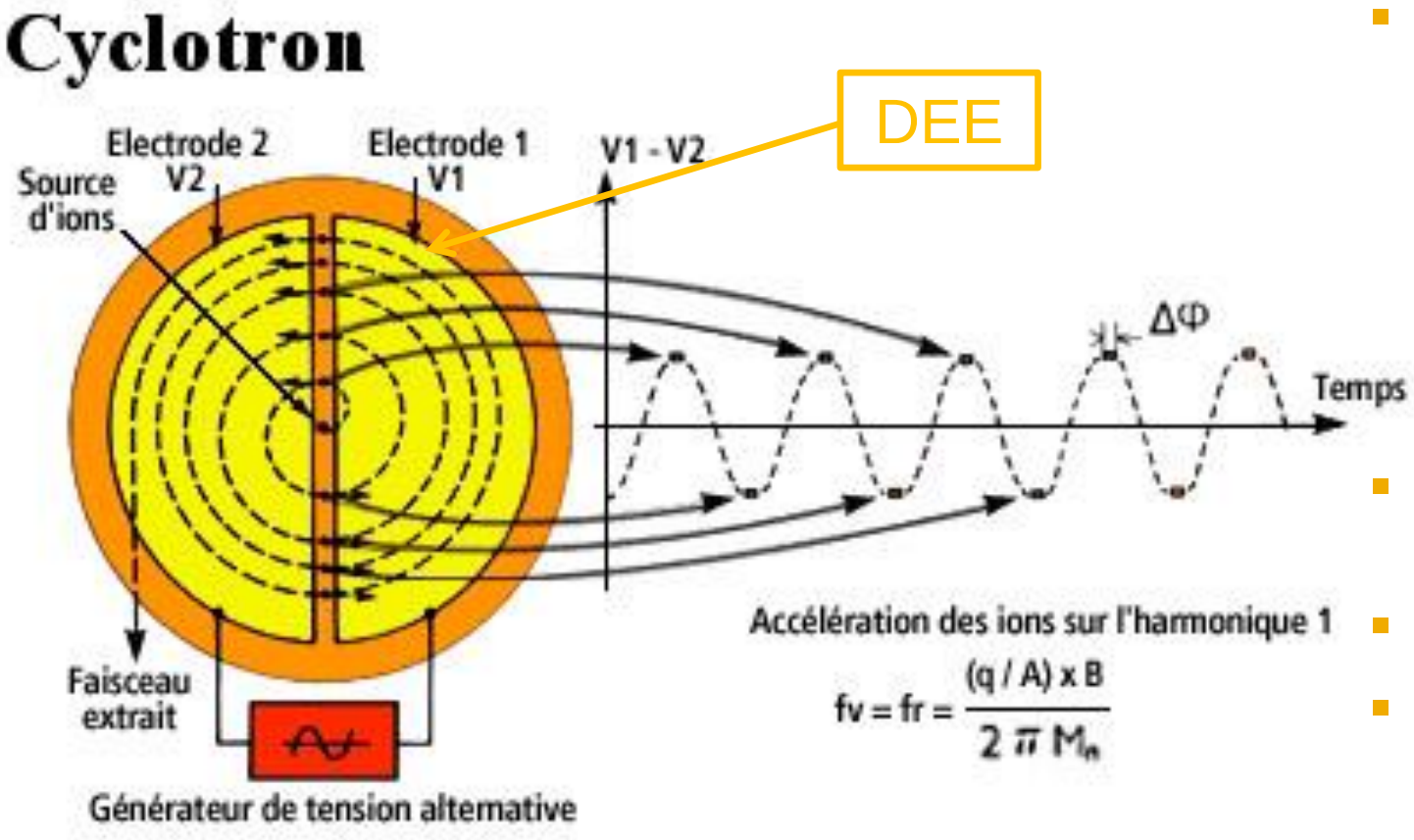


- Particles can be accelerated with an RF-system operating at constant frequency:

$$F_{RF}(MHz) = 15.2\ h(Z/A)\ B$$

- Orbit starts in center (injection) and spirals outward towards pole radius (extraction)
- Magnet field is constant in time

Consequences of constant cyclotron frequency



Classical cyclotron:
Lawrence and Livingston, Phys. Rev. 40 (1932) 9

- Particles can be accelerated with an RF-system operating at constant frequency:

$$F_{RF}(MHz) = 15.2\ h(Z/A)\ B$$

- Orbit starts in center (injection) and spirals outward towards pole radius (extraction)
- Magnet field is constant in time
- RF and magnetic stucture are completely integrated => **Same RF structure accelerates many times** =>compact and cost-effective
- Continuous Wave operation

Classical cyclotron: where is the problem?

i. In a uniform magnetic field there is no vertical focusing (**metastable**)

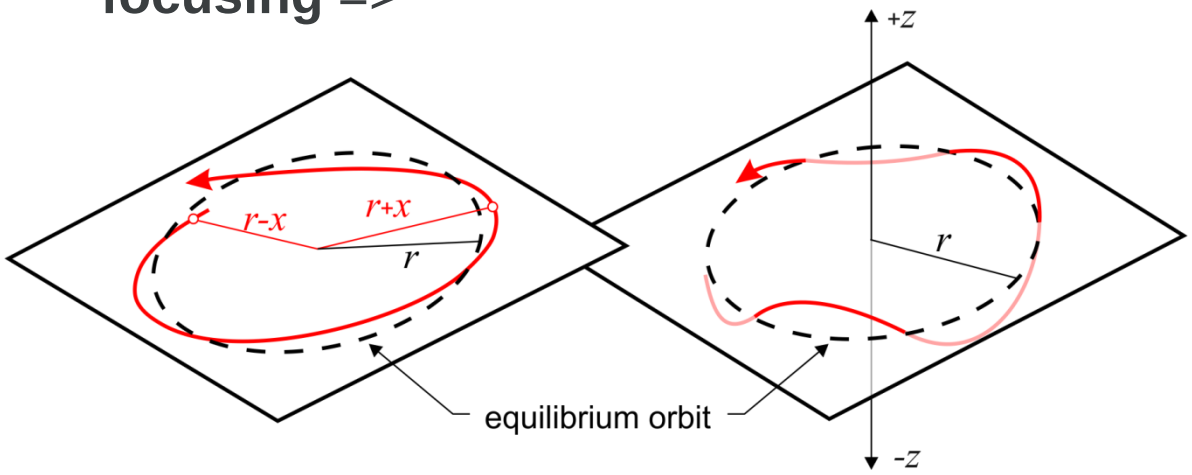
ii. During acceleration, due to relativistic mass increase, revolution frequency decreases in a uniform magnetic field
=> loss of resonance between RF and the beam
=> loss of isochronism

$$\omega = \frac{qB}{m_0}\sqrt{1-\left(\frac{v}{c}\right)^2}$$

iii. just increasing the magnetic field with radius is not possible
=> **vertically unstable**

A solution: **the synchro-cyclotron**

- Let the magnetic field gradually decrease with radius in order to obtain **weak vertical focusing** =>



$$v_z = \sqrt{n} \Rightarrow n = -\frac{r}{B}\frac{dB}{dr}$$

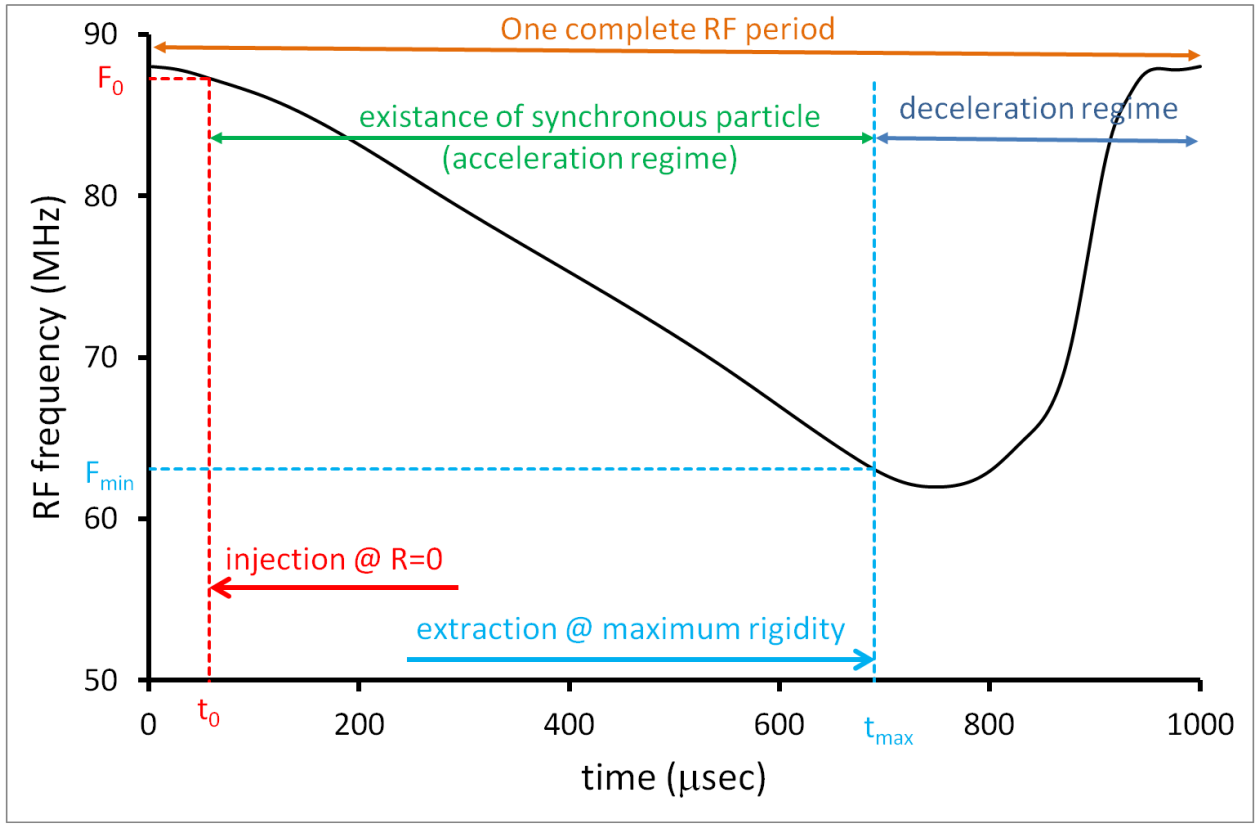
- Let the RF frequency gradually decrease with time in order to compensate for the drop of the magnetic field and for the increase of the mass

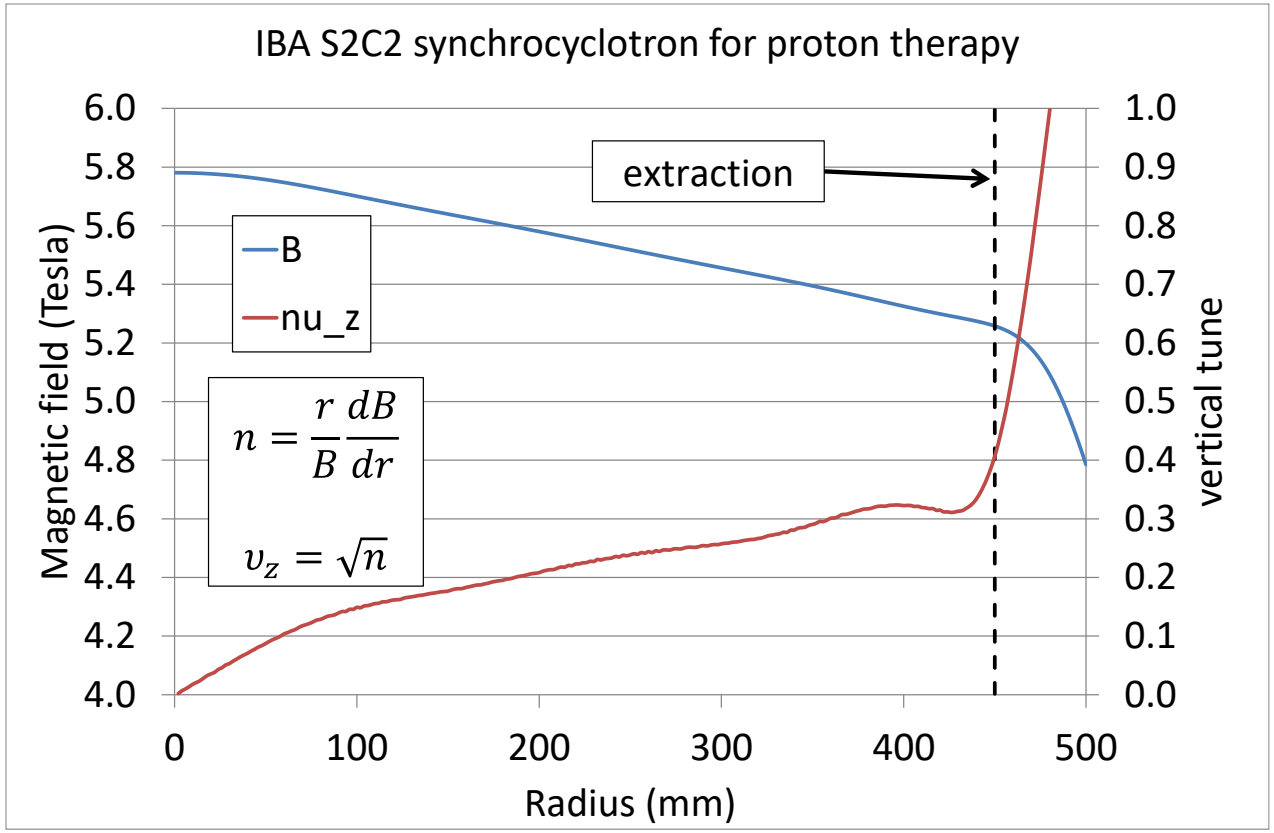
$$\omega = \frac{qB}{m}$$

Veksler, J. Phys. USSR **9** (1945)153
McMillan, Phys. Rev. **68** (1945)143L

Note: synchrotron was invented at the same time

Example: the IBA S2C2 for proton therapy

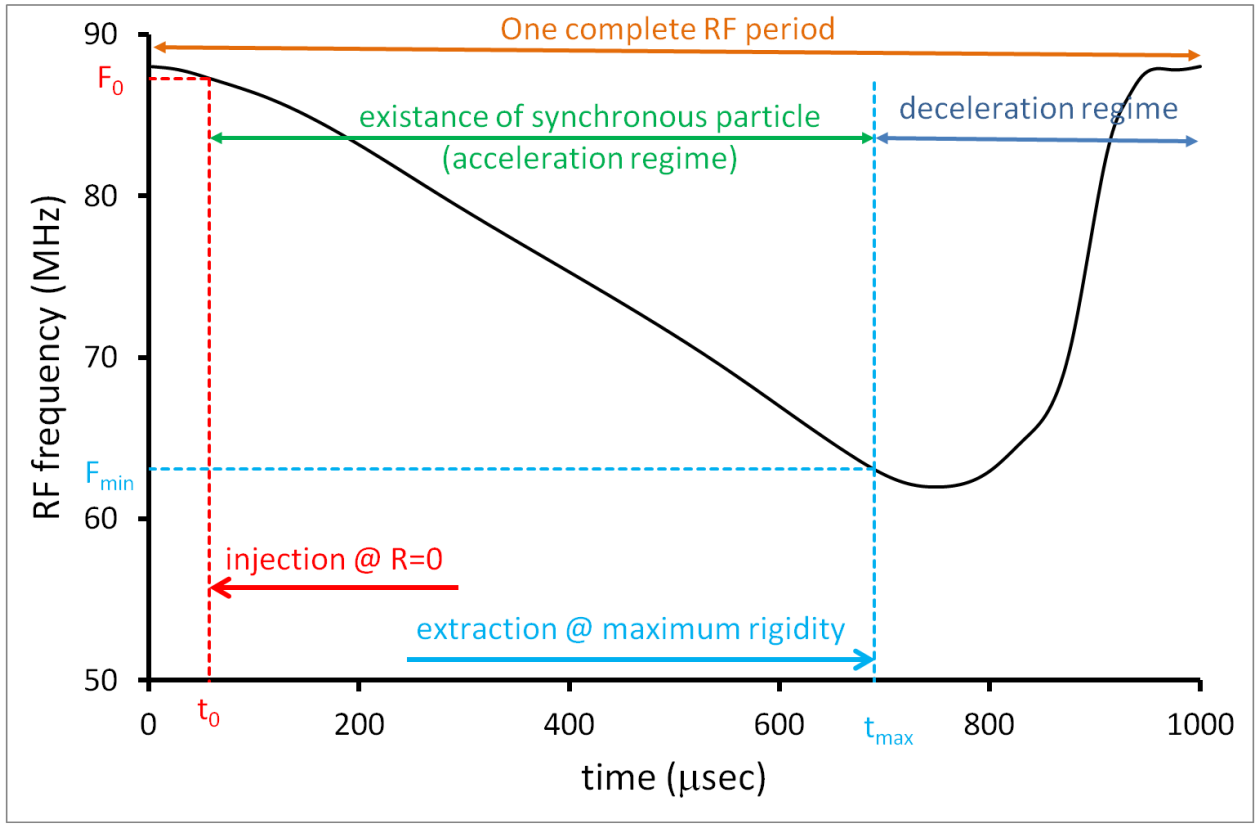




Repetition rate = 1 kHz
Duty cycle about 100

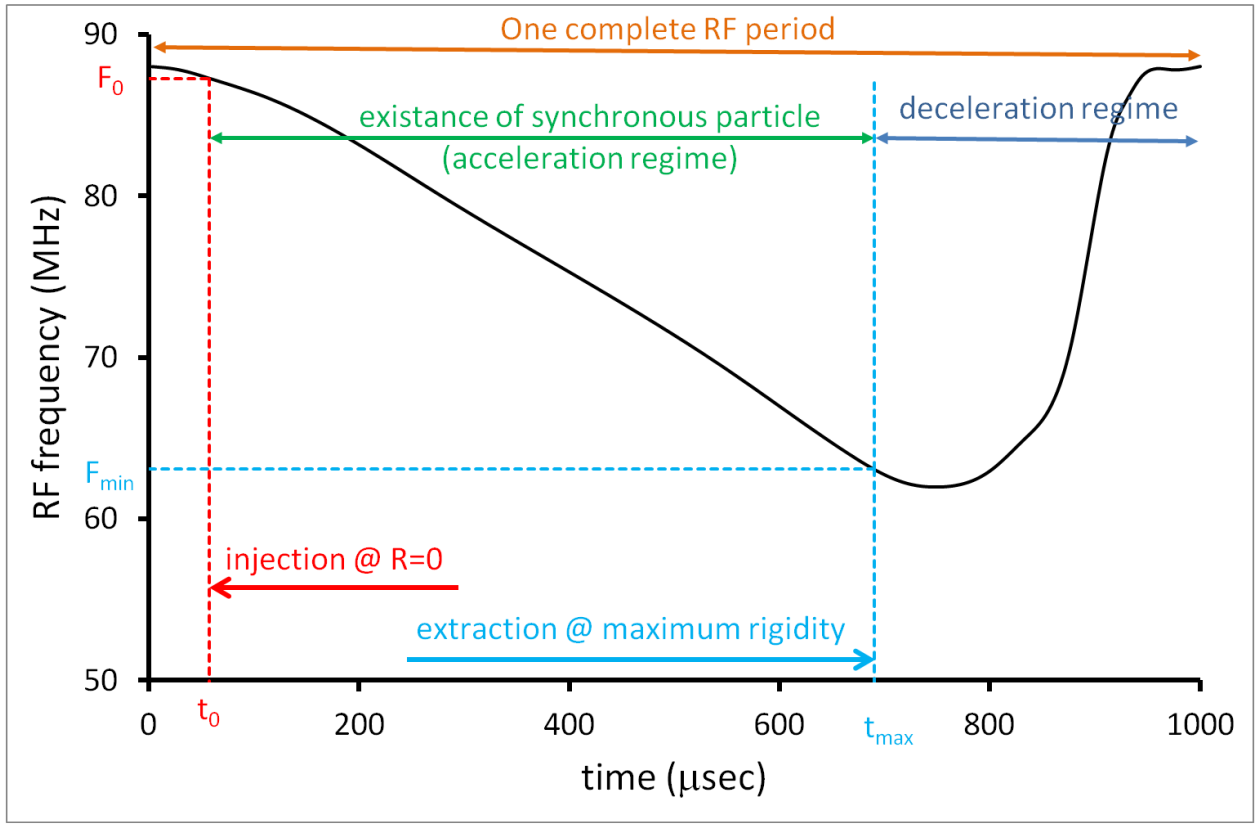
Superconducting synchro-cyclotron
Extraction energy 230 MeV

Some consequences



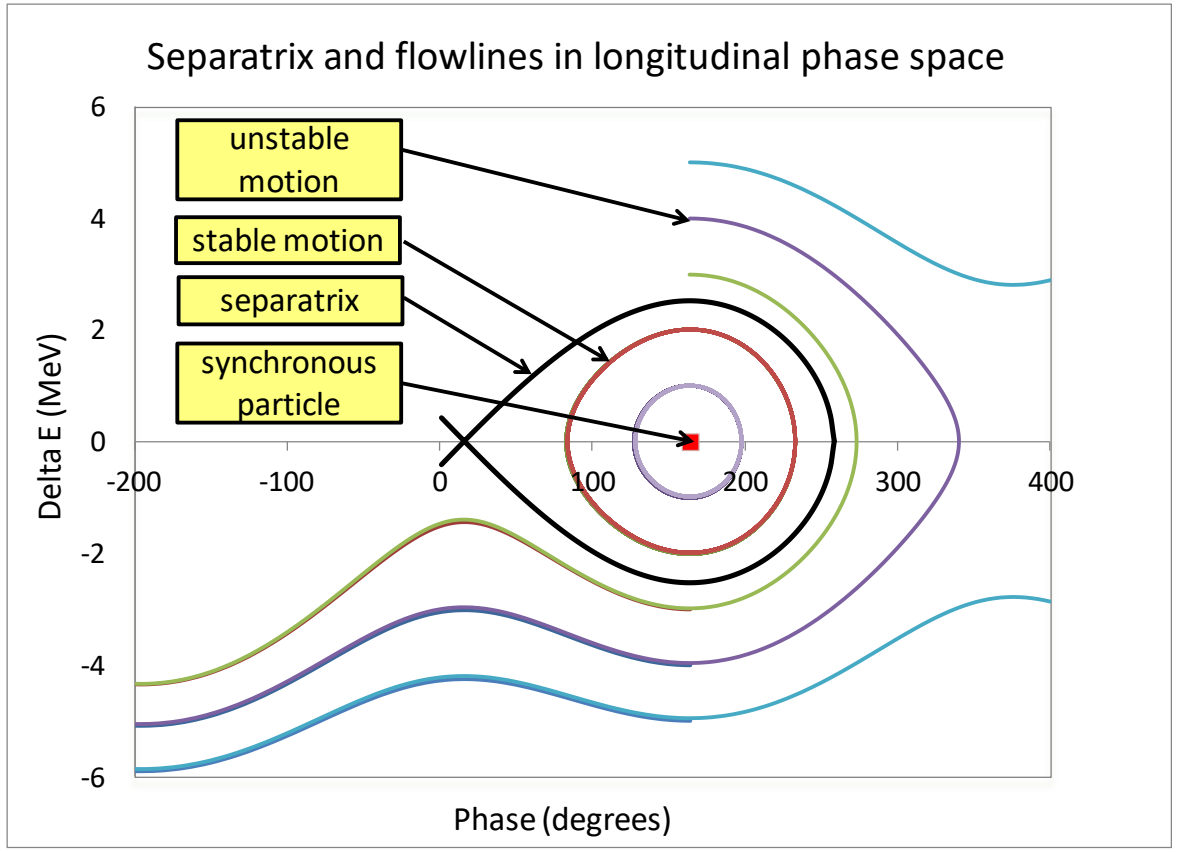
1. magnetic field B = const = B(r) ≠ B(t), but RF (and beam) is pulsed
2. Only during a short time-window, beam can be captured in the center
3. The mean beam intensity is much lower => OK for proton therapy

Some consequences



1. magnetic field B = const = B(r) ≠ B(t), but RF (and beam) is pulsed
2. Only during a short time-window, beam can be captured in the center
3. The mean beam intensity is much lower => OK for proton therapy
4. A more complicated RF because of the required frequency variation
5. RF frequency cannot be varied very fast (rotating capacitor) and therefore the acceleration must be slow
   => many turns up to extraction
   => little RF power needed
6. => very small turn-separation
   => special extraction method

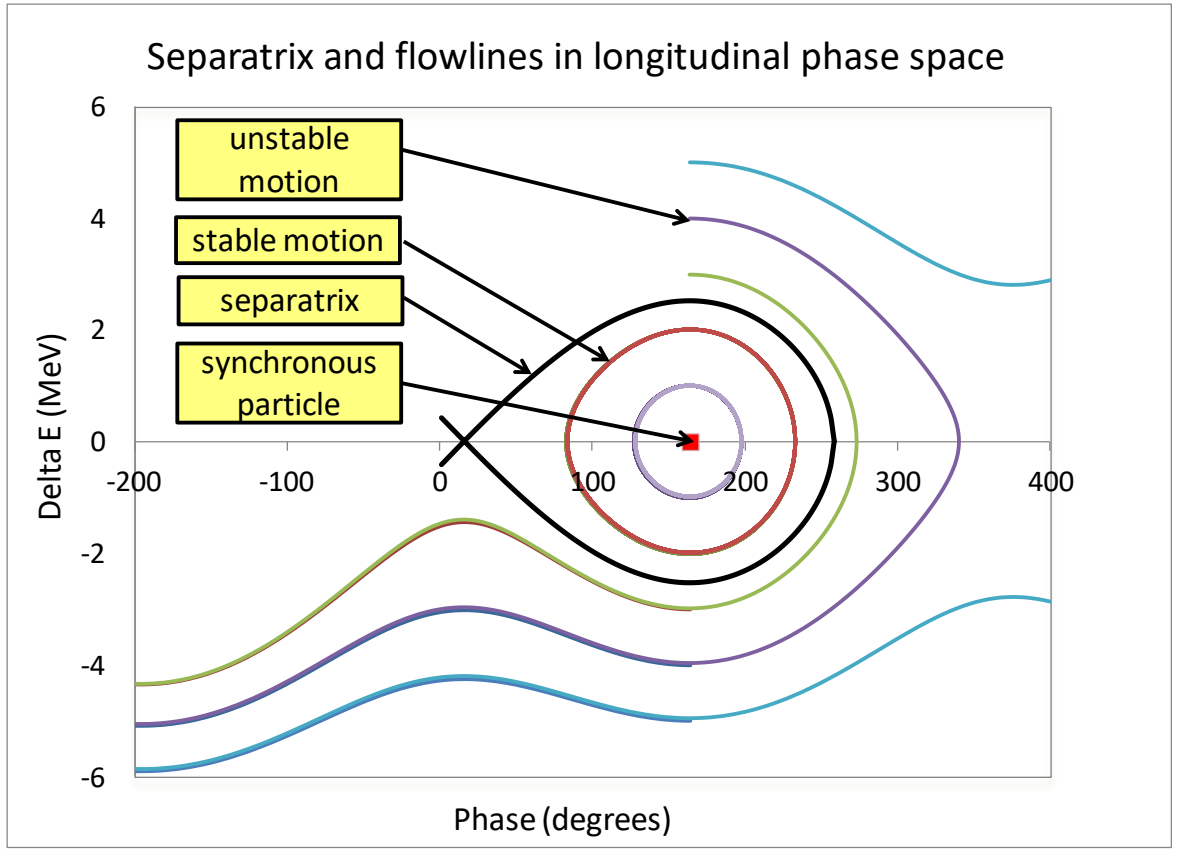
Longitudinal dynamics in a synchro-cyclotron



There is longitudinal beam dynamics similar to that of the synchrotron

- Synchronous particle: everywhere in synchro-cyclotron, at any moment in time, revolution frequency of particle is equal to RF frequency

Longitudinal dynamics in a synchro-cyclotron



There is longitudinal beam dynamics similar to that of the synchrotron

- Synchronous particle: everywhere in synchro-cyclotron, at any moment in time, revolution frequency of particle is equal to RF frequency
- There are oscillations (in energy and phase) of real particles around the synchronous particle
- There is a stability zone for these oscillations defined by a separatrix in the longitudinal phase space
- This separatrix is filled during the beam capture in the cyclotron center

Another solution: **the isochronous cyclotron**

- Two contributions to vertical focusing:

$$F_z = q(\vec{v} \times \vec{B})_z = -q(v_\theta B_r - v_r B_\theta)$$

  - $v_\theta B_r$
    => obtained in the radially decreasing rotationally symmetric magnetic fields as in the classical cyclotron and the synchro-cyclotron

  - $v_r B_\theta$
    => requires an azimuthal modulation of the magnetic field
    => introduce sectors (hills) with high field and valleys with low field
    => the field variation creates the non-circular orbit

    => azimuthally varying field cyclotron

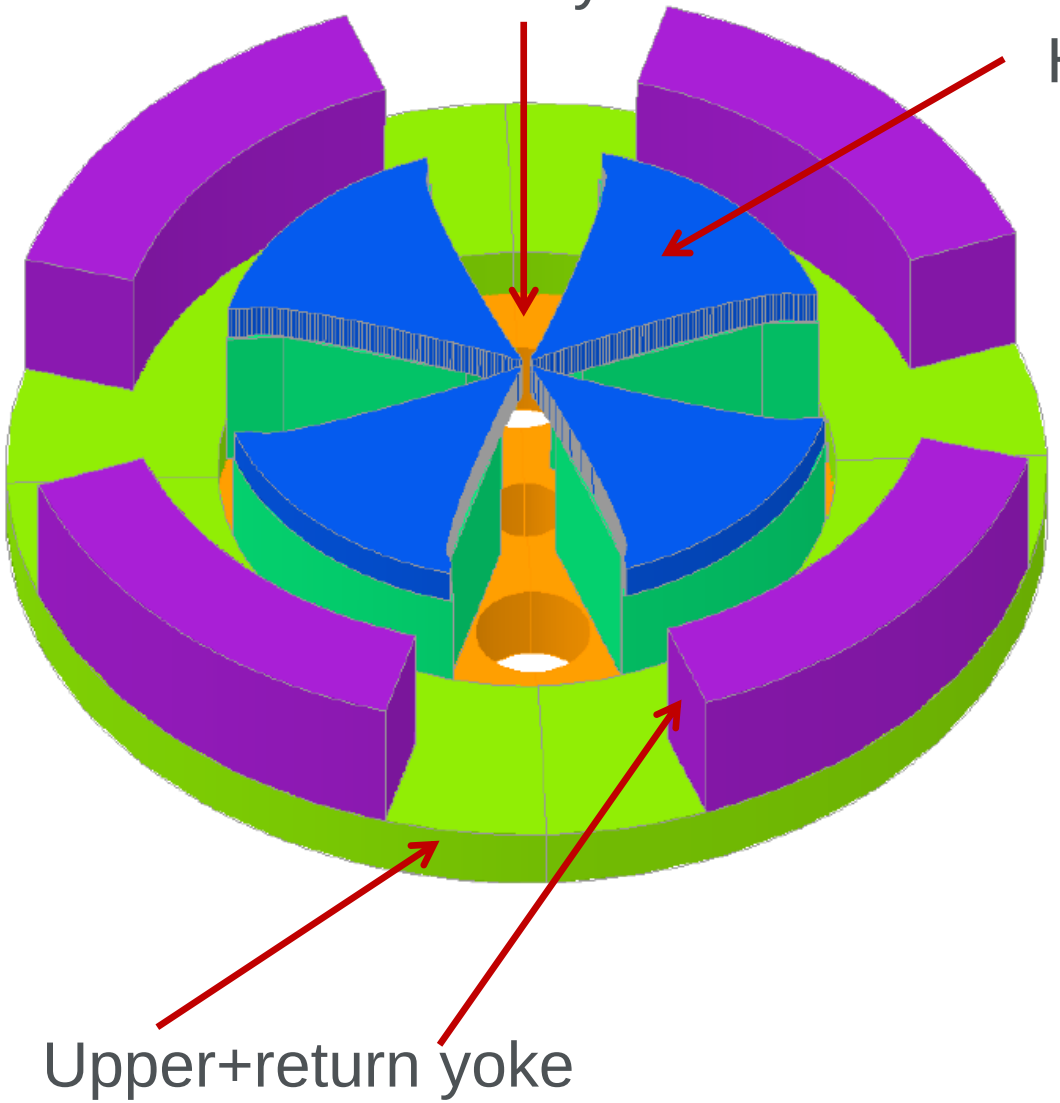
QUIZ #1

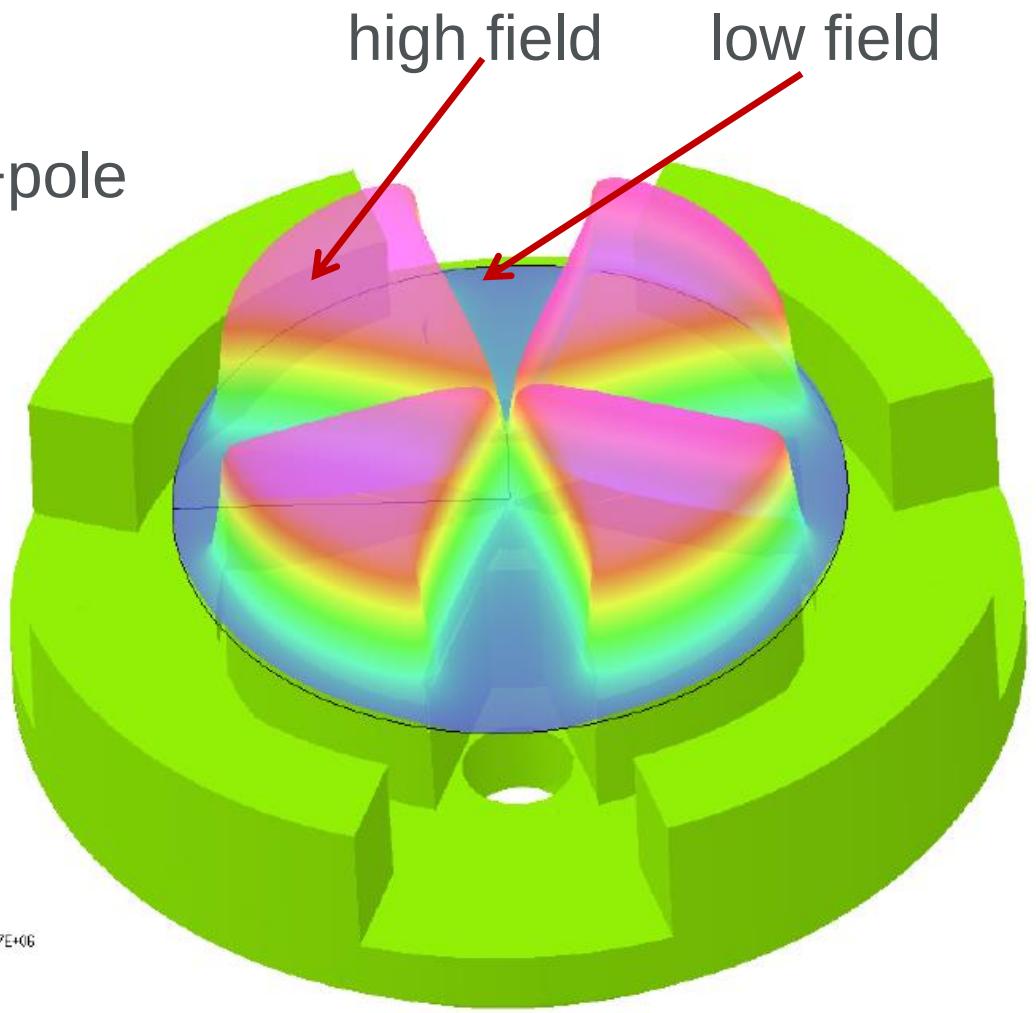
In the IBA synchro-cyclotron, does B increase or decrease for higher R ?

1. Increase
2. Decrease

Some aspects of vertical focusing and isochronism

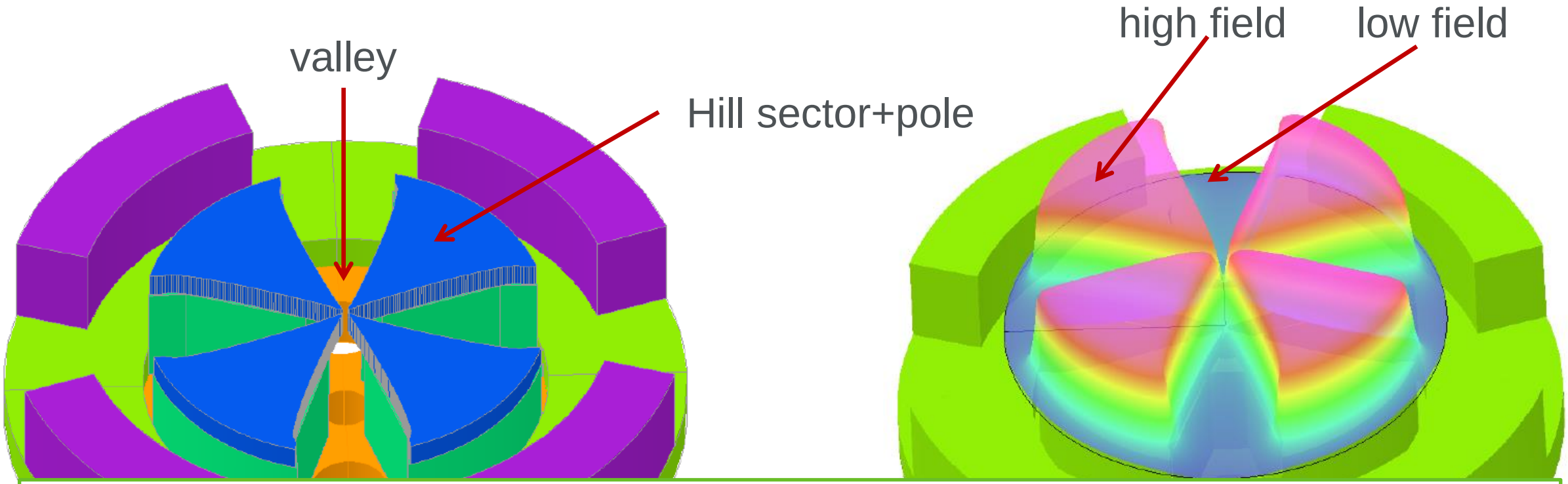
The Azimuthally Varying Field (AVF) cyclotron





Magnetic field on hills and valleys

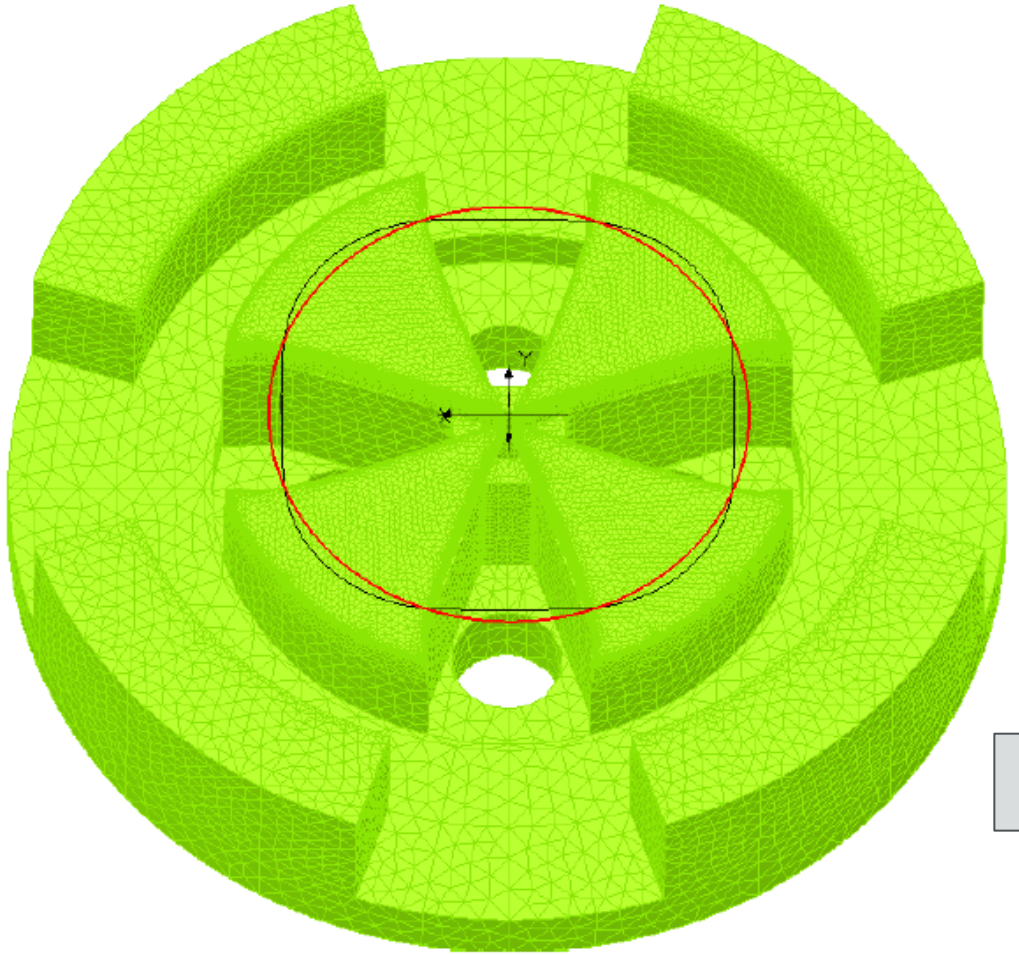
The Azimuthally Varying Field (AVF) cyclotron



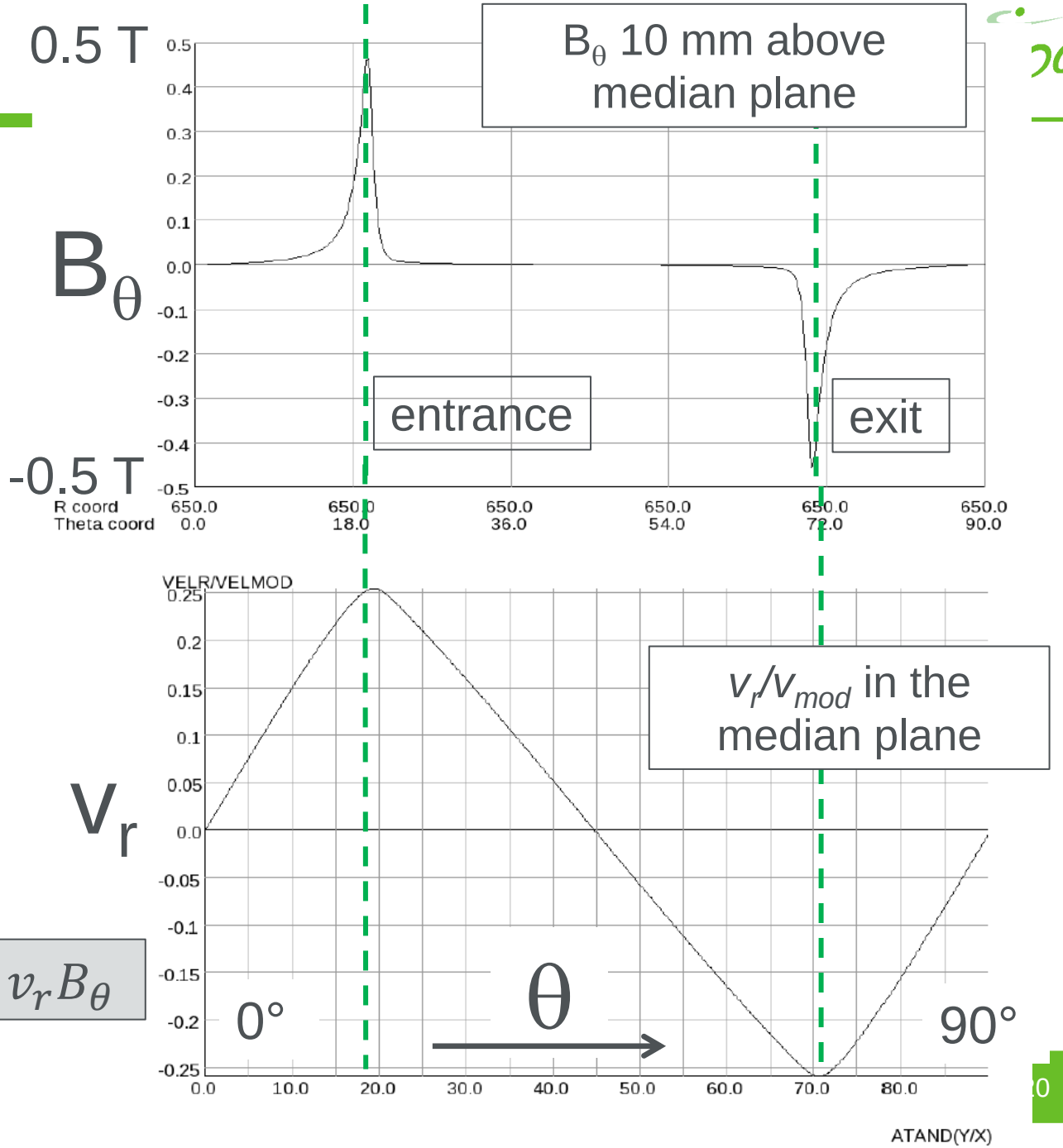
- $v_r B_\theta$
  => requires an azimuthal modulation of the magnetic field
  => introduce sectors (hills) with high field and valleys with low field
  => the field variation creates the non-circular orbit

scaloping of the orbit

=> Vertical focusing



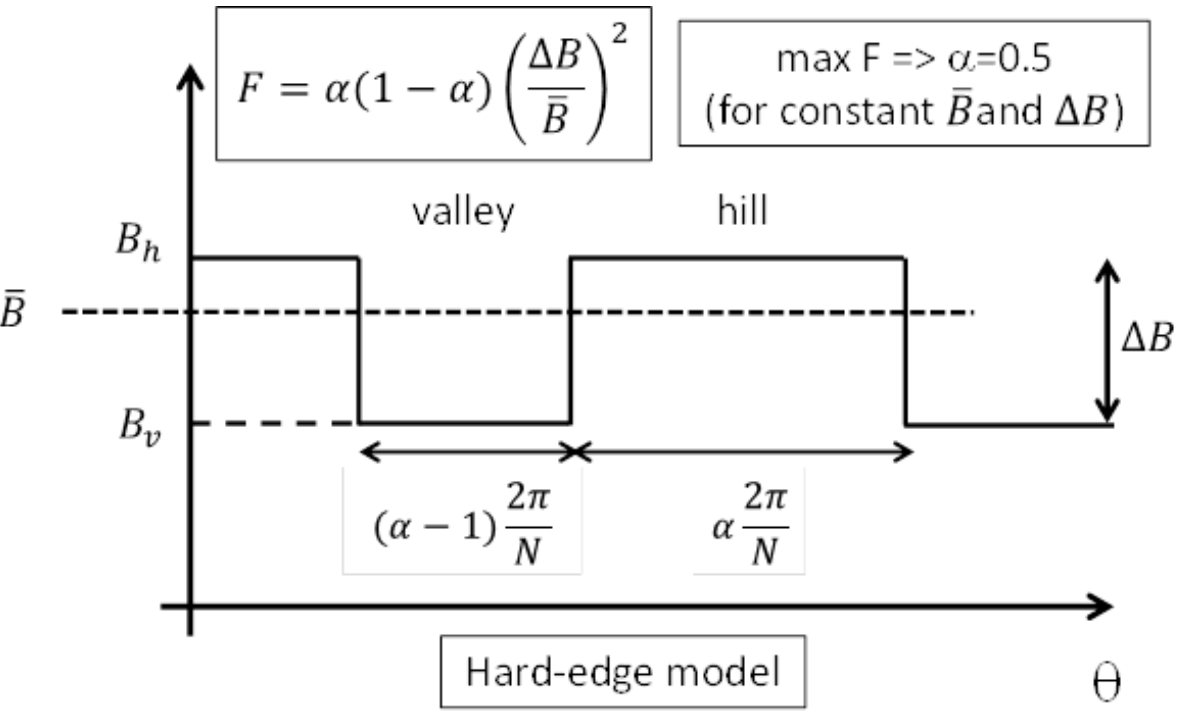
$F_z \propto v_r B_\theta$



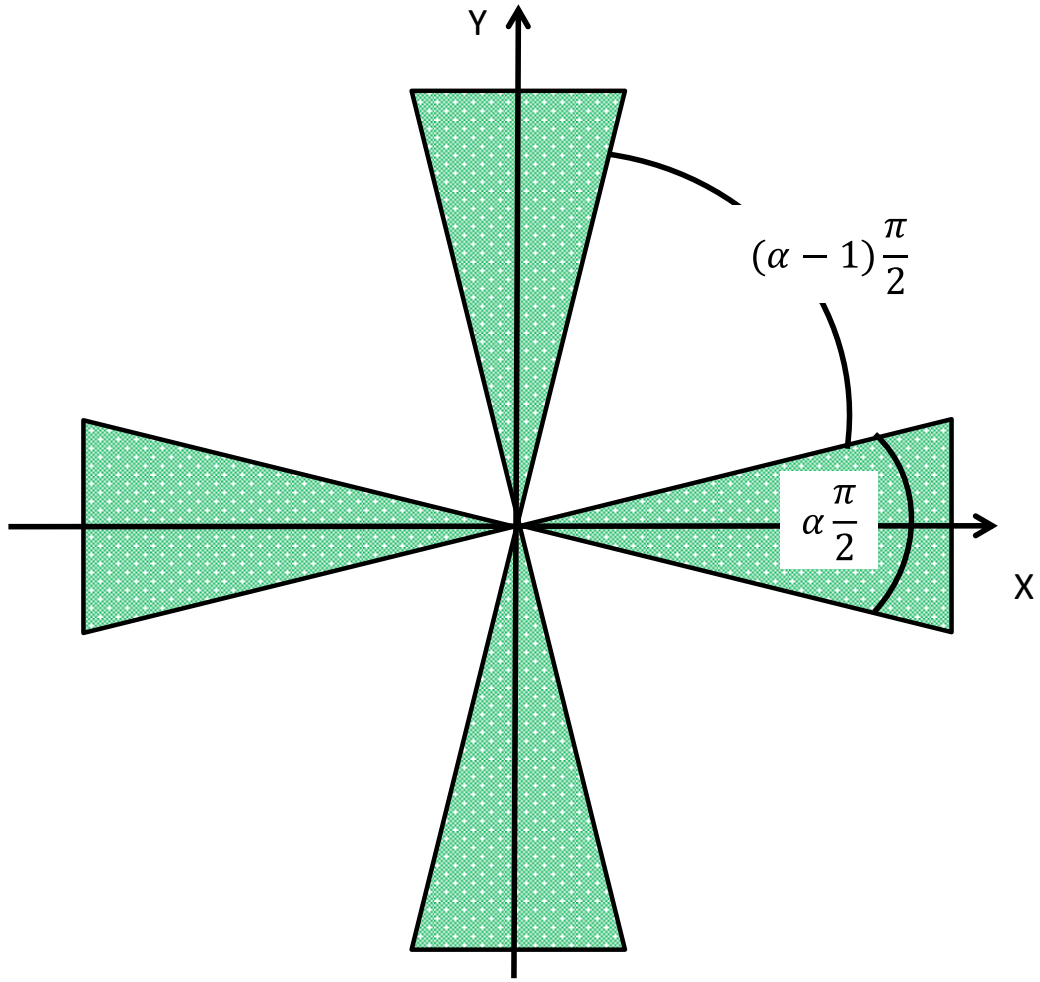
Flutter: a measure for the azimuthal field variation

- Average of the field modulation

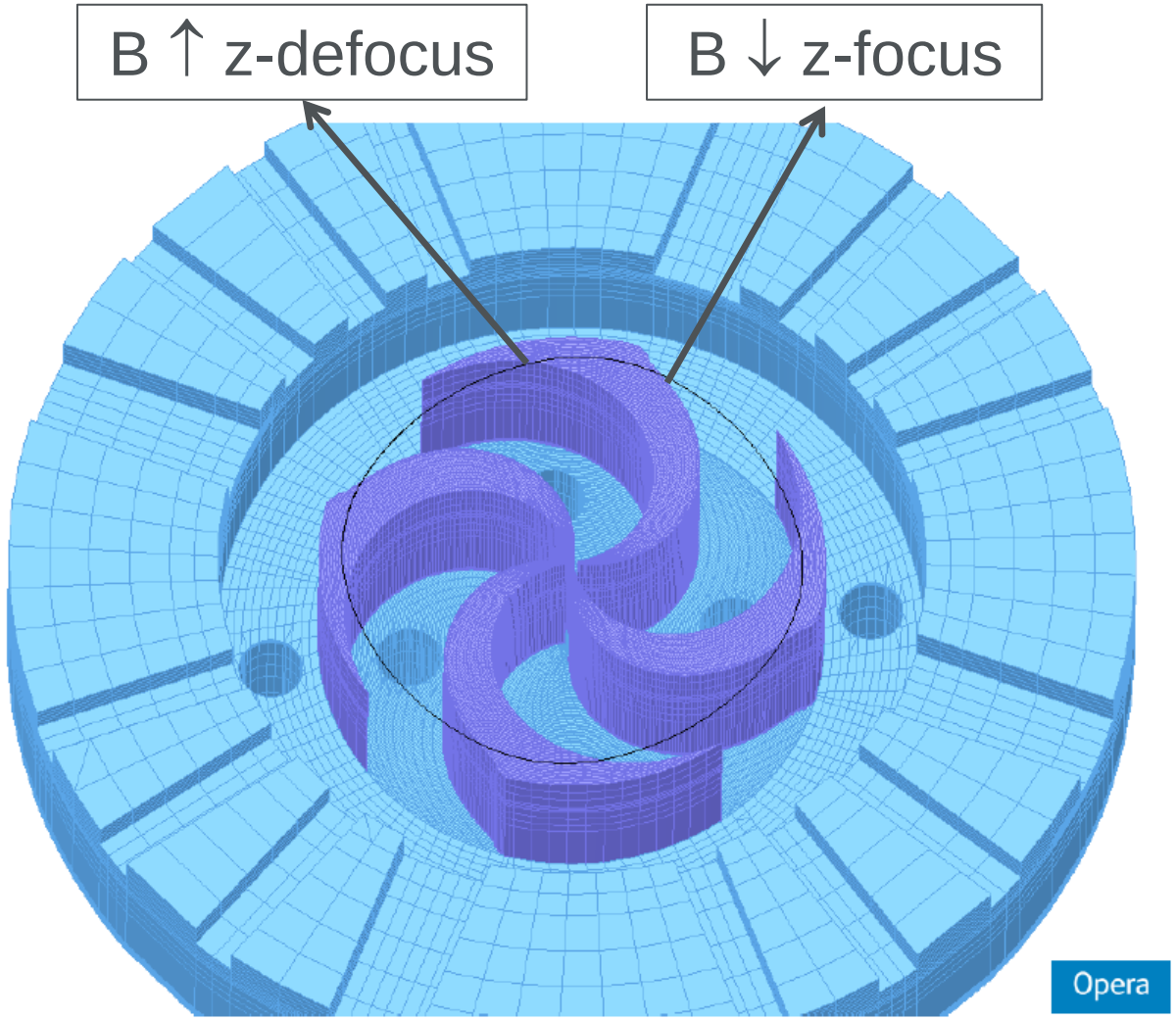
$$F(r) = \frac{\overline{B^2} - \bar{B}^2}{\bar{B}^2}$$



N=number of sectors
α='filling factor'



More vertical focusing => pole spiraling



- For straight sectors: equal vertical focusing at entrance and exit of sector
- Spiraling of pole changes focusing strength at entrance and exit of the sector:
  - Entrance: strong B-decrease => strong z-foc.
  - Exit: strong B-increase => strong z-defoc.

ALTERNATING FOCUSING
This may give a very large contribution

Formulas for focusing in an AVF cyclotron

$$\upsilon_z{}^2 = k + \frac{N^2}{N^2 - 1} F$$

$$\upsilon_r{}^2 = (1 - k) + \frac{3N^2}{(N^2 - 1)(N^2 - 4)} F$$

F = flutter

N = number of sectors

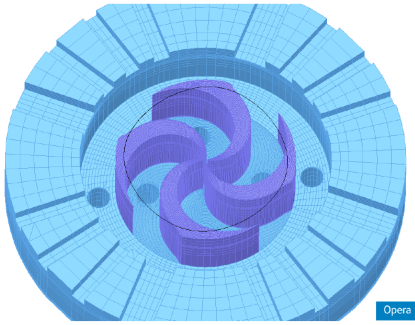
k = field index = $-\frac{r}{\bar{B}} \frac{d\bar{B}}{dr}$

NB - for an isochronous cyclotron:

$$k = 1 - \gamma_{\mathrm{rel}}{}^2$$

$$\upsilon_r \cong \gamma_{\mathrm{rel}}$$

Formulas for focusing in an AVF cyclotron



$$\upsilon_z{}^2 = k + \frac{N^2}{N^2-1} F(1 + 2\tan^2\xi)$$

$$\upsilon_r{}^2 = (1-k) + \frac{3N^2}{(N^2-1)(N^2-4)} F(1 + \tan^2\xi)$$

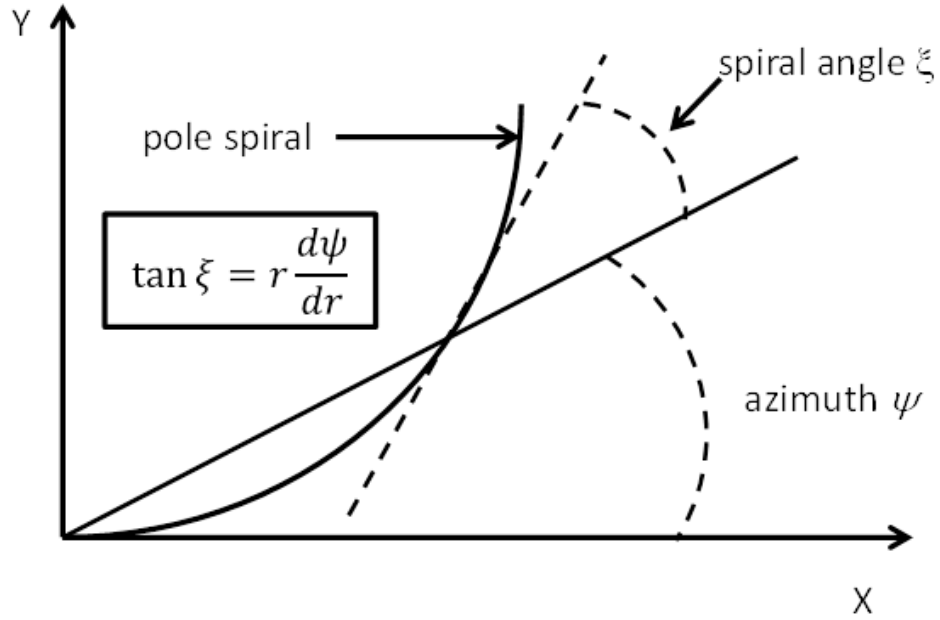
F = flutter

N = number of sectors

ξ = spiral angle

k = field index = $-\frac{r}{\bar{B}}\frac{d\bar{B}}{dr}$

Definition of spiral angle



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F = flutter

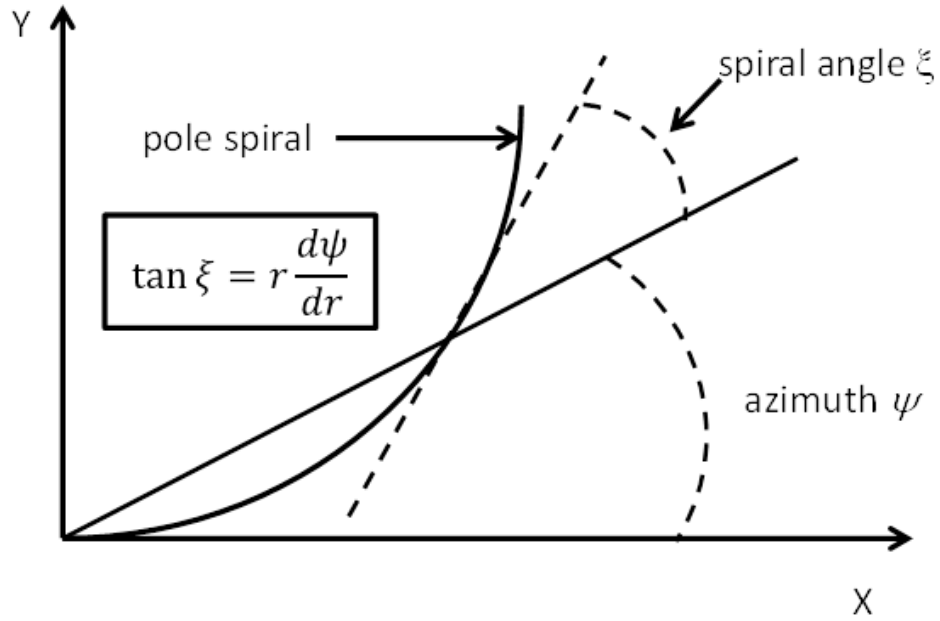
N = number of sectors

ξ = spiral angle

k = field index = $-\frac{r}{\bar{B}}\frac{d\bar{B}}{dr}$

This is an approximation: There is also some dependency on radial gradients of the flutter.
See: Hagedoorn and Verster, NIM **18,19** (1962) 201-228

Definition of spiral angle



NB - for an isochronous cyclotron:

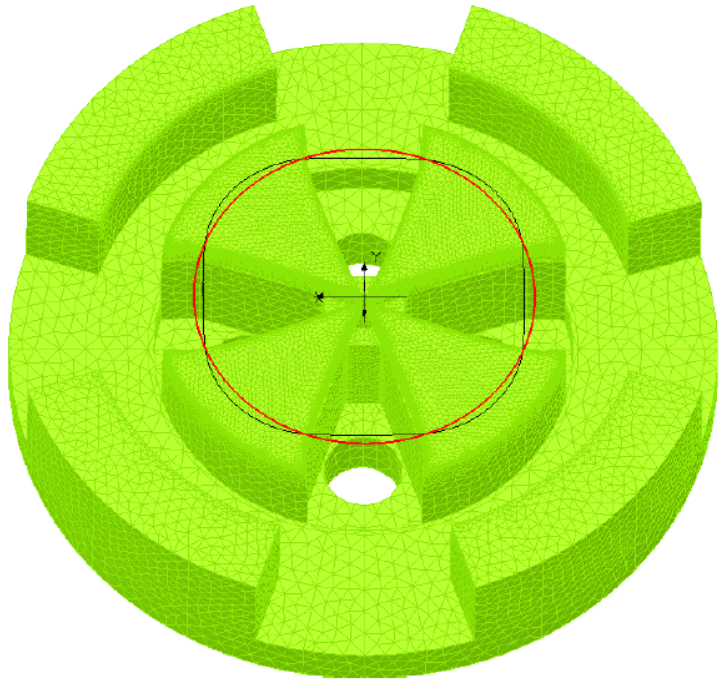
$$k = 1 - \gamma_{\mathrm{rel}}{}^2$$

$$\upsilon_r \cong \gamma_{\mathrm{rel}}$$

Isochronism

- Isochronism => revolution frequency of particle is constant everywhere in the cyclotron independent of the energy of the particle

- Isochronous cyclotrons have to be isochronized by correct shaping of the average magnetic field as a function of radius

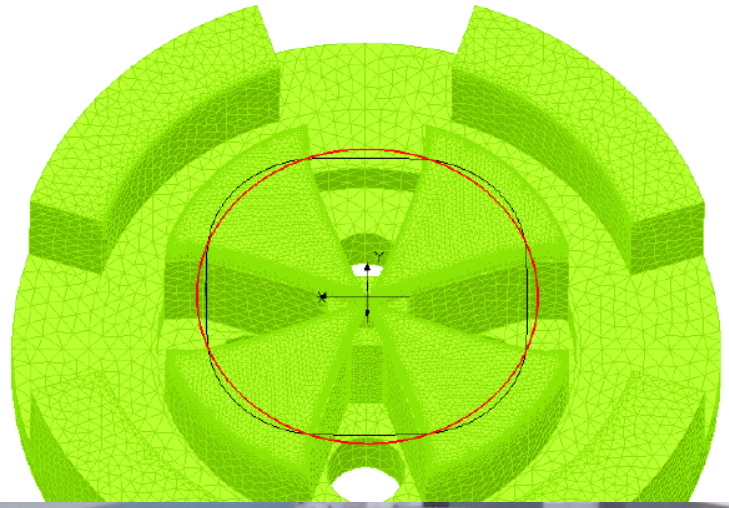


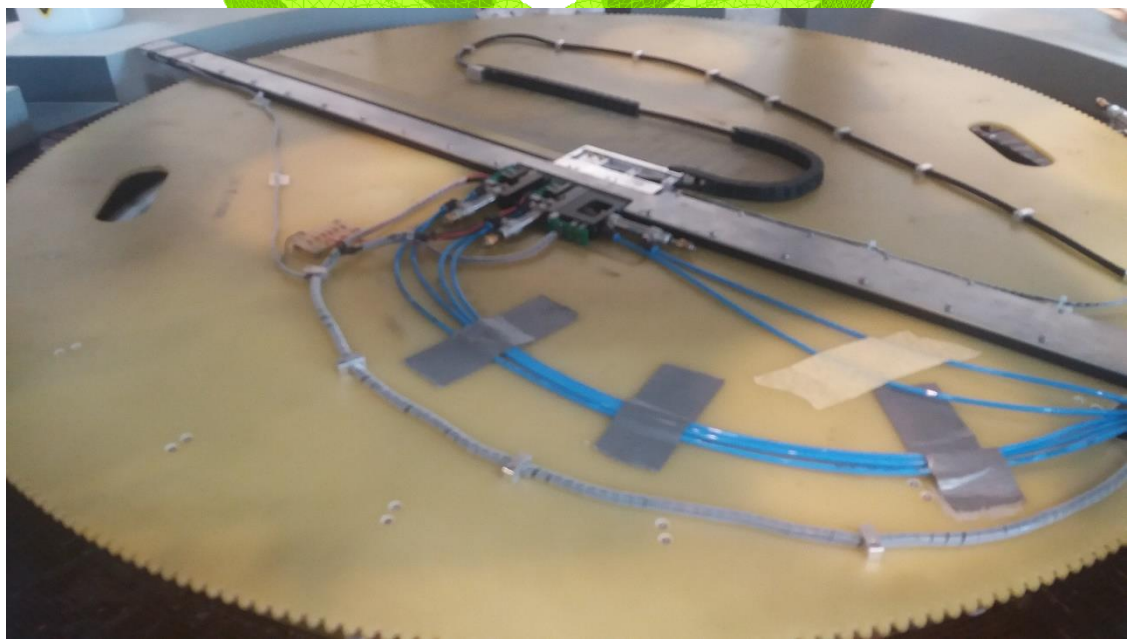
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- It is not possible to obtain isochronism just from the design
  => required precision <B> ~ 10^{-4} to 10^{-5}

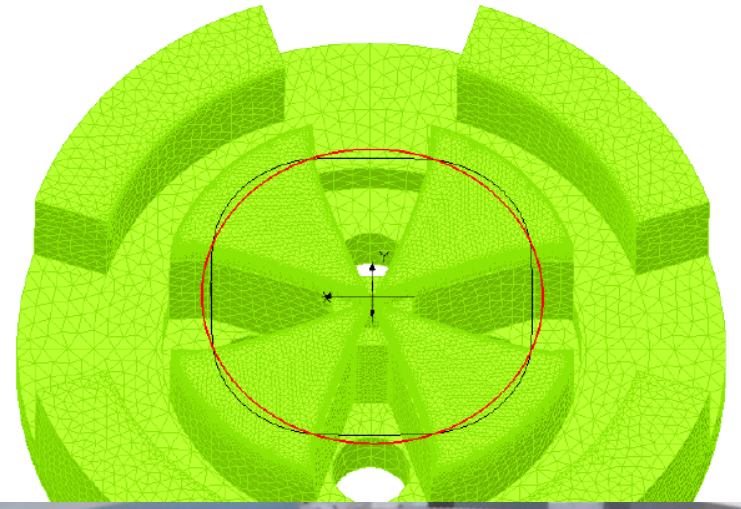


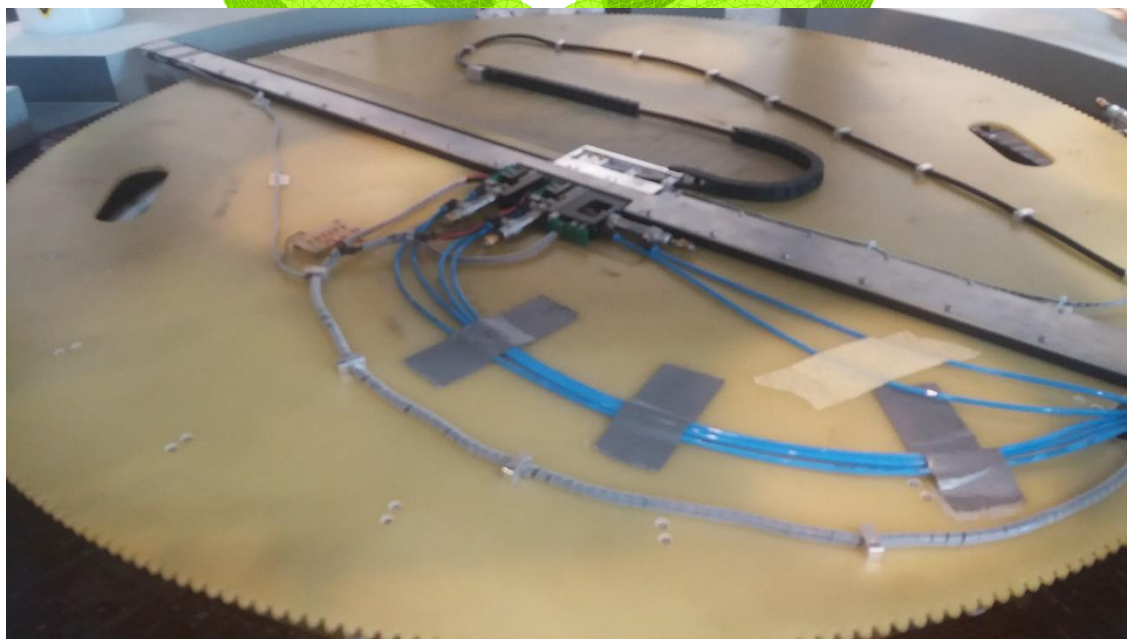


All cyclotron B-fields are mapped in median plane

- Precise mapping and iron pole shimming is needed in order to isochronize the magnetic field

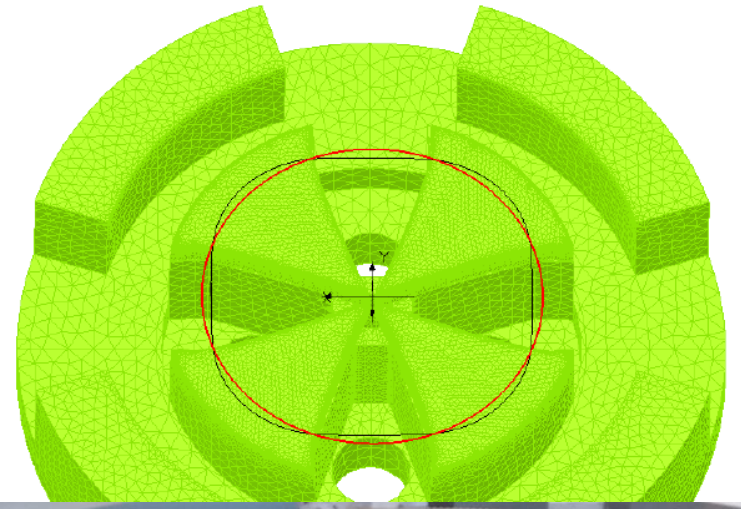
- Move Hall-probe or a search coil (S2C2) on a 2D polar grid to obtain a full field-map, by automized and computer controlled system
- Analyse magnetic field on equilibrium orbits in order to evaluate isochronism
- Shim the iron in order to improve the isochronism (reduce the RF phase slip)

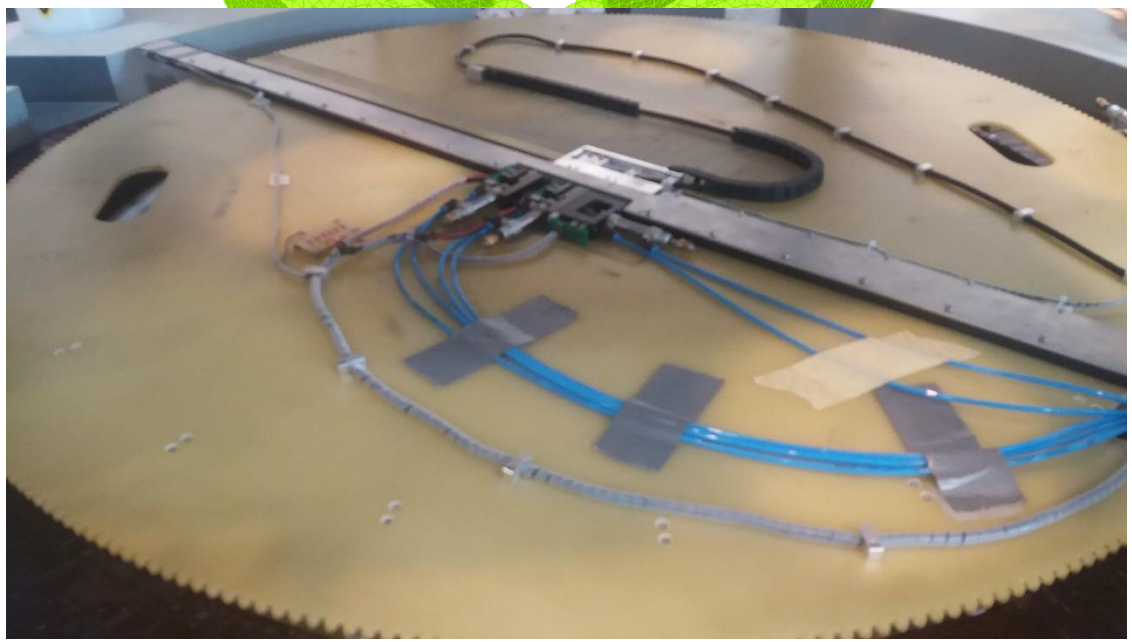




Essential information of a cyclotron field map

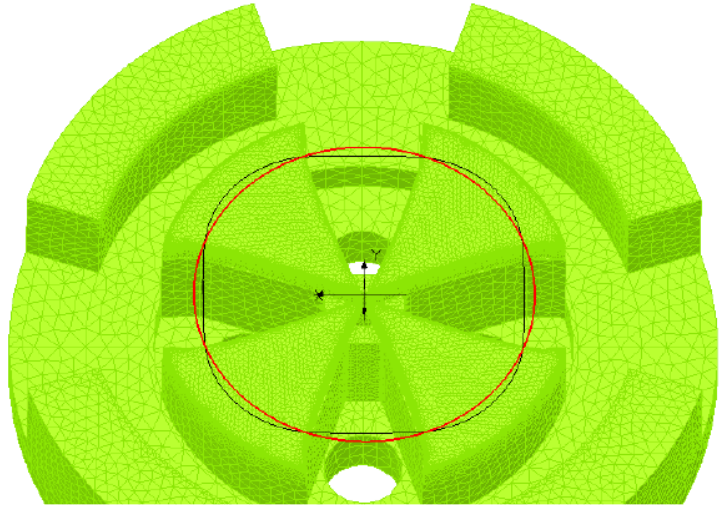
1. Level of isochronism
   => integrated RF phase slip
2. Transverse optical stability
   => tune functions
3. Crossing of dangerous resonances
   => operating diagram
4. Magnetic field errors
   - First and second harmonic errors
     => resonance drivers
   - Median plane errors
     => very difficult to measure
5. …



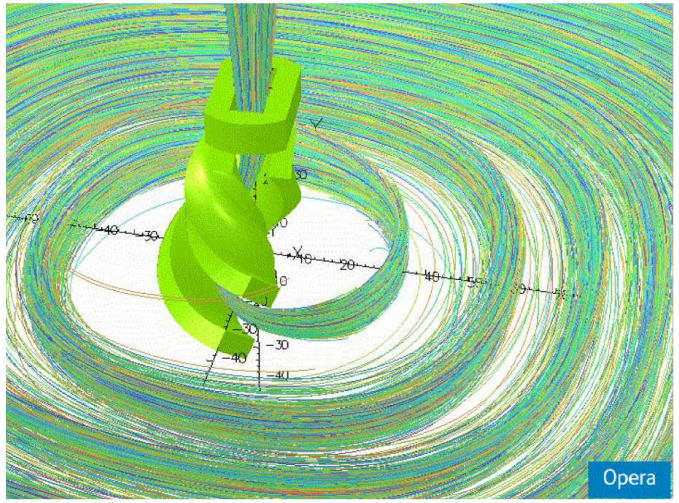


Closed Orbit Calculation vs Accelerated Orbits

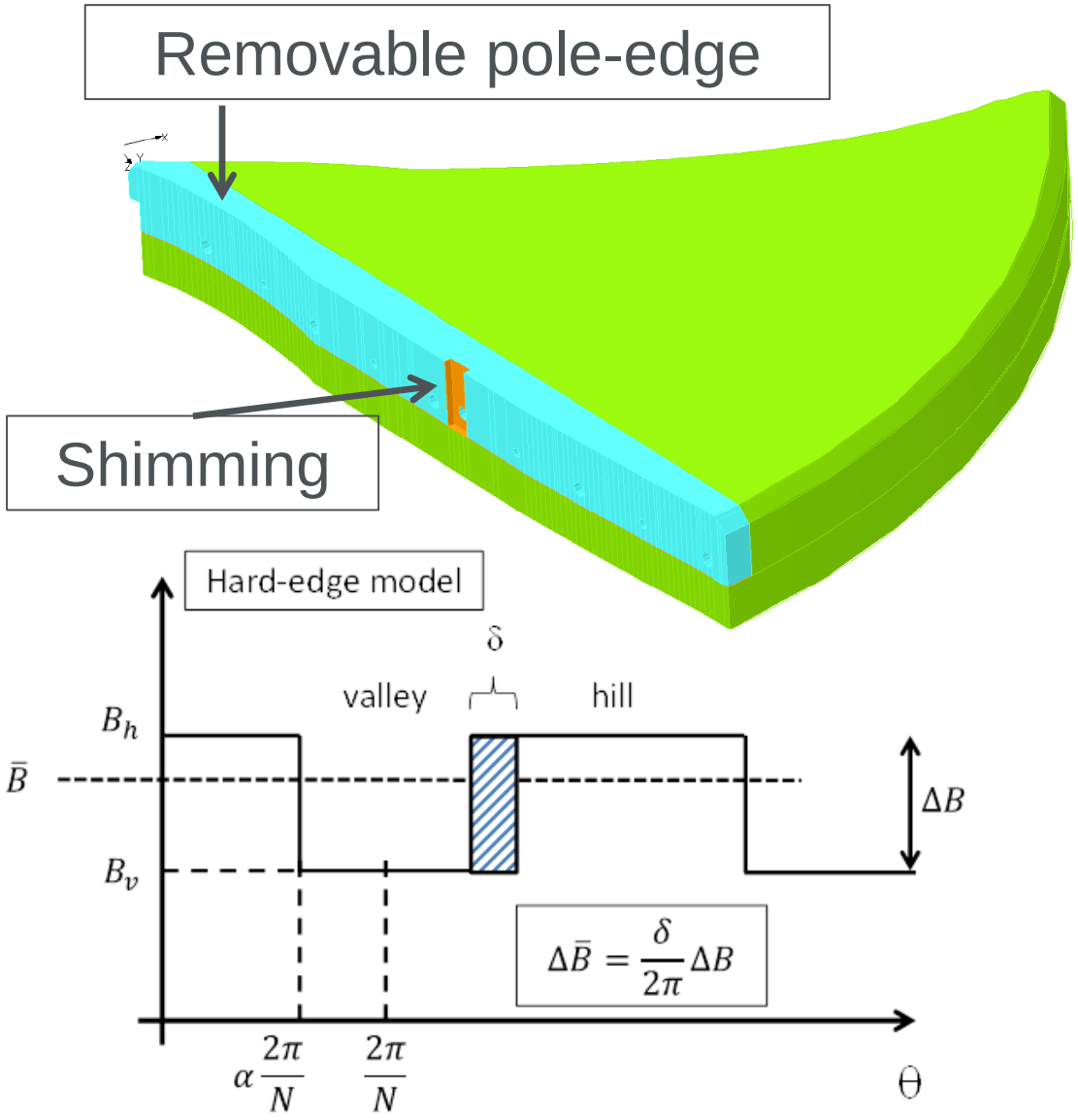
1. Static analysis
   => Acceleration is turned off
   - Computation of a family of closed orbits and their properties



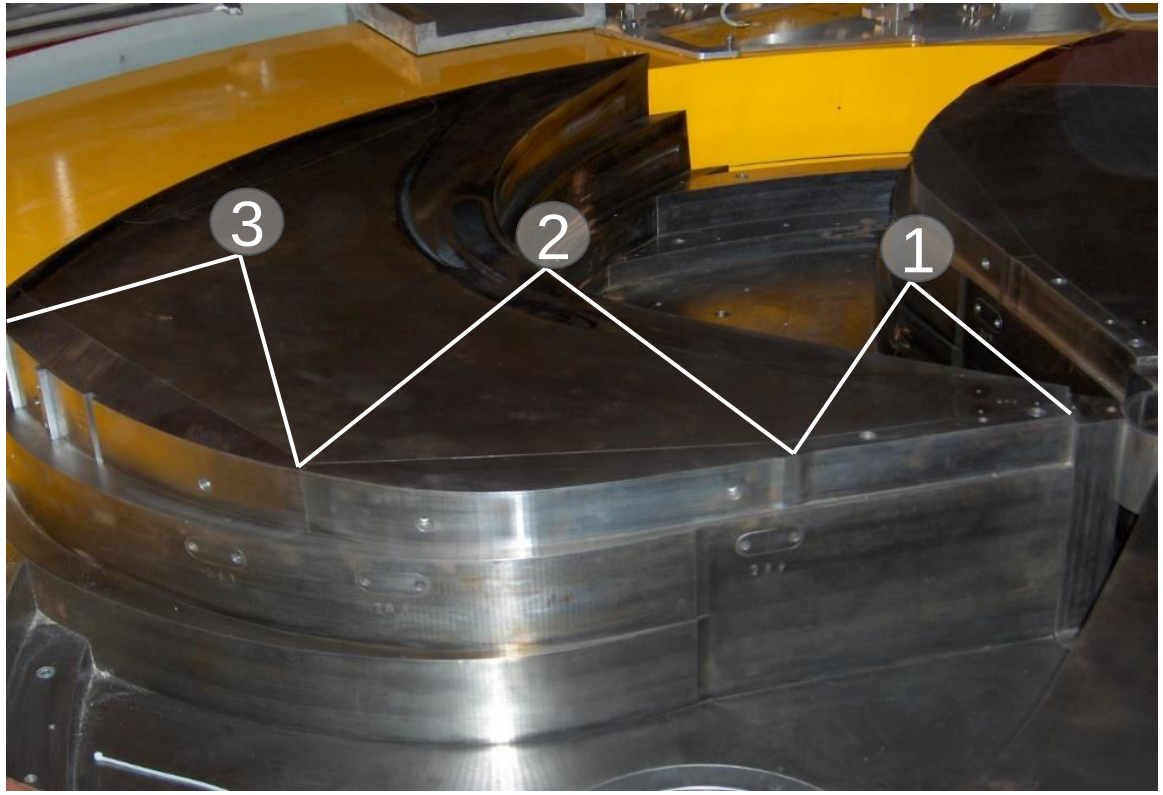
2. Accelerated orbits
   => for special problems
   - Central region studies and design
   - Extraction studies and design
   - Study of resonance crossings
   - …
     ➔ Full transmission



Isochronization by pole shimming



Removable pole edges in C235

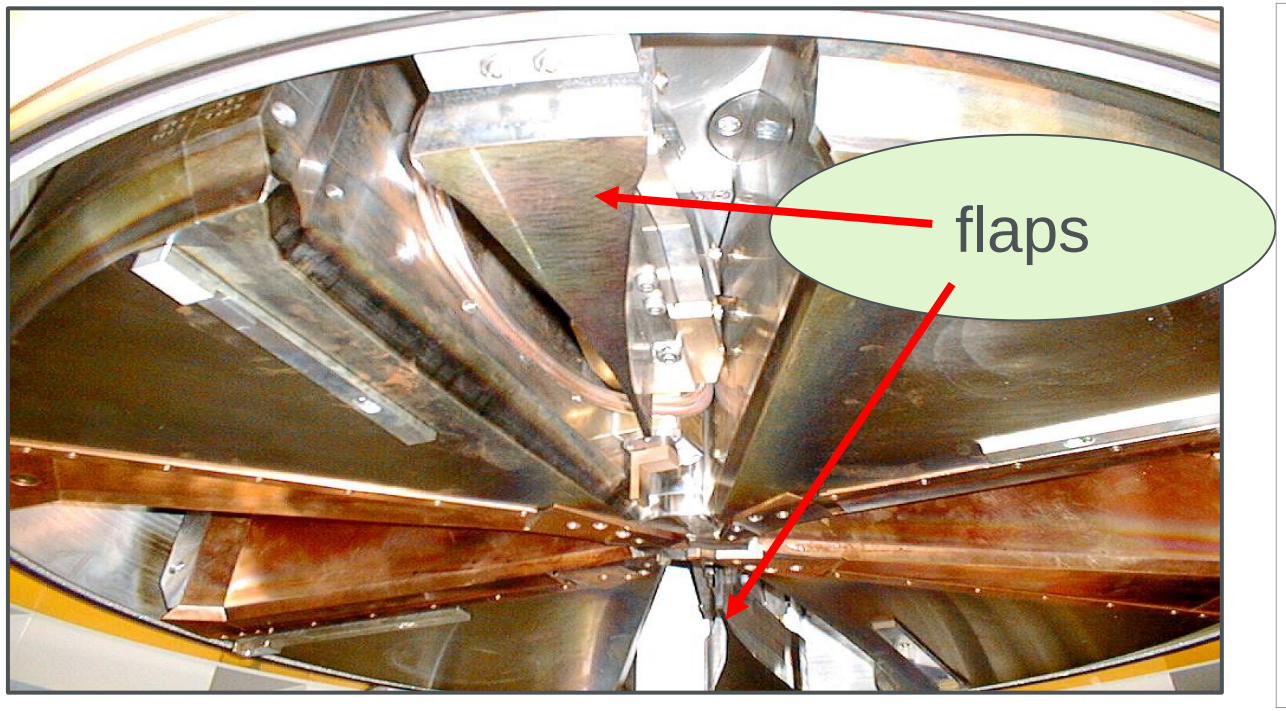


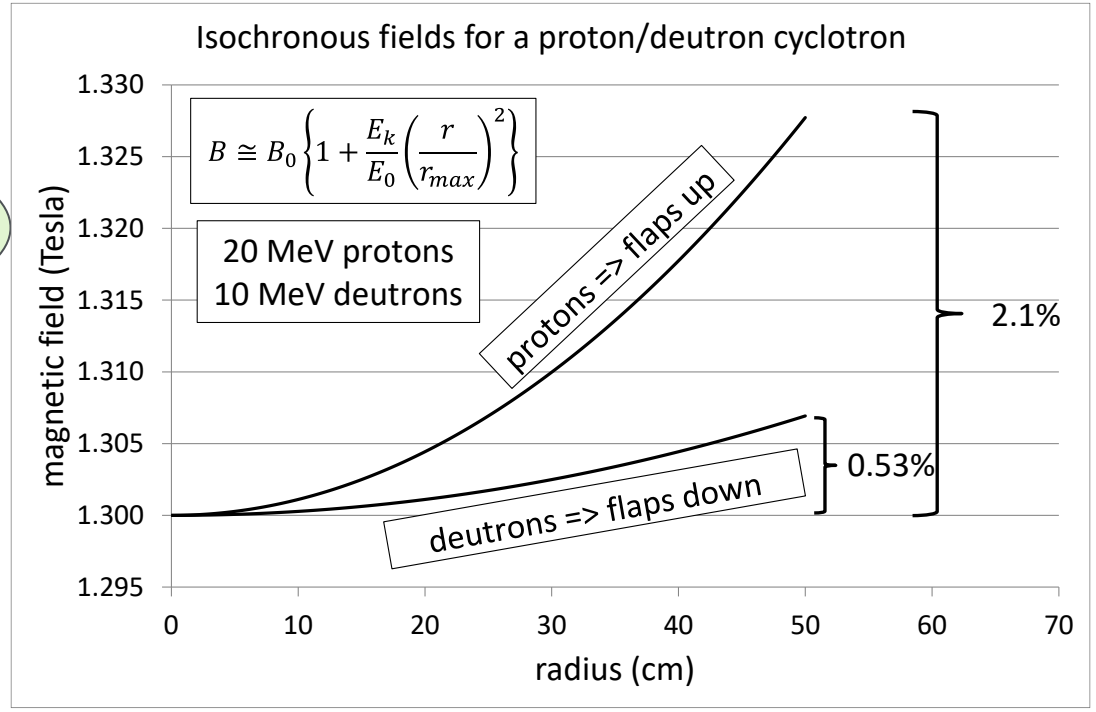
In this PT cyclotron, there are 3 removable pole edges (per pole) for shimming the average field as needed for isochronism

Isochronization for two different particles

Example: a proton/deuteron isotope production cyclotron

- Special iron pieces (flaps) in the valleys can be moved vertically
- Closer to the median plane (for protons) or further away from the median plane (for deuterons)





QUIZ #2

Select one or more of the following answers. A closed orbit analysis gives us:

1. injection efficiency
2. isochronism
3. optical stability
4. transmission

Injection

Injection into a cyclotron

- Transfer of the beam from the ion source onto the equilibrium orbit in the center of the cyclotron, two appoaches:

1. Internal Ion Source:
   - Ion source placed in the center of the cyclotron
   - Source is ‘integrated part’ of the accelerating stucture
   - Is used in PT cyclotrons as well as isotope production cyclotrons

Injection into a cyclotron

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   - Source is ‘integrated part’ of the accelerating stucture
   - Is used in PT cyclotrons as well as isotope production cyclotrons
2. External Ion Source:
   - Ion source placed outside of the machine
   - An injection line with magnets and electrostatic inflector is needed
   - Used in high intensity isotope production cyclotrons (and in IBA C400)

Injection: some important design goals

1. Vertical beam centering with respect to the median plane
2. Horizontal beam centering : Place the beam on the correct equilibrium orbit given by the injection energy

Injection: some important design goals

1. Vertical beam centering with respect to the median plane
2. Horizontal beam centering : Place the beam on the correct equilibrium orbit given by the injection energy
3. Longitudinal matching
   => bunching
   => compressing the DC beam from the ion source into shorter packages at the frequency of the RF
4. Matching of the beam phase space into the cyclotron acceptance or eigenellipse (if possible)

➢ Preserve as well as possible the beam quality with minimum losses between the ion source and the cyclotron center

Injection: internal ion source

Some advantages

- Simple and cost-effective: simple ion source; no injection line needed
- Compact:
  - two ion sources can be placed simultaneously
  - Can be used in the high-field (6 to 9 Tesla) superconducting cyclotrons

Injection: internal ion source

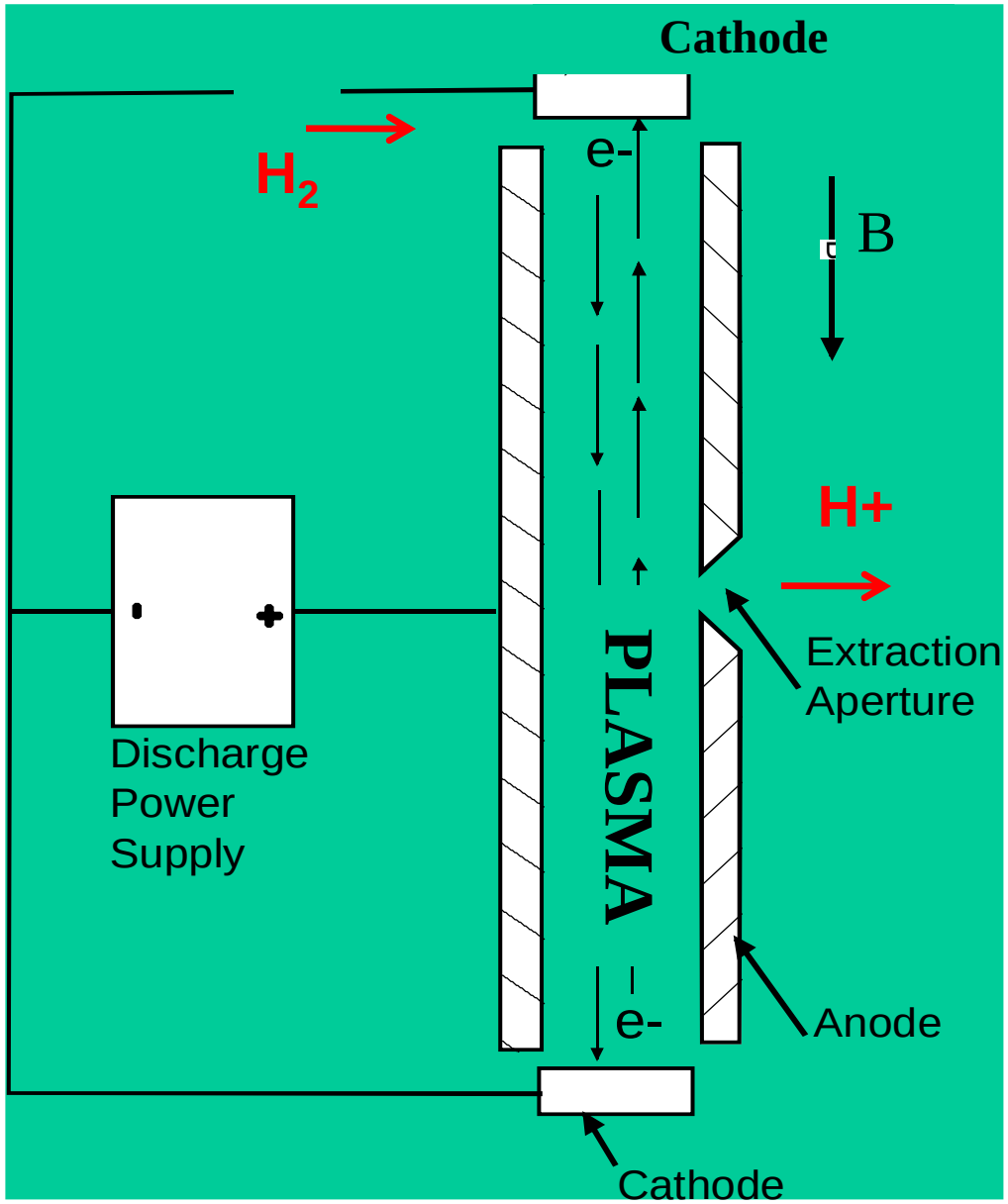
Some advantages

- Simple and cost-effective: simple ion source; no injection line needed
- Compact:
  - two ion sources can be placed simultaneously
  - Can be used in the high-field (6 to 9 Tesla) superconducting cyclotrons

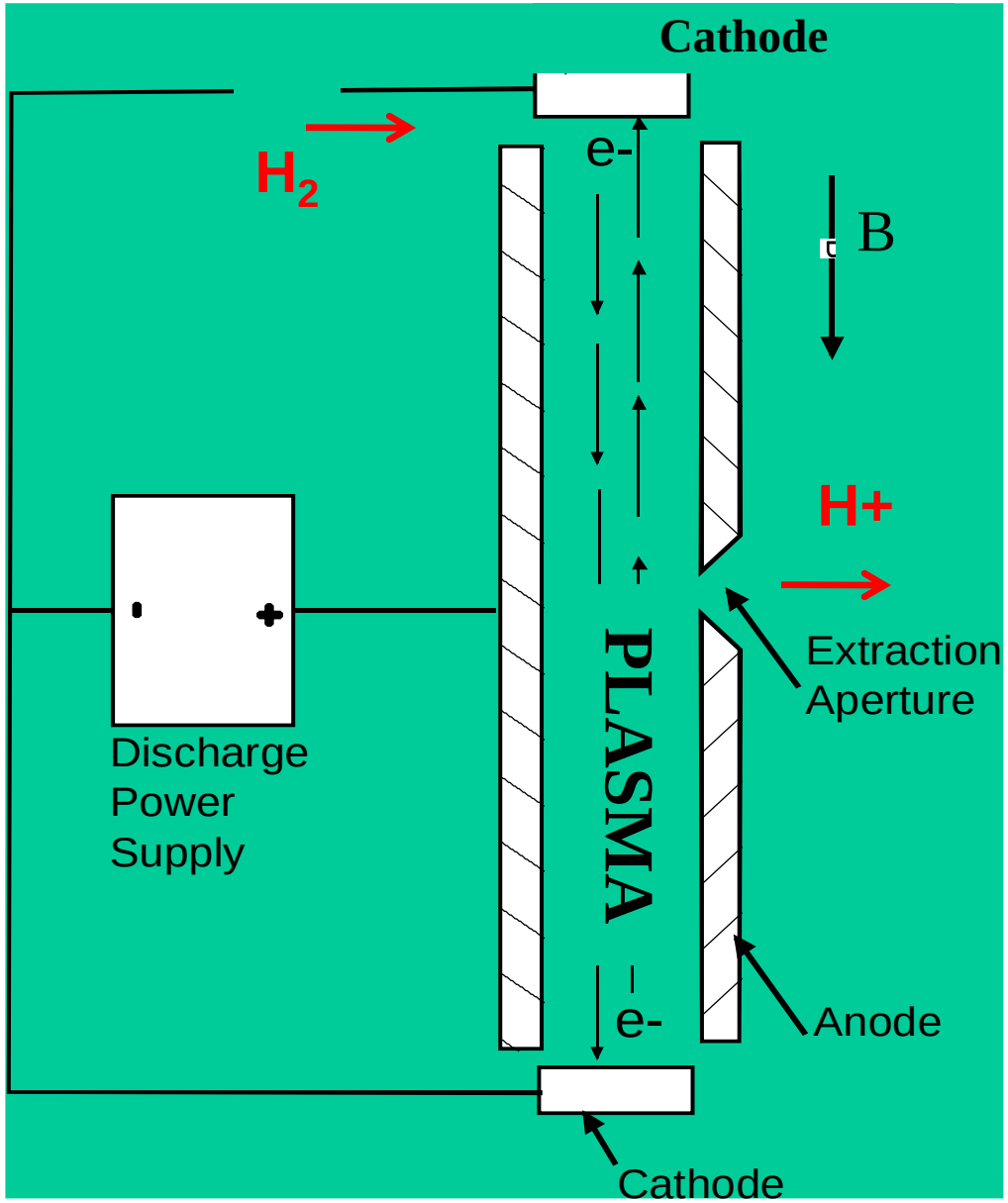
Some disavantages/limitation

- Low to moderate beam intensities
- Simple ion species (H^+,H^-,deuterons,He-3, He-4)
- Beam matching/bunching/manipulation not possible
- Gas-leak directly into the cyclotron (bad for negative ions)
- Machine has to be vented for ion source maintenance

Injection: cold cathode PIG ion source

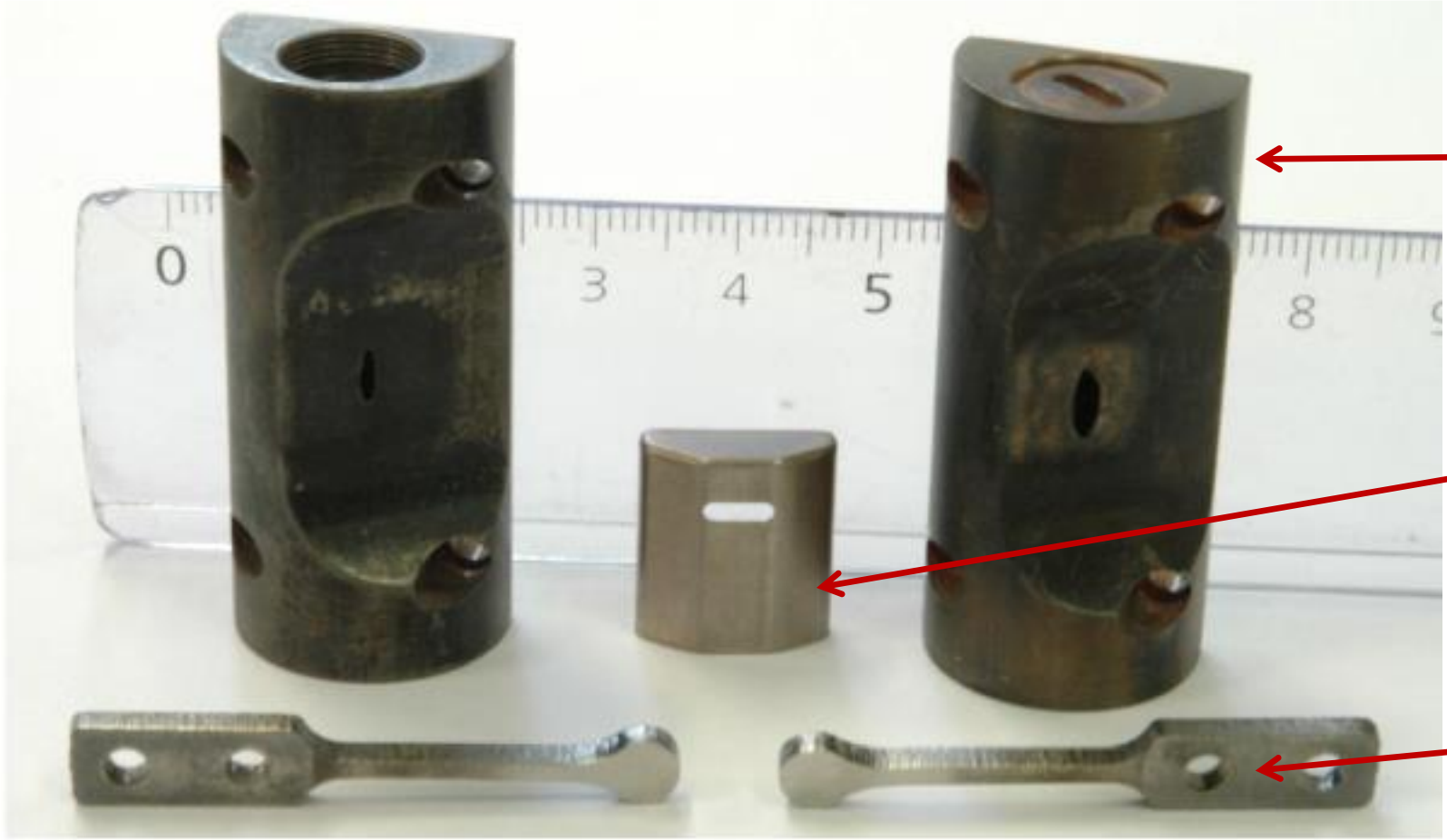


Injection: cold cathode PIG ion source



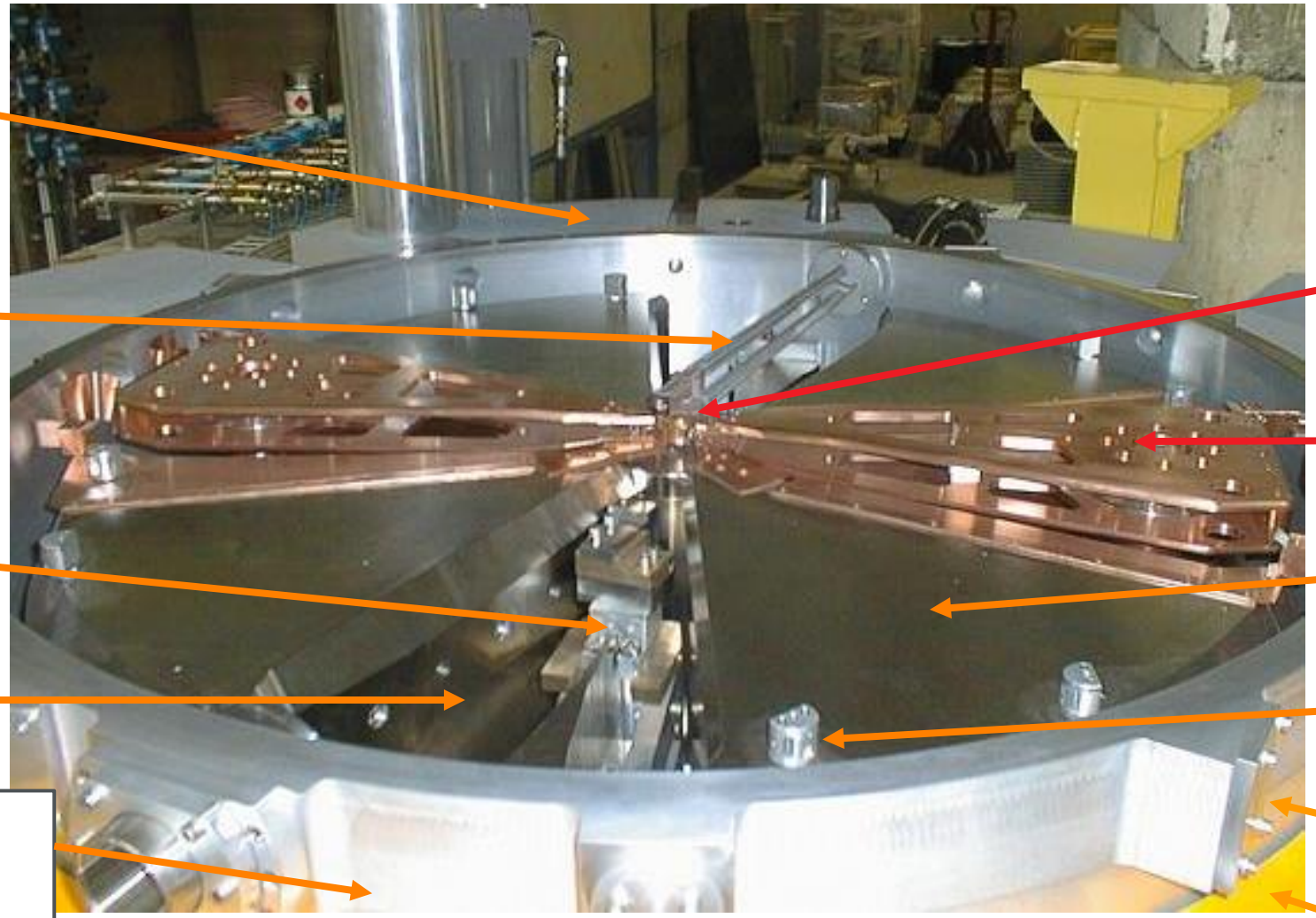
- Electron emission due to electrical potential on the cathodes
- Electron confinement due to the magnetic field along the anode axis
- Electrons produced by thermionic emission and ionic bombardment
  - Start-up: 3 kV to strike an arc
  - At the operating point : 100 V
- cathodes heated by the plasma (100 V is enough to pull an outer e- off the gas atoms)
- Hot cathode PIG => heated with filament

Chimney, cathodes and puller

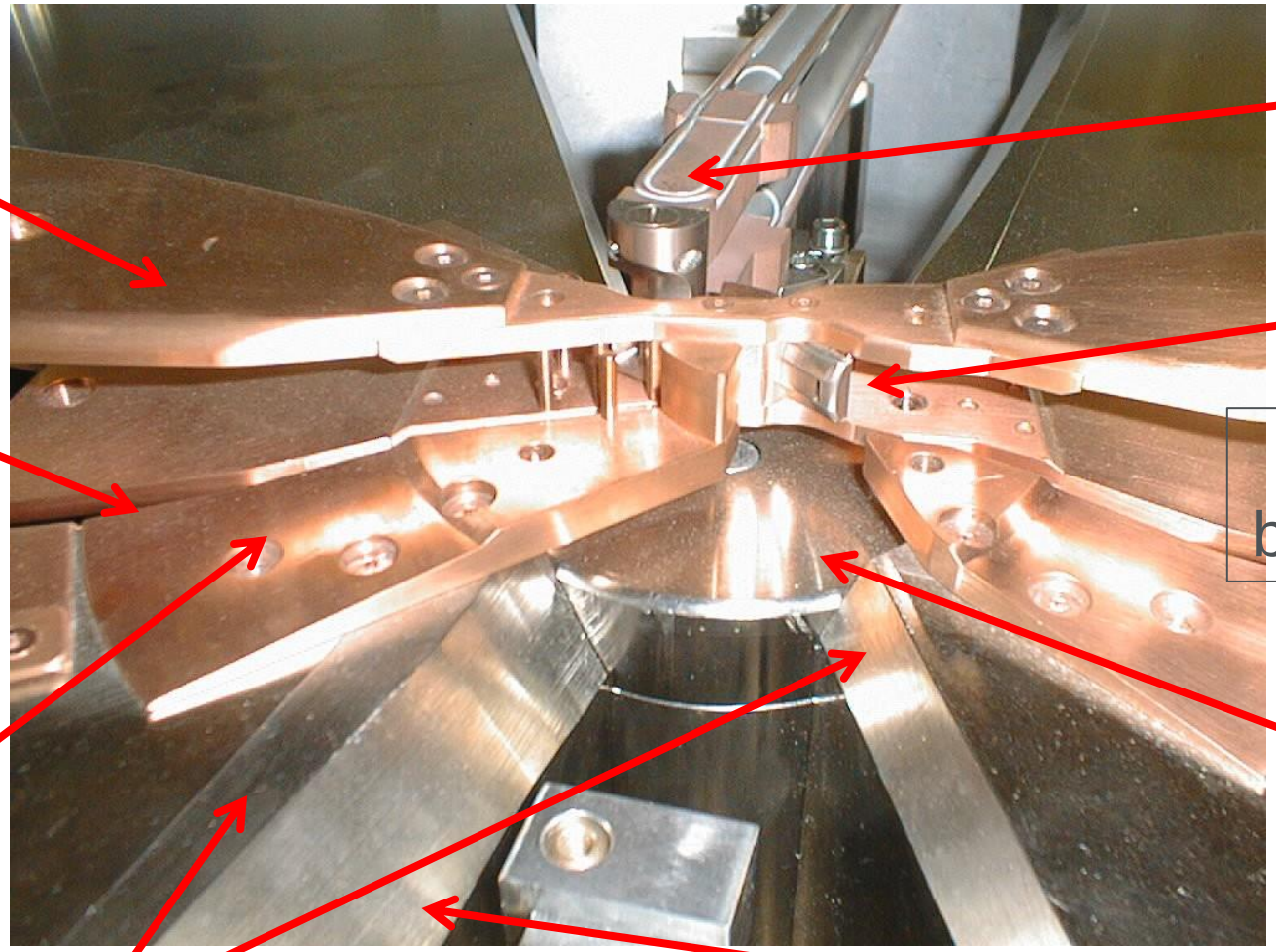


- Chimney: copper-tungsten ⇒ good heat properties; machinable
- Cathodes: tantallum ⇒ high electron emission; shaped to reduce heat conduction

Compact Deep-valley Cyclotron Design



Example: central region of a compact cyclotron



2 Dees at V_{dee}

4 accelerating gaps

dummy dees at ground

4 poles

2 ion sources (H^- and D^-)

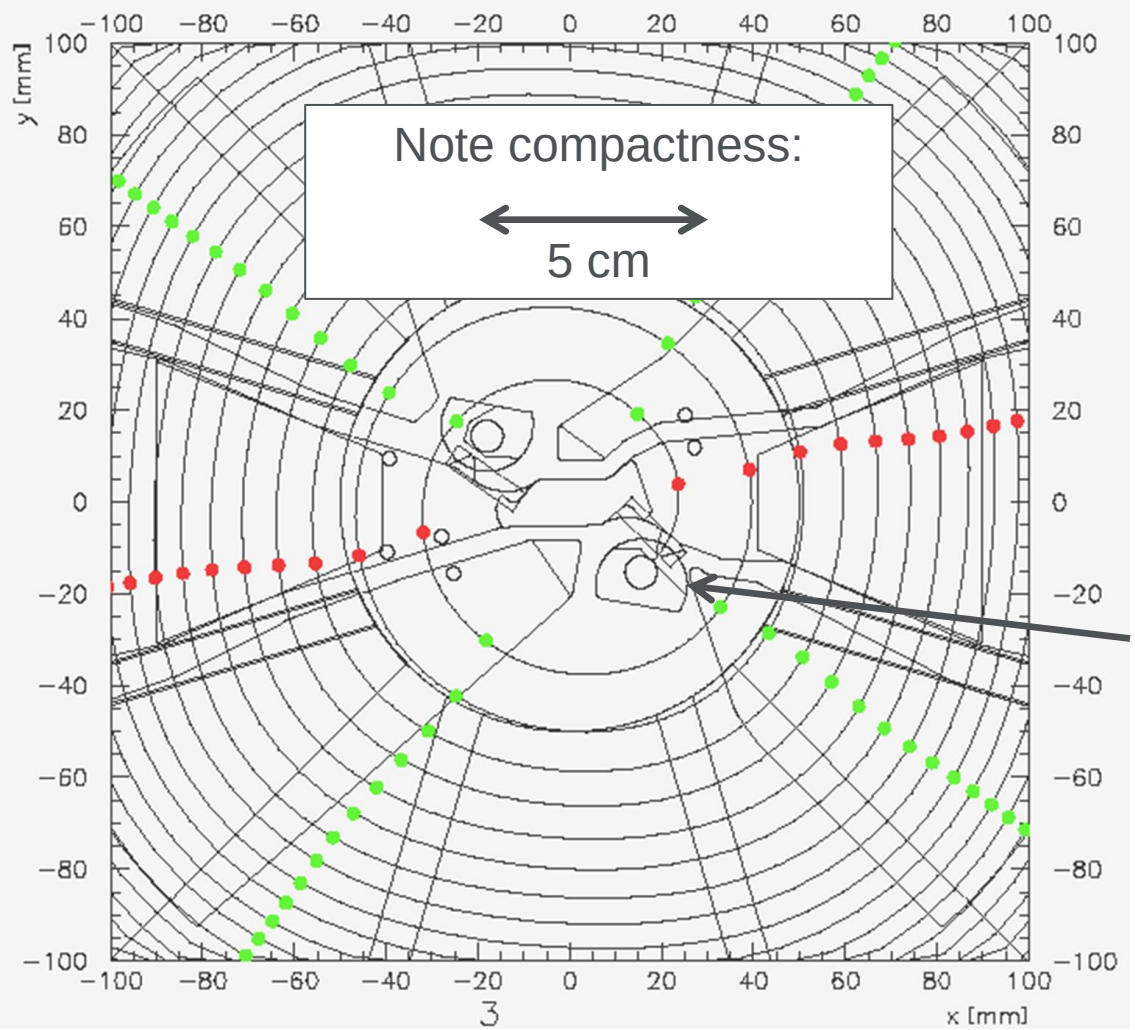
Puller at V_{dee}

Small gap (≅1.5 mm) between chimney and puller

Central plug to adjust field in the center

4 removable pole edges for shimming of isochronism

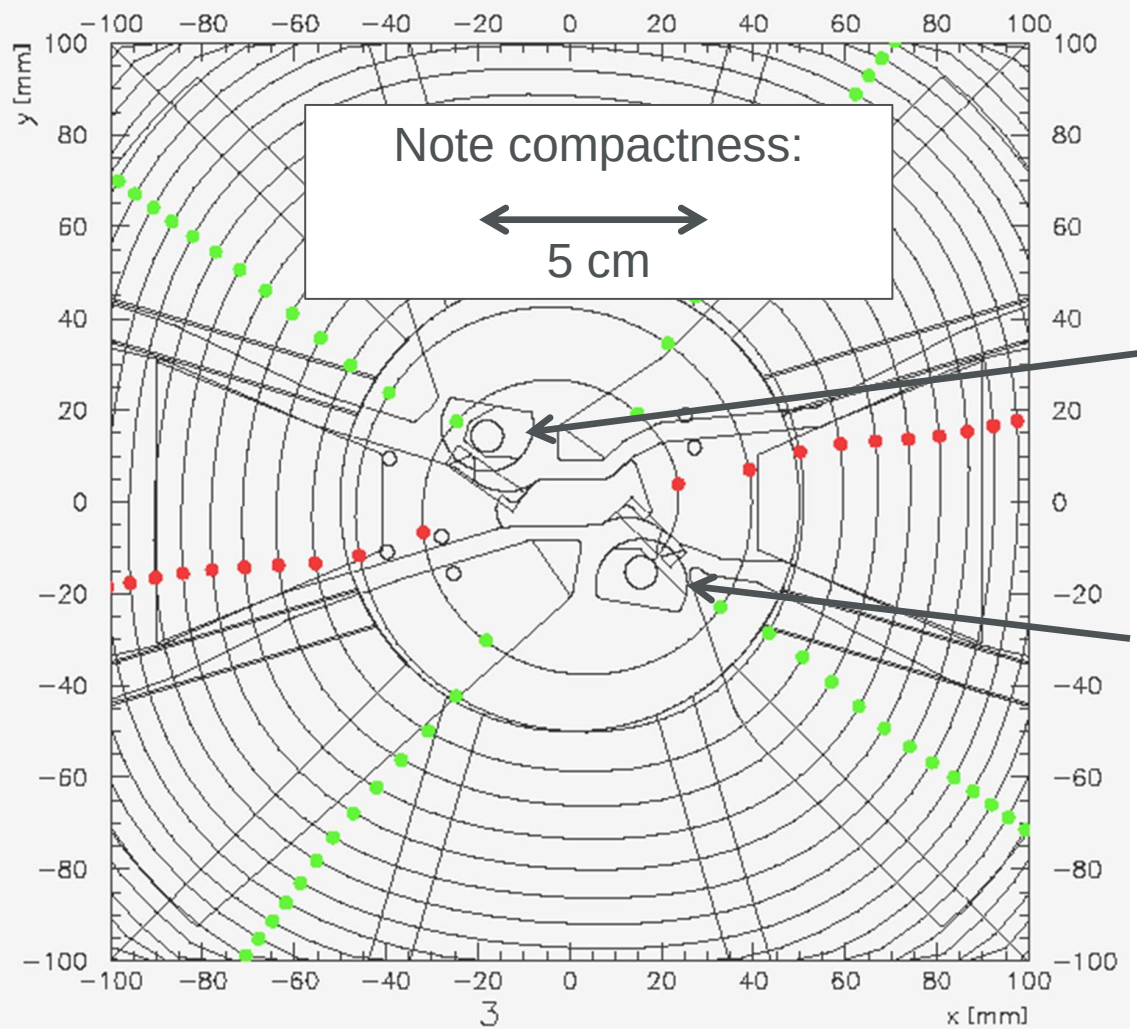
Orbit tracking in C18/9 isotope production cyclotron



H^- source
h=2

- Red dots: position of particle when V_{dee} =0
- Green dots: position of particle when V_{dee} =V_{max}

Orbit tracking in C18/9 isotope production cyclotron



E-fields => from Opera
B-fields => measured or Opera
Tracking code => AOC

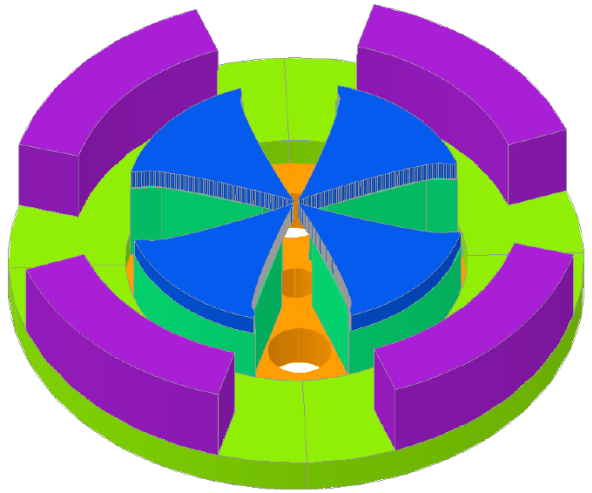
D^- source; h=4

H^- source h=2

- D^- source is placed further out because of larger orbit
- Cut D^- chimney for H^- passage
- Red dots: position of particle when V_{dee} =0
- Green dots: position of particle when V_{dee} =V_{max}

Vertical focusing in the center

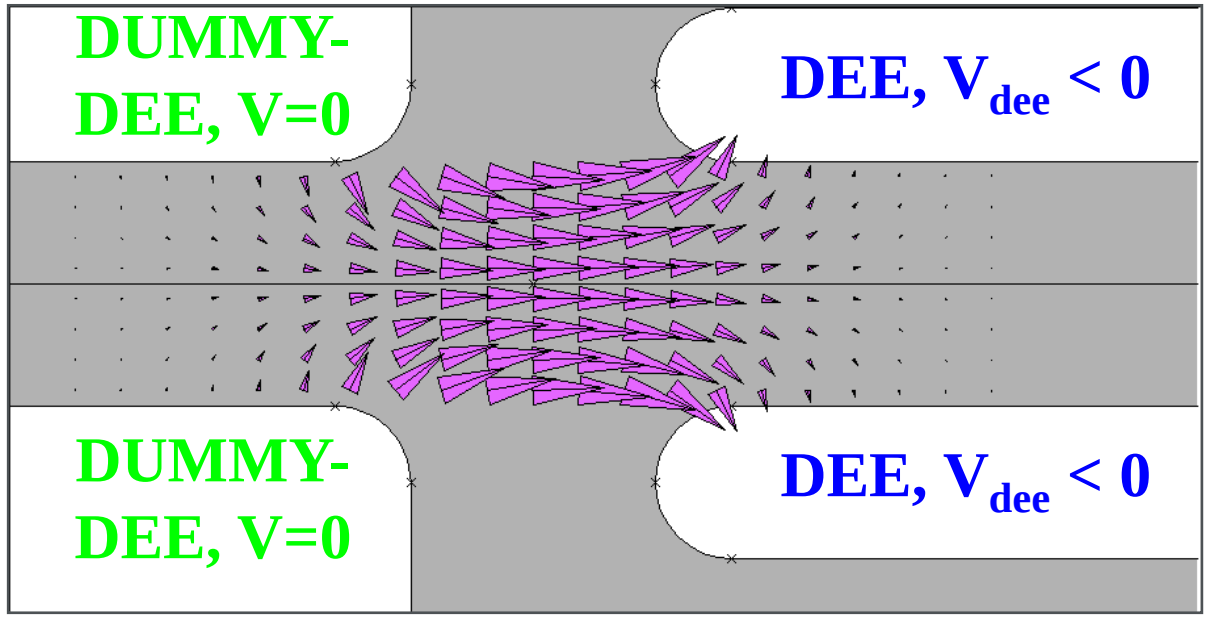
Azimuthal Field Variation (AVF) goes to zero in the cyclotron center
⇒ magnetic vertical focusing disappears



- Two remedies
  - Add a magnetic field bump in the center
    ⇒ negative field gradient creates vertical focusing: field bump of a few hundred Gauss ⇒ central plug
  - The first few accelerating gaps provide electrical focusing
    ⇒ proper positioning of accelerating gaps during the design to get some phase focusing

Vertical Electrical Focusing in accelerating gap – 2 factors

Vertical cross section:

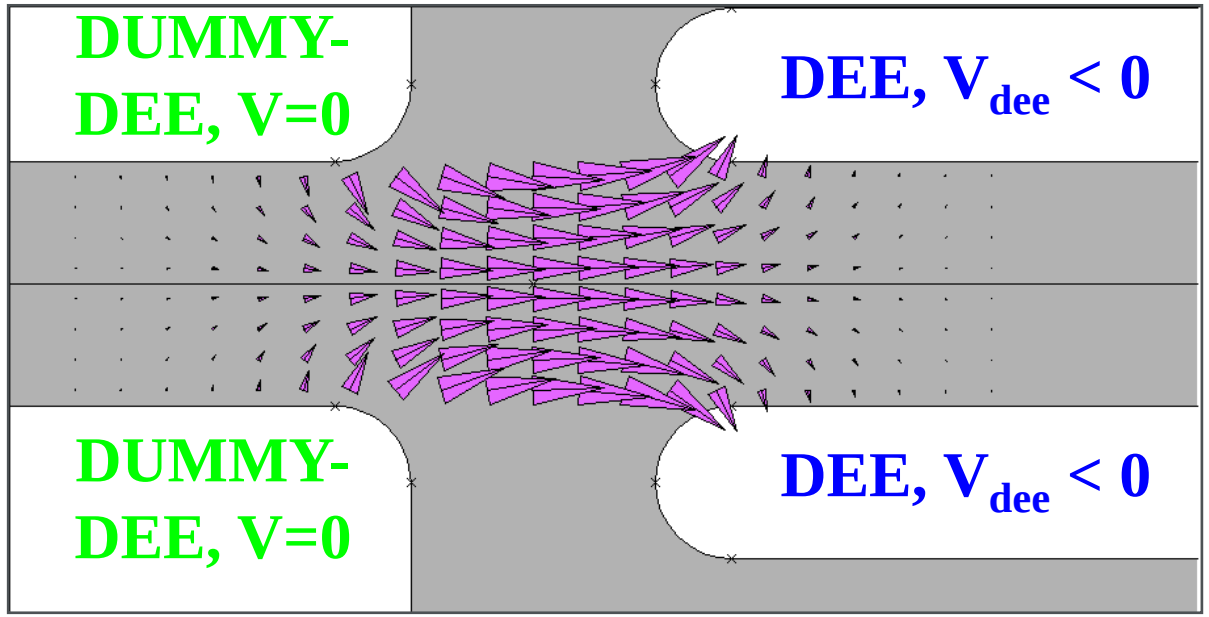


1. Due to the shape of electric field lines in the gap: first half is focusing and second half is defocusing => total effect is focusing => comparable to **Einzel lens**

1st half => focusing	2nd half => defocusing

Vertical Electrical Focusing in accelerating gap – 2 factors

Vertical cross section:

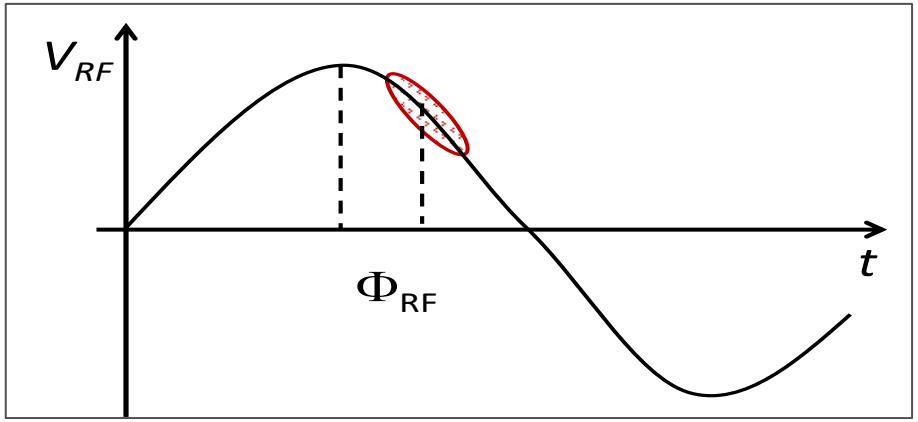


1. Due to the shape of electric field lines in the gap: first half is focusing and second half is defocusing => total effect is focusing => comparable to **Einzel lens**
2. Due to RF effect: If E-field is decreasing in time at moment of acceleration => falling slope of RF sine wave => second defocusing half is less important => net focusing (phase focusing)

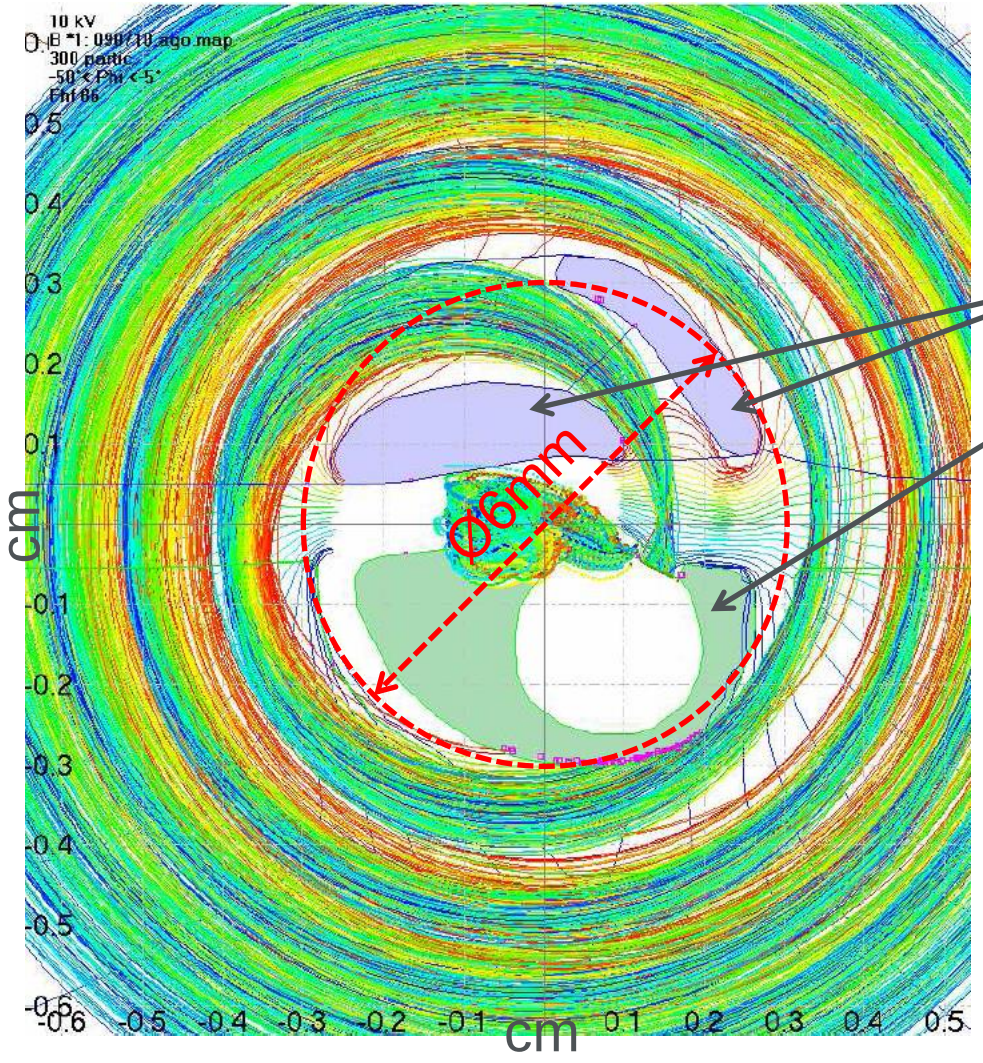
1st half => focusing

2nd half => defocusing

Falling slope of RF ⇒ net focusing (phase focusing)



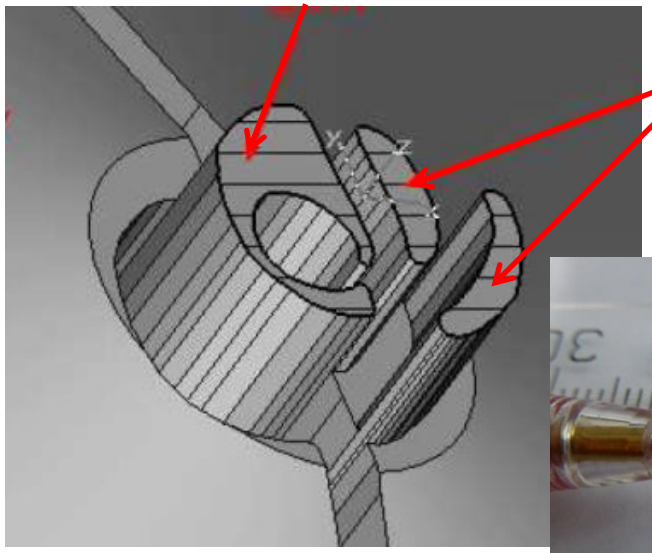
Example: Ion source and central region of the S2C2



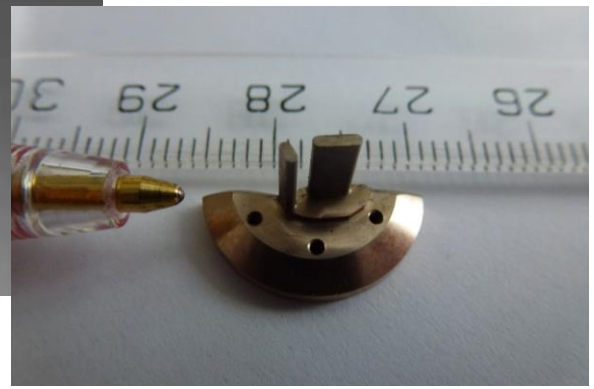
Puller

Chimney

Chimney



Puller



- Due to high field (5,74T) and low dee voltage (11 kV), Central region size is very compact:
  - Cold cathode PIG source diameter < 5 mm
  - Vertical gap in the center 6 mm
  - First 100 turns within a radius of 3 cm

By the way: why a SC synchrocyclotron for PT

- An isochronous cyclotron needs flutter
- Flutter can only be created by the iron (not by the coil)
- Maximum achievable field modulation about 2 Tesla
- If average field is pushed too far up (using a SC coil) then no longer enough flutter => not enough vertical focusing
- In a synchro-cyclotron this problem does not occur

➢ In a synchrocyclotron you can fully exploit the potential offered by superconductivity

QUIZ #3

The main force used for vertical focusing at injection is:

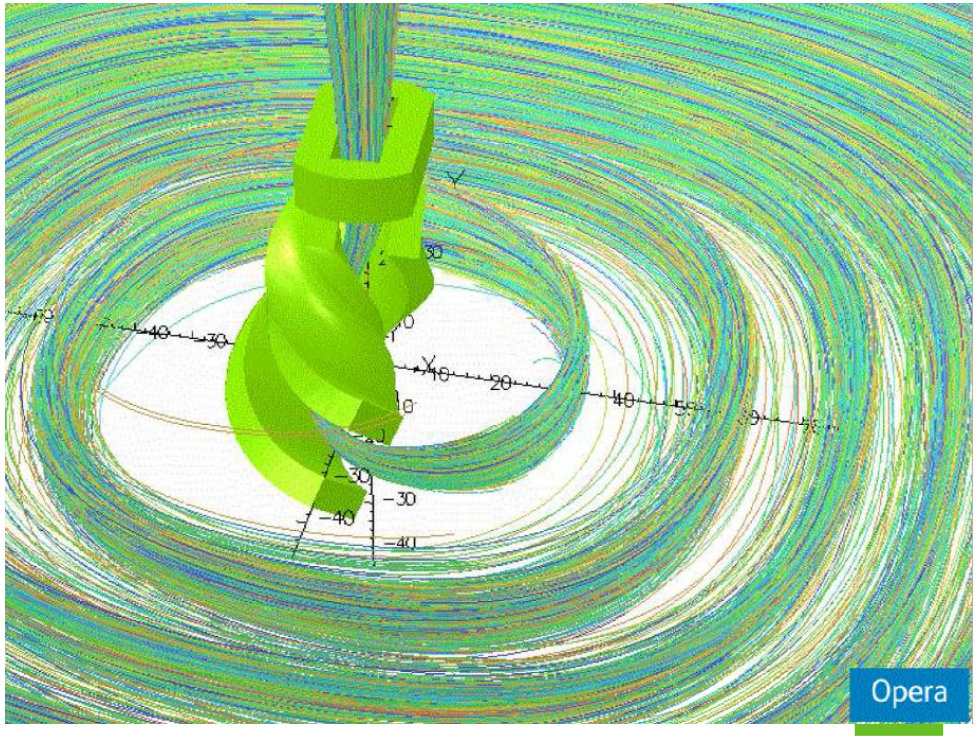
1. Electric
2. magnetic

Axial Injection

- Axial injection
  - Along the vertical symmetry axis of the cyclotron
  - In the center, the beam is bent by 90° into the median plane
  - For this an electrostatic inflector device is used

- The E-field between 2 electrodes bends the beam 90° from vertical to horizontal. The presence of the cyclotron B-field creates a 3D orbit

- Spiral inflector is basically a cylindrical capacitor which is gradually twisted in order to take into account spiraling of trajectory induced by vertical magnetic field



Spiral inflector for Axial Injection

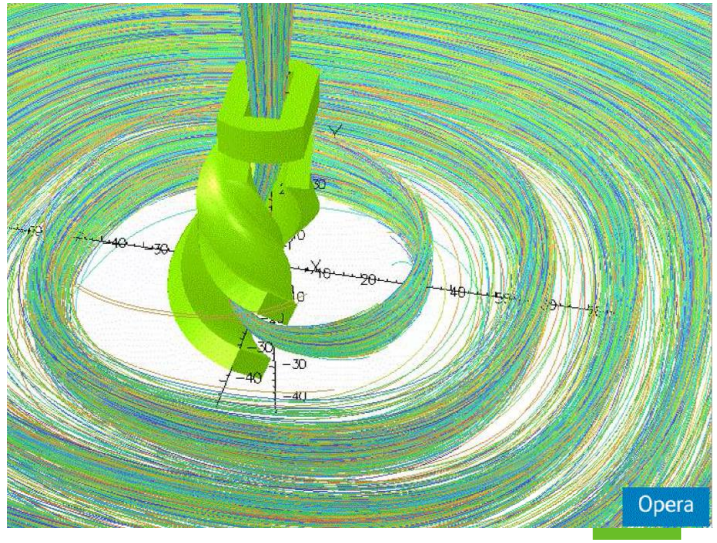
- Electric field always perpendicular to velocity ⇒ orbit on equipotential ⇒ this allows for low electrode voltage

$$\frac{qV}{E} = \frac{2d}{A}$$

E=energy, V=voltage, d=electrode spacing

- Two free design parameters available to obtain orbit centering

1. Electric radius A
   (equivalent to height of inflector)
2. Tilt parameter k'
   (equivalent to a change of magnetic field)



- Very compact geometry
- Complicated electrode structure needs a 5 axis milling machine

spiral inflector

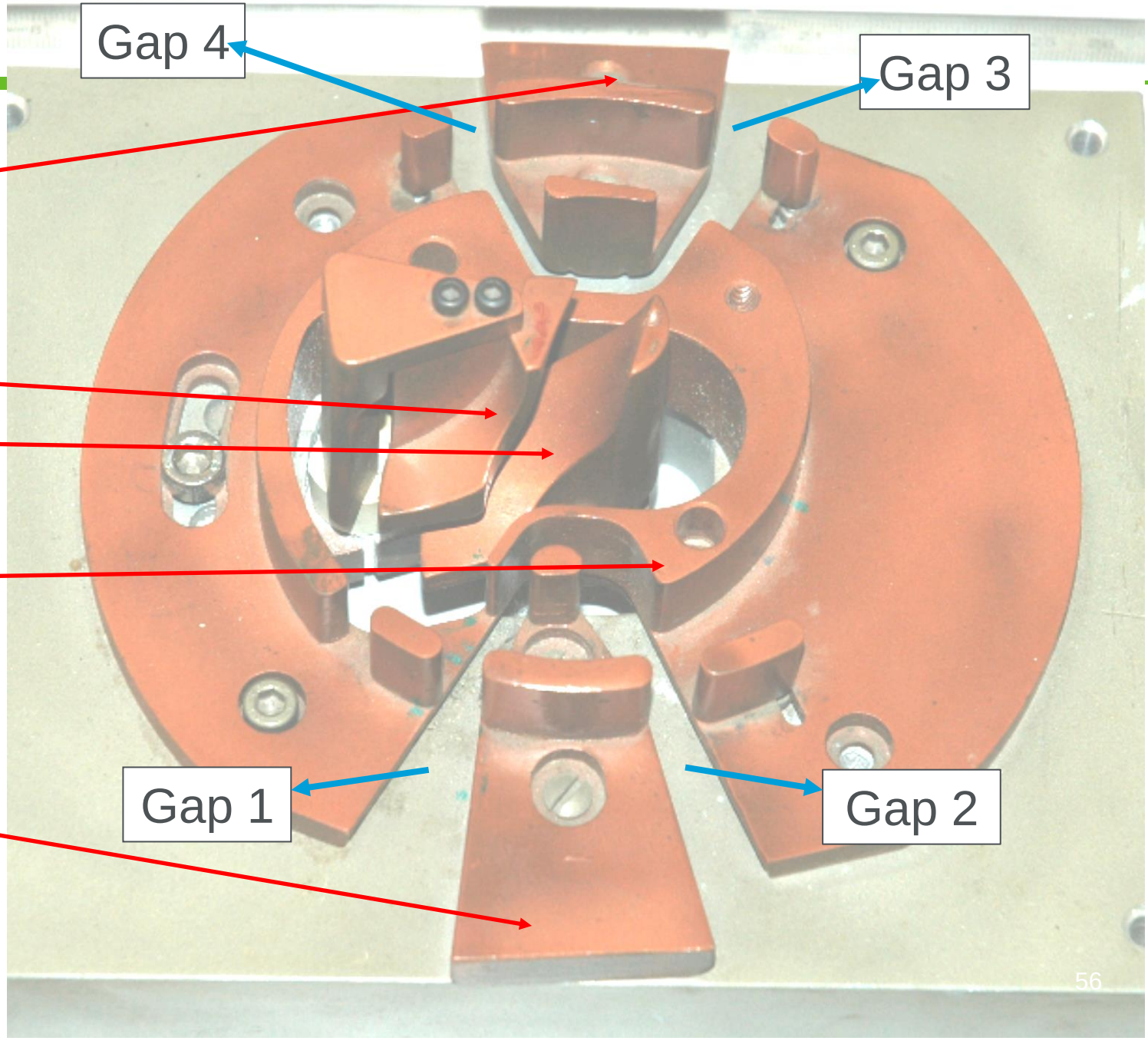
Left Dee tip

Upper electrode

Lower electrode

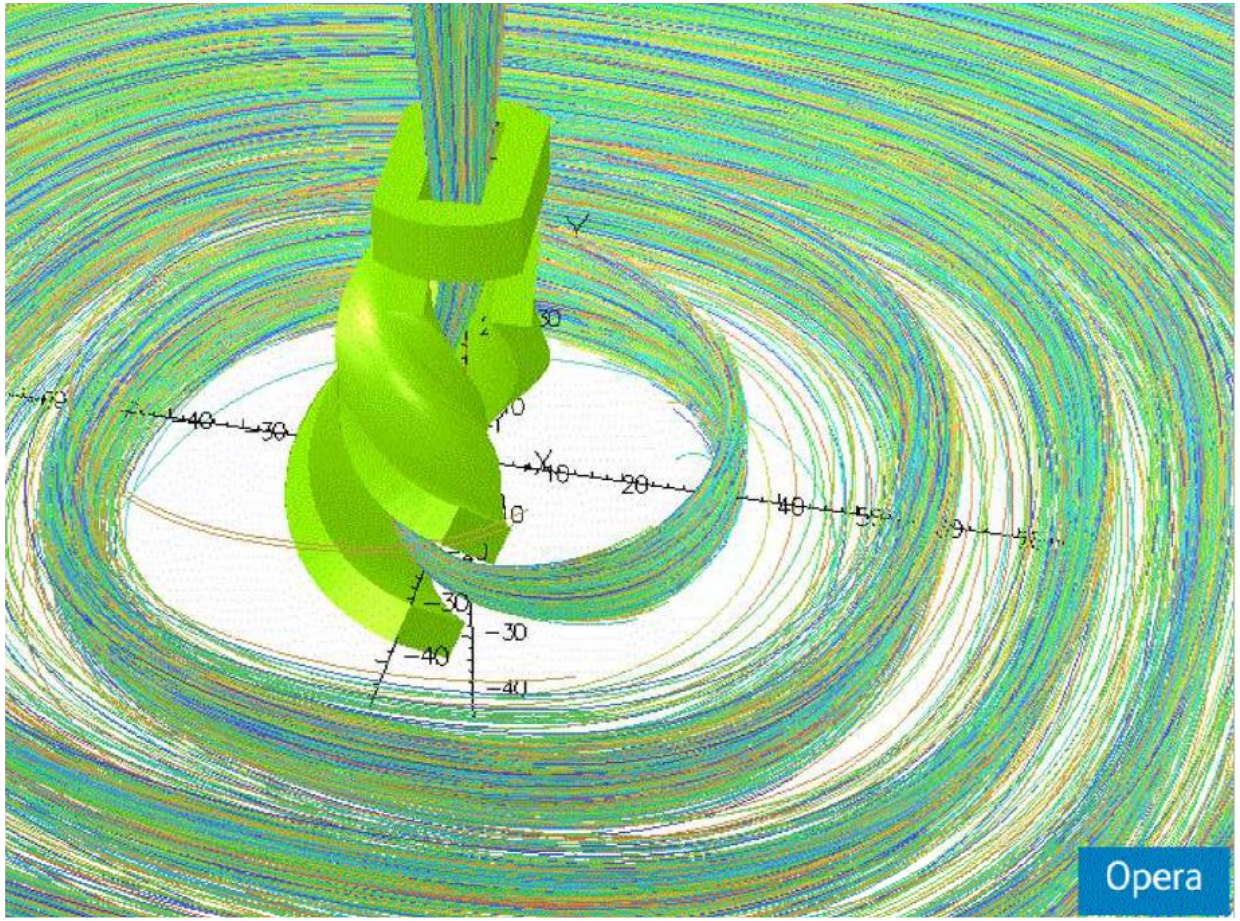
Housing

Right Dee tip



Inflector simulations

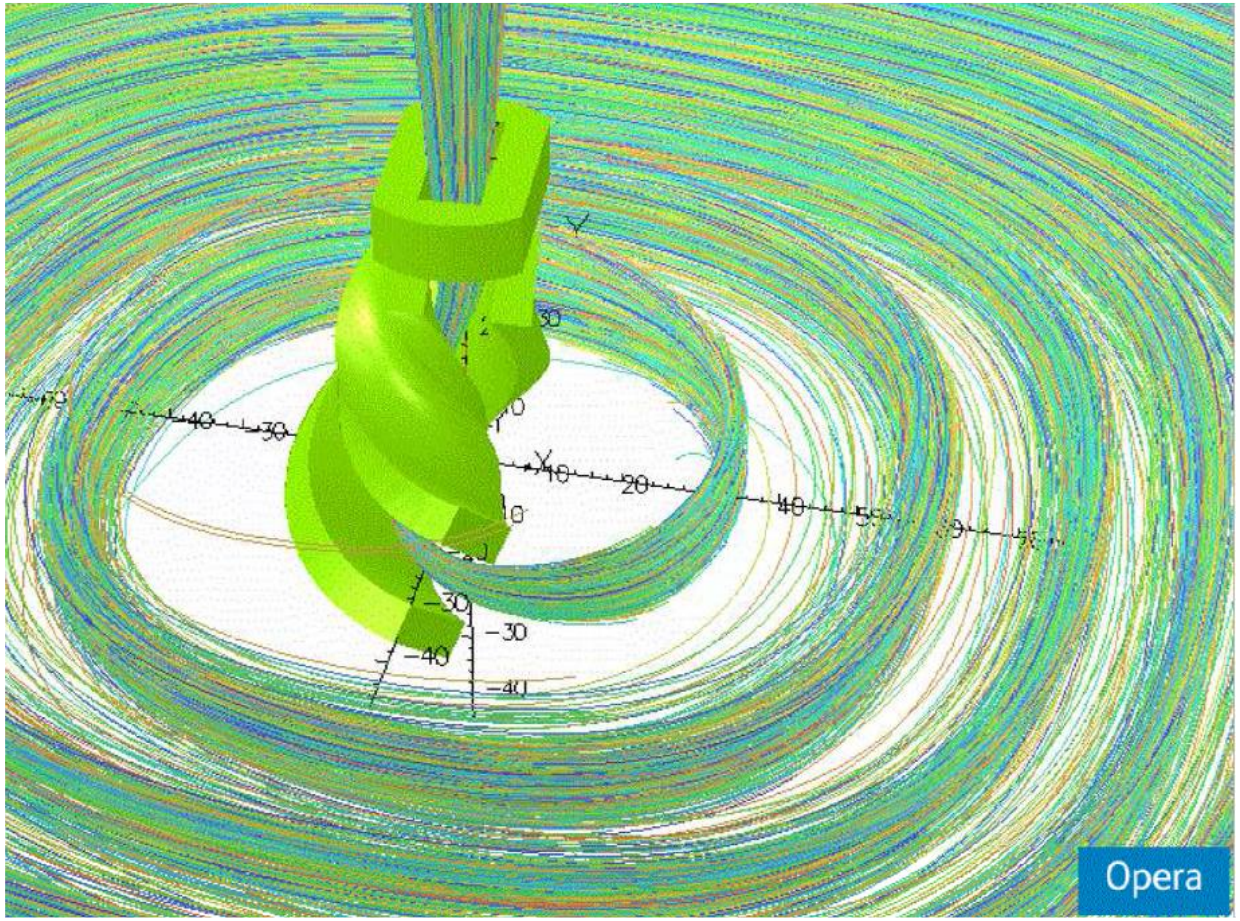
Calculated orbits imported in Opera3D:



- Spiral inflector is a complex 3D problem
- 3D fields (B,E) are needed => Opera3d
- In house developed tracking code
- Calculated orbits are imported in Opera3d post-processor
- Tilt k' is seen as the electrode-rotation at the exit

Inflector simulations

Calculated orbits imported in Opera3D:



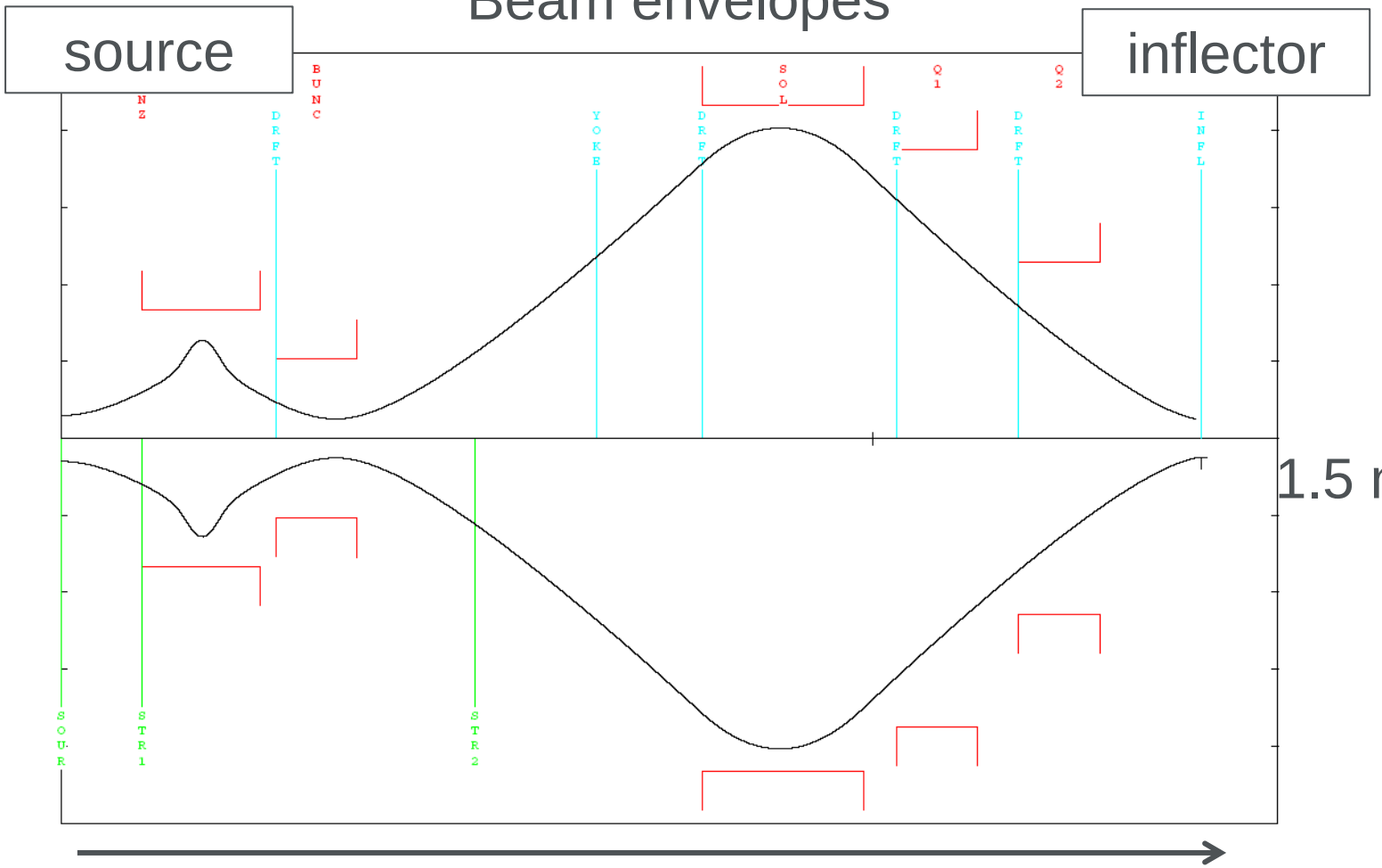
C70-example

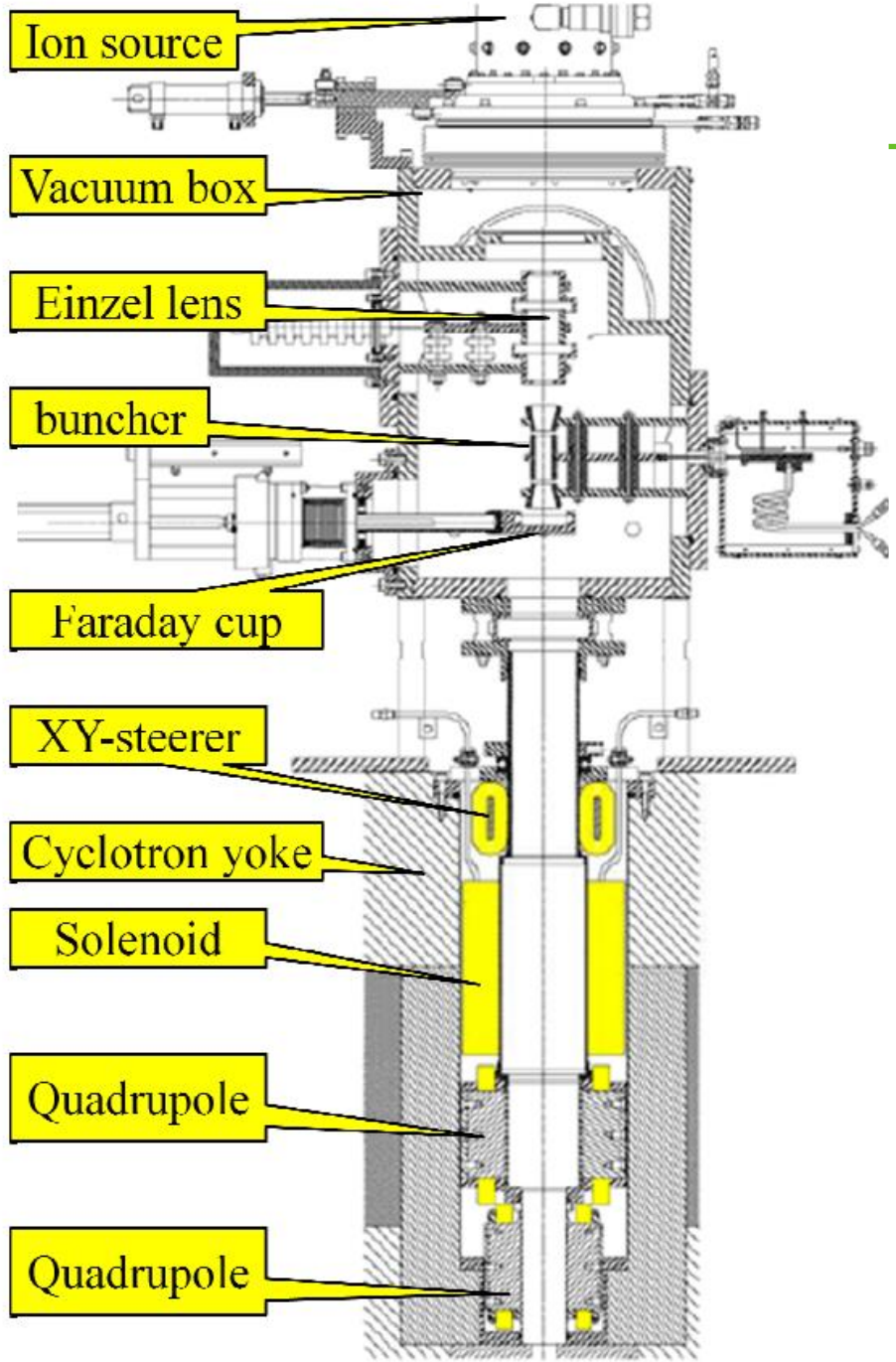


An additional horizontal deflector at tension is needed for multi-particle cyclotron

Injection line

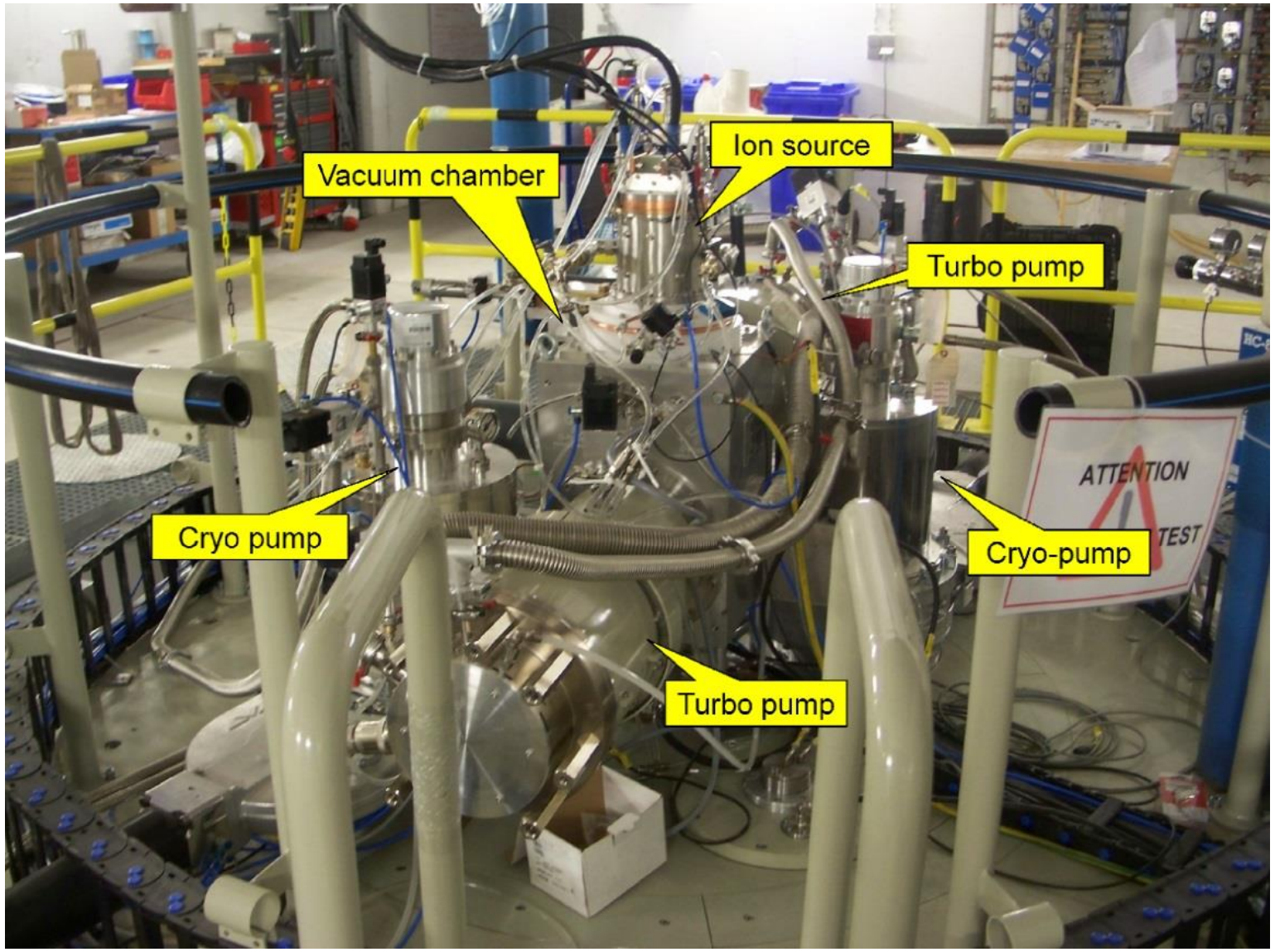
Beam envelopes

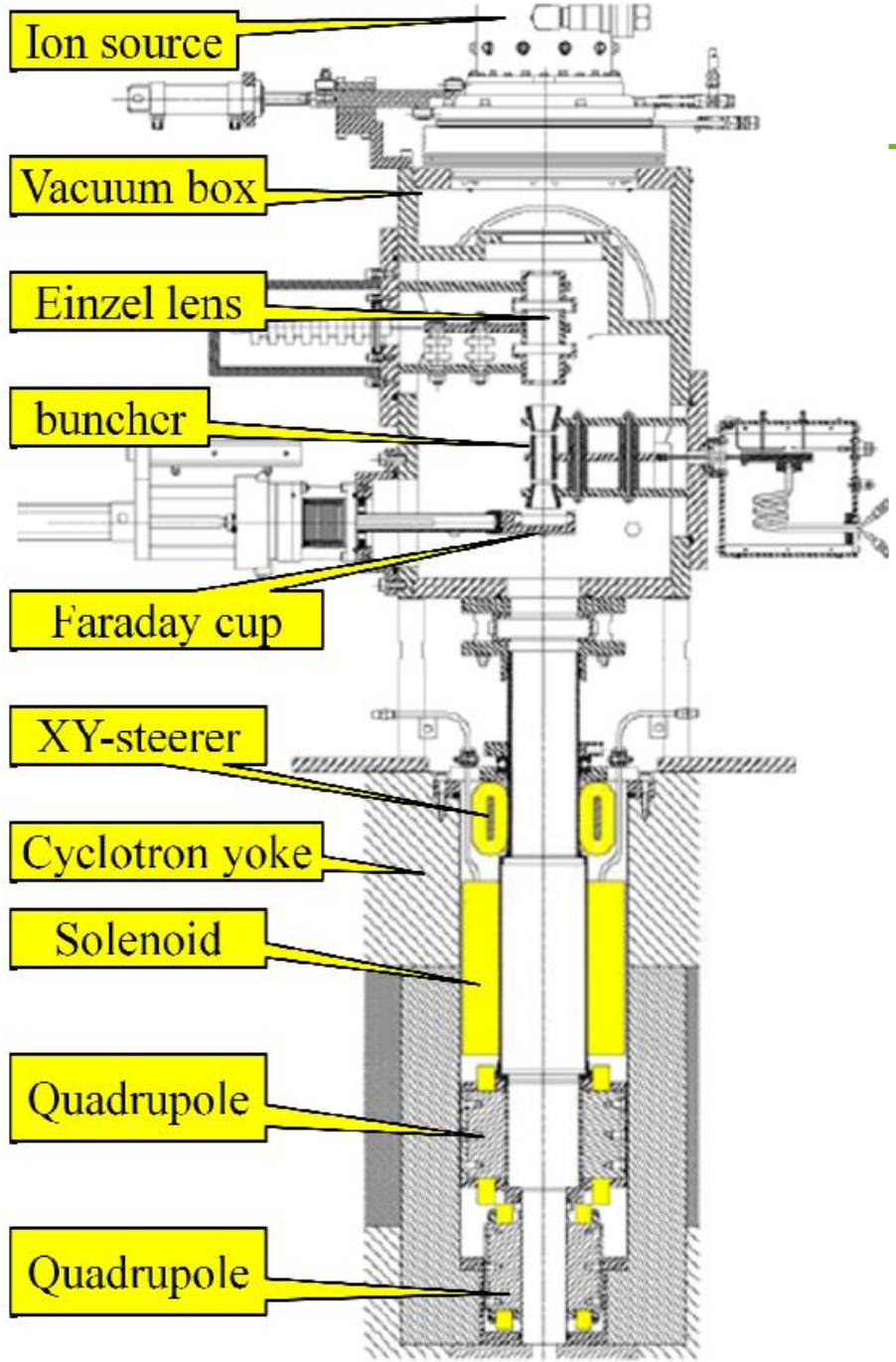




More than 2mA injected (30 MeV cyclotron)

Injection line





Extraction

Extraction from a cyclotron

- Extraction: transfer of the beam from an internal orbit to application outside

- Often a difficult process. Why?
  1. The magnetic field acts like a trap: When the particle enters into the radial fringe field of the pole, it runs out of RF phase and will be decelerated ⇒particle is « reflected » inwards (if nothing is done to prevent this)
  2. The orbits pile up at high radii ⇒ smaller and smaller turn-separation

$$R \infty \sqrt{E}$$

  3. The beam quality can be quickly destroyed in the non-linear fringe field

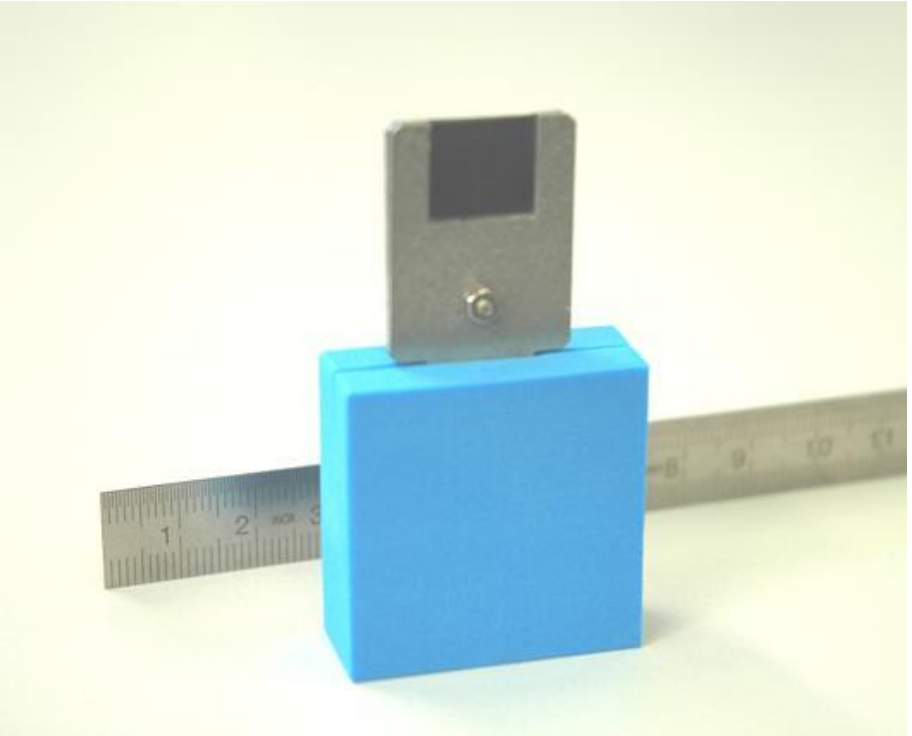
Different ways of extraction

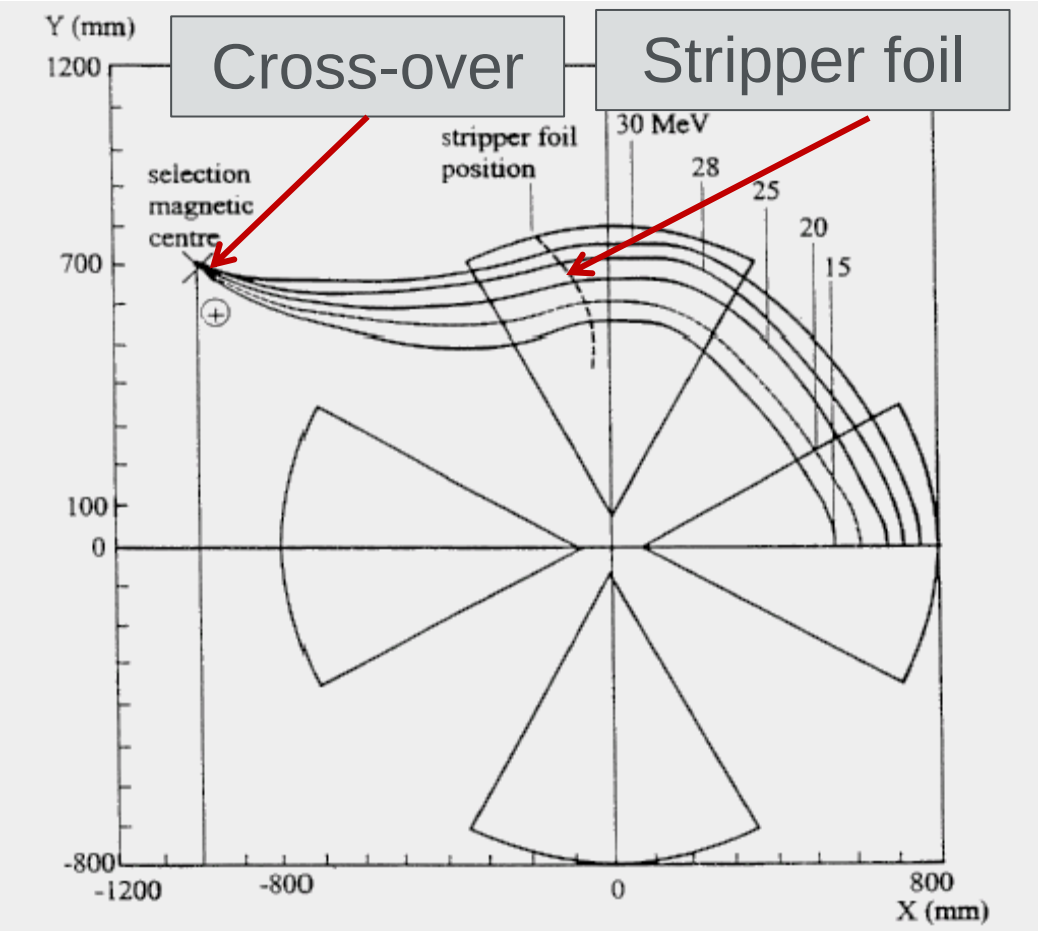
1. No extraction at all => place an internal target
   - Can be done for isotope production (a little bit dirty)
2. Stripping extraction (H^- cyclotrons; or H_2^+)
   - Isotope production cyclotrons
3. Extraction with an electrostatic deflector (ESD)
   - Proton therapy cyclotrons (Varian, IBA, SHI)
4. Regenerative extraction => synchrocyclotron
   - Proton therapy cyclotrons (Mevion, IBA)
5. Self-extraction => suitable shaping of the magnetic field
   - One IBA prototype cyclotron but needs further improvement

Cases 3 and 4 require some way to increase the turn separation before extraction

H^- stripping extraction

- Stripper foil removes two electrons of H^- ion and orbit curvature changes sign
- Energy variation by moving stripper position
- All energies go to one crossover point by proper foil azimuthal position
- Place combination magnet at crossover
- Ideal solution for industrial cyclotrons

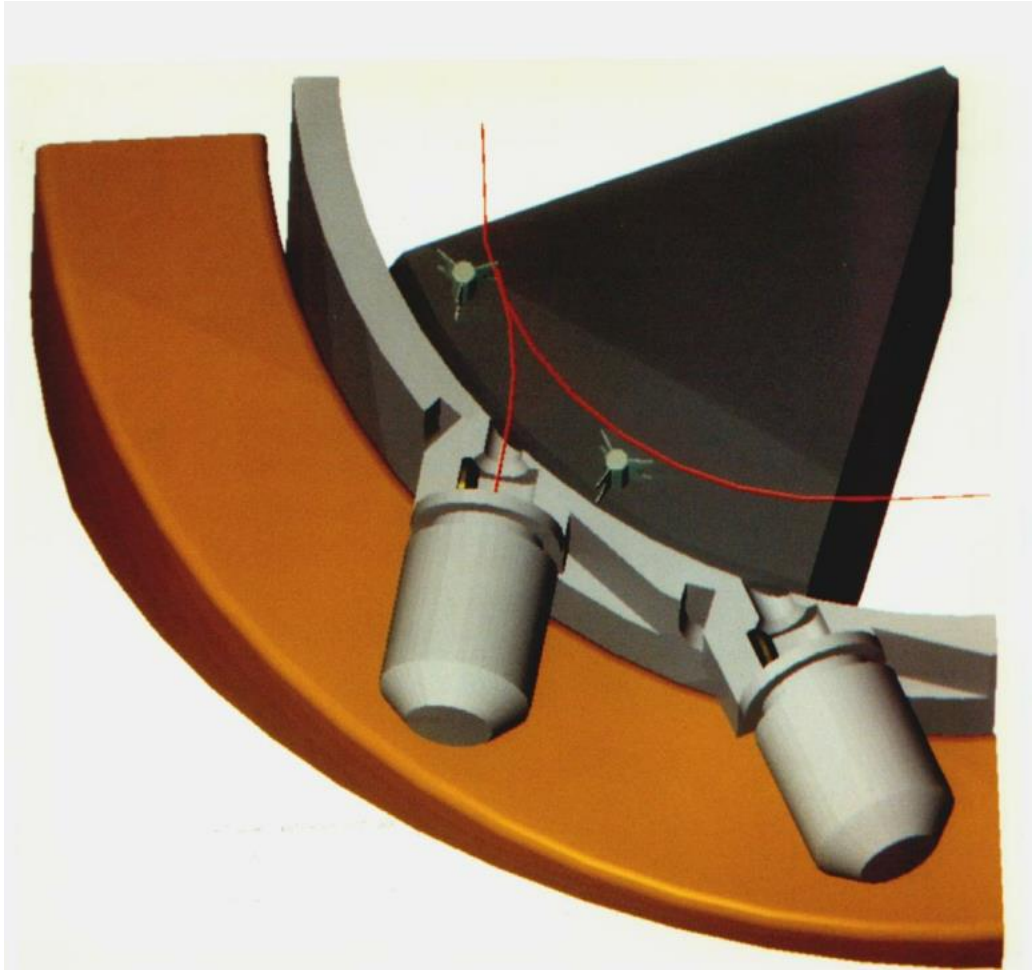




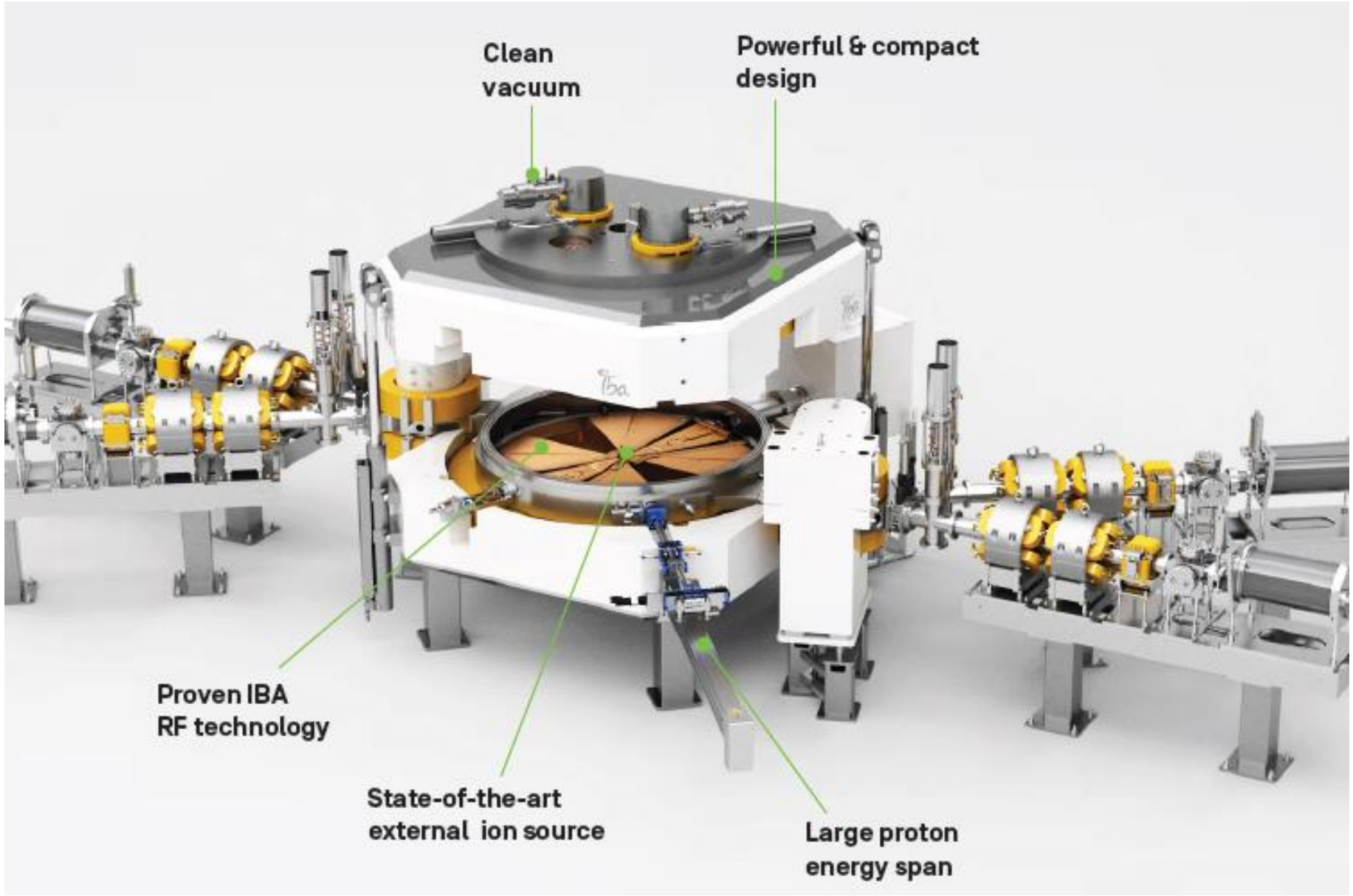
Stripping Extraction

- Other advantages
  - Simple and 100 % extraction efficiency
  - Multiple targets around the machine
  - Dual beam extraction
  - Good extracted beam optics

- Limitations due to stripping losses
  - Low B-field
    ⇒large magnet (Triumf 500 MeV/3 kG)
  - Good vacuum required (expensive)
  - OK for isotope production but not for proton therapy



C30 with four beamlines for isotope production

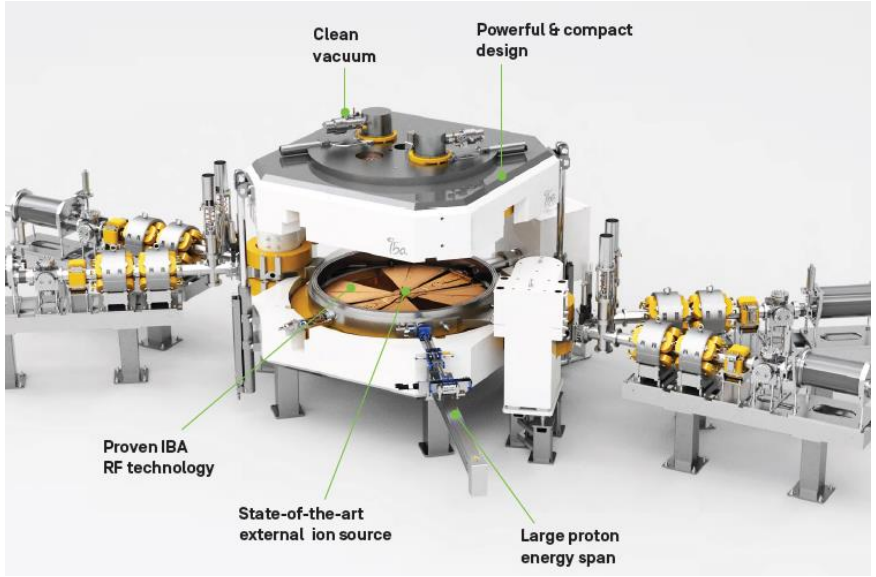


C30 with four beamlines for isotope production

Old vs new:

- 15 to 30 Mev, 12 to 30MeV
- 55 ton vs 30 ton
- R*h = 2.8m * 2m, vs 2.2*2.2*1.46m
- Current is equal :
  1.5mA extraction, 1.2mA target





c18: More compact & production capacity

Cyclone®	18/9	18P
Shape	Circle	Square
Height (m)	2.2	1.85
Dimensions (m) (overall)	Ø 2.3	1.95 x 1.95
Median plan height (m)	1.25	1.12
Minimum vault dimensions (WxLxH)	4 x 4 x 3	3.7 x 3.7 x 2.5
Accelerated particles	Proton/ deuteron	Proton ONLY
Rigging hole	2.5 x 2.5	2.2 x 2.2
Production capacity (Ci)	16	Up to 30
Weight (tons)	25	17





A side step: why cyclotrons for isotope production?

- Cost-effective machines for achieving:
  - required energies (<100 MeV) and
  - high currents (upto 1 to 2 mA)
- Efficient use of RF power => same accelerating structure used multiple times
- Compact =>
  - magnet and RF integrated into one system
  - Single stage => no injector accelerator needed
- Moderate magnetic fields: 1 to 2 Tesla (no SC needed)
- Simple RF system:
  - Constant RF-frequency (10-100 MHz) => CW operation
  - Moderate voltages (10-100 kVolt)
- Relative easy injection (internal ion source or axial injection)
- Simple extraction (stripping for H^- ions with carbon foil => 100% efficiency)

IBA was founded in 1986.
Since then **>500 isotope production cyclotrons** have been sold by IBA;
Many more by competitors

Some commercial cyclotron vendors/manufacturers

RP = Radio Pharma; PT = Proton Therapy

SIEMENS — Germany (RP)

GE, USA (RP)

Best — Canada (RP)

ADVANCED CYCLOTRON SYSTEMS, INC. — Canada (RP)

VARIAN medical systems — USA (PT)

MEVION medical systems — USA (PT)

SUMITOMO — Japan (RP+PT)

C235

iba Particle Therapy — Belgium (RP+PT)

QUIZ #4

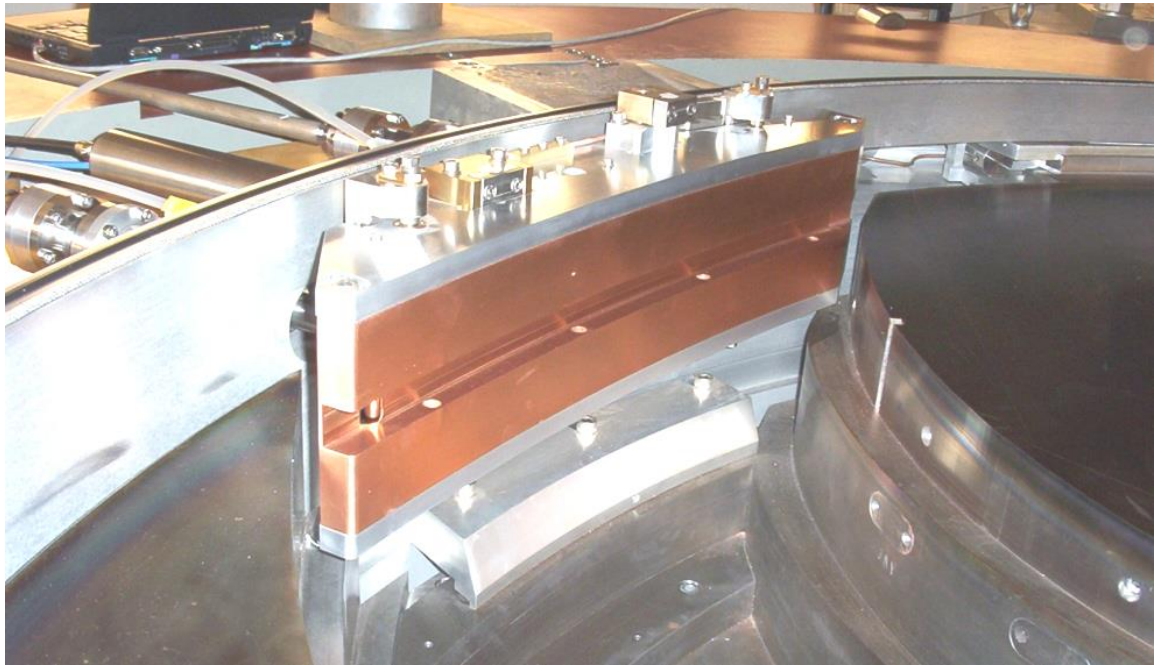
Which is the easiest way of extraction a cyclotron:

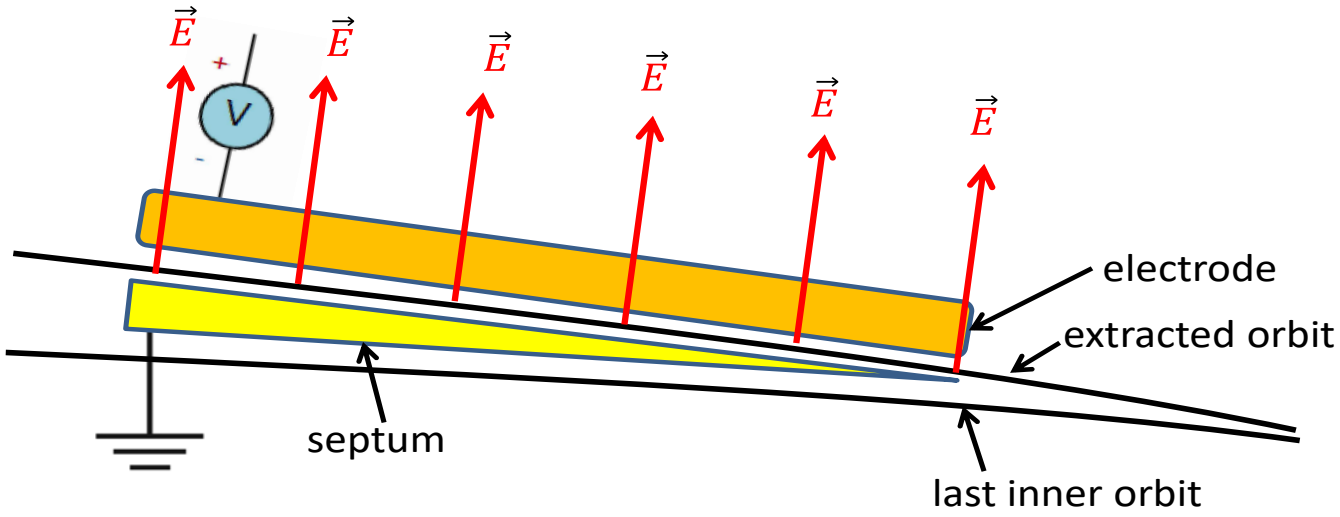
1. Stripping
2. Electro-static deflection
3. Regenerative extraction
4. Self-extraction

Electrostatic Deflector

- Constant radial E-field creates initial angular kick to deflect beam
- Inner electrode (septum) on ground potential
  - No disturbance on inner orbits
  - Knife thin (0.1 mm) and
  - V-shape at entrance (distribute heat)
  - Water cooled ⇒ limitation for maximum beam intensity

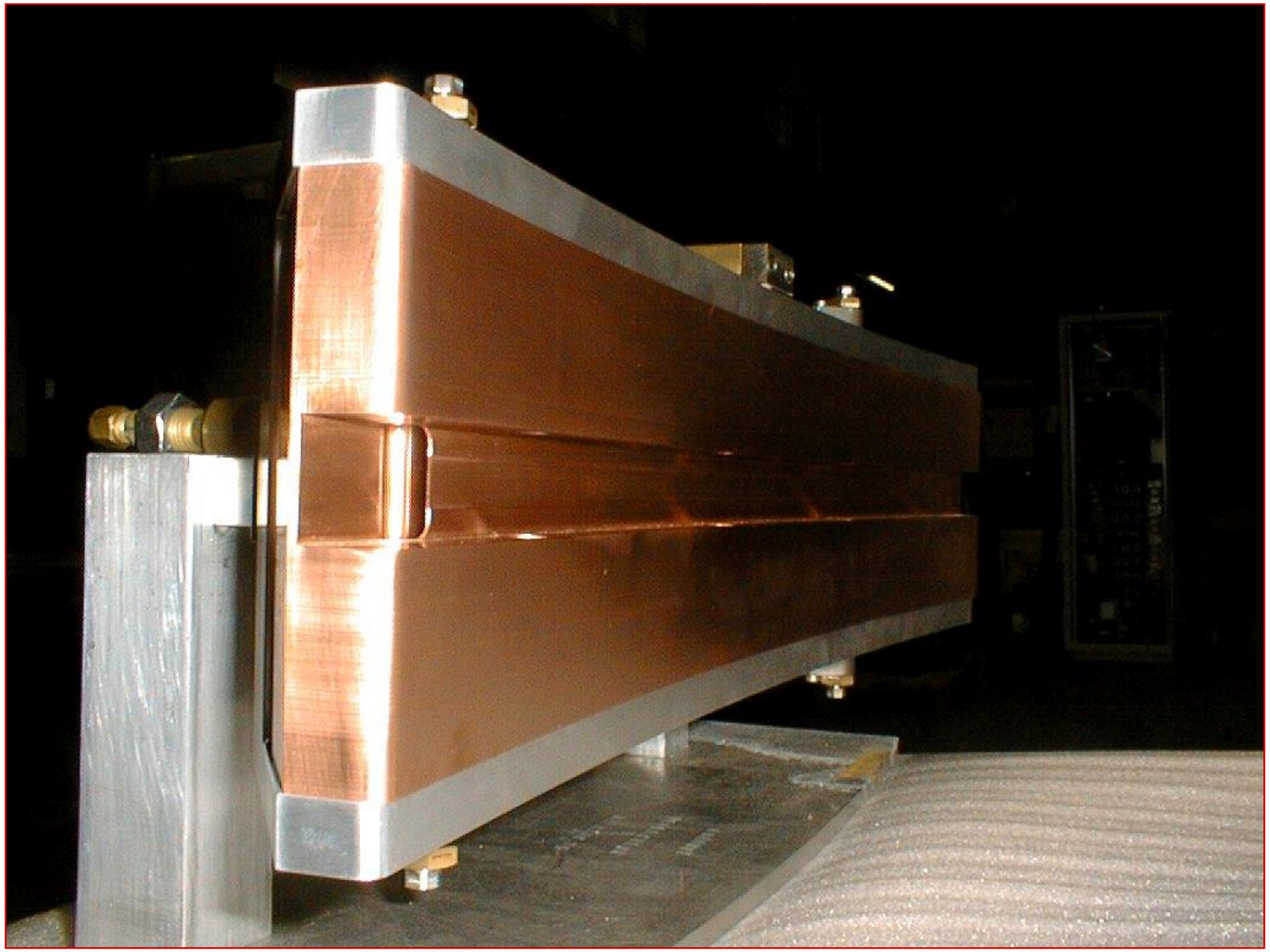
ESD for IBA C235

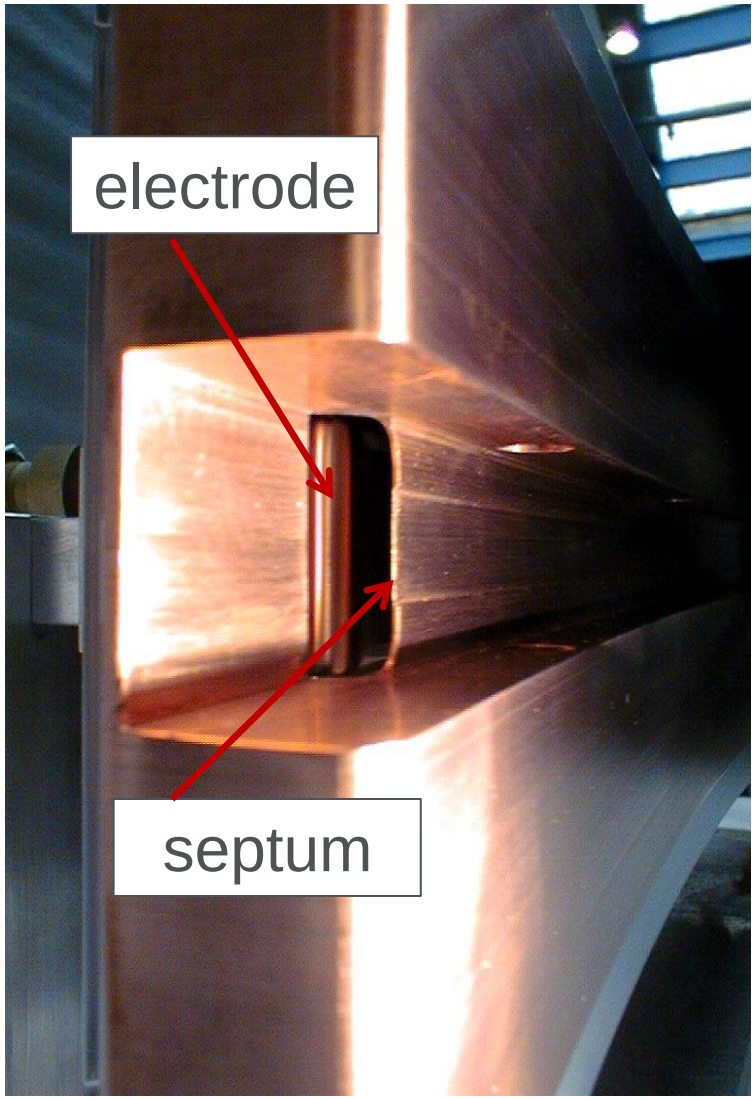




- Outer electrode on negative potential
- Electrode shape = orbit shape

C235 Electrostatic deflector



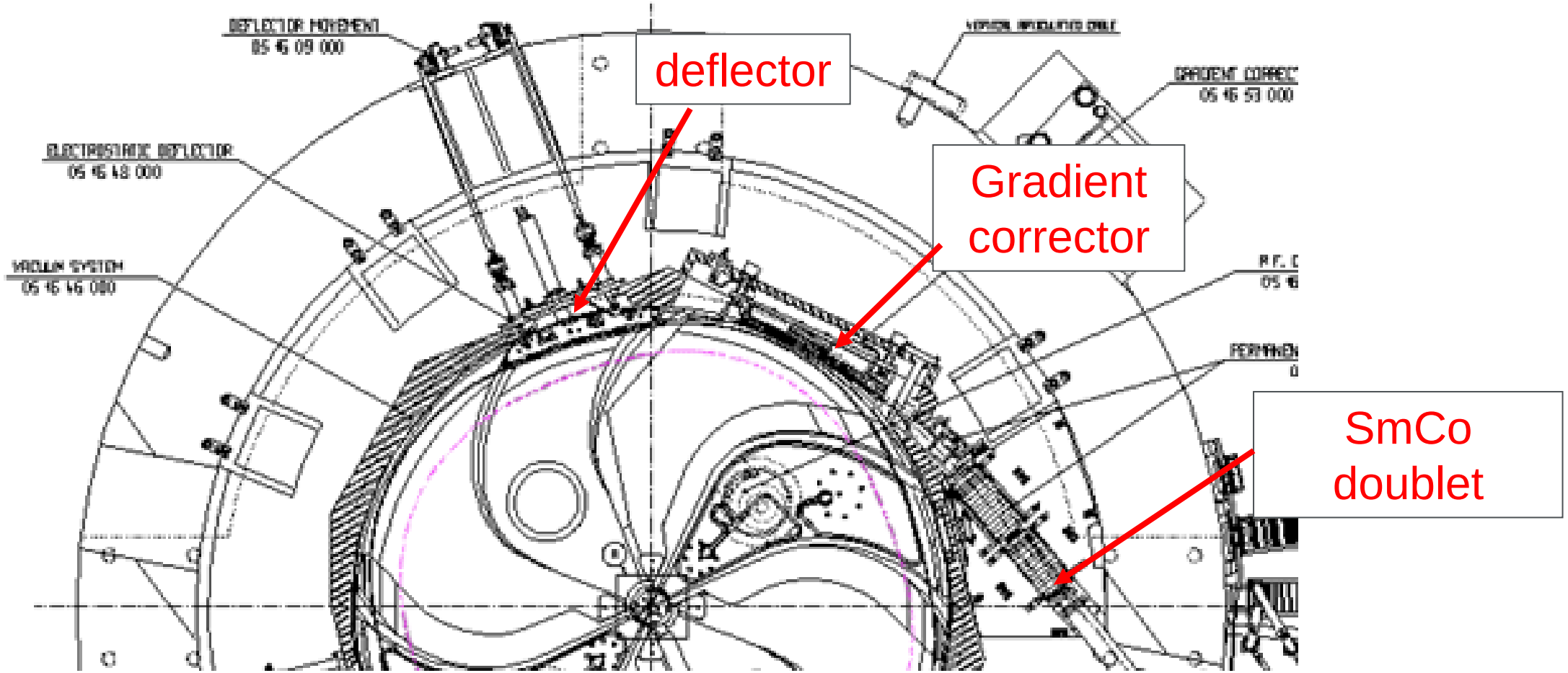


Deflecting and guiding the beam out

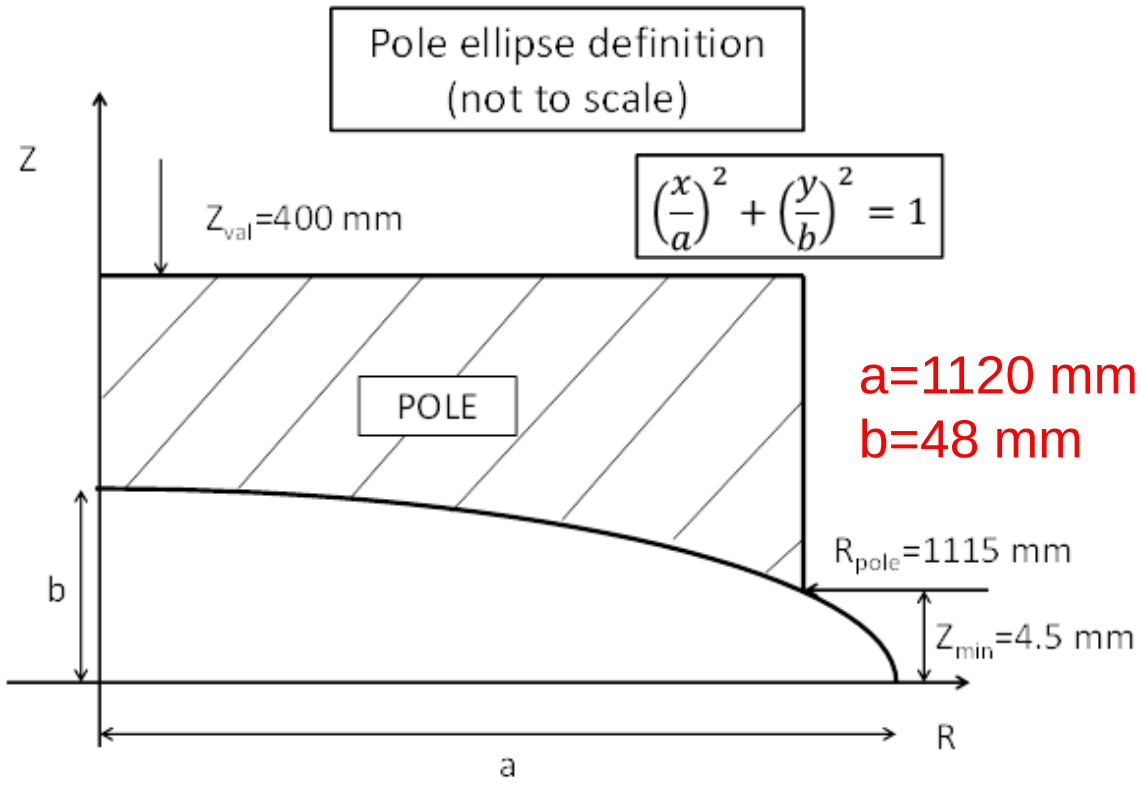
Generic method extraction :

1. Obtain turn-separation
2. Provide an initial radial kick
   - Electostatic deflector ESD (peel off last turn)
3. Reduce and shape B-field and minimize optical damage when passing fringe field ⇒ Gradient corrector
4. Re-focus beam as quickly as possible to handle beam divergencies created in fringe field
   - First quadrupole doublet (in return yoke)

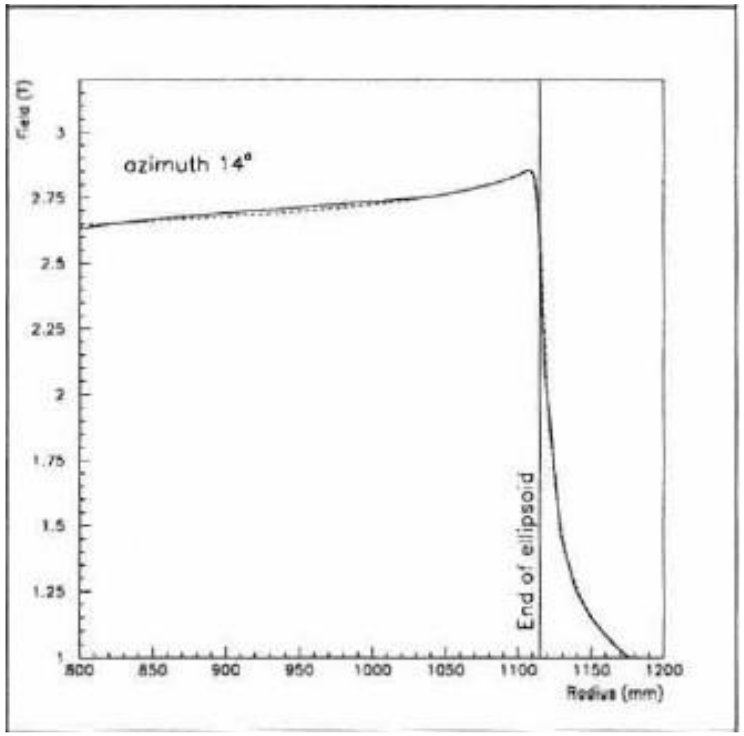
C235 Extraction Scheme



Extraction in the IBA C235



Sharp transition from stable to unstable:



Pole gap in the C235 has an elliptical form.

- This allows to obtain a good field region very close to the radius of the pole
- Therefore particles can be accelerated very close to the radius of the pole

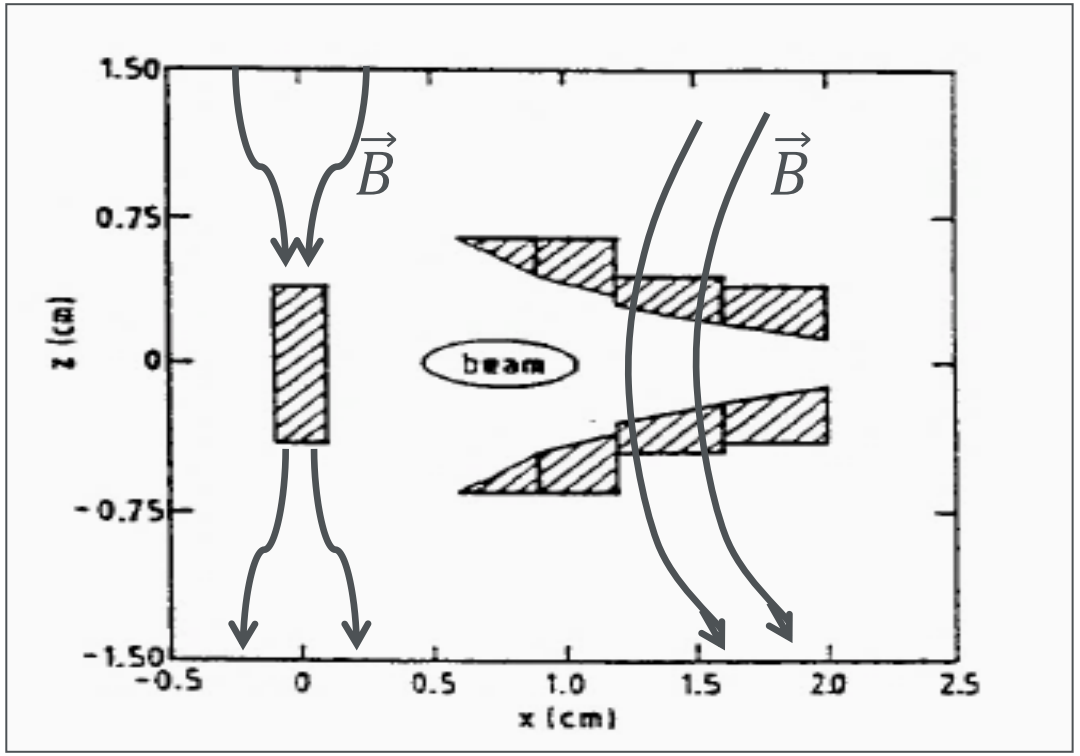
➢ Only a small kick is needed to extract beam => orbit is extracted in ¼ turn

Gradient Corrector focusing Channel

- Goal:
  - Guide the beam through the fringe field
  - Lower magnetic field on extraction path
  - reduce vertical/increase radial focusing through fringe field

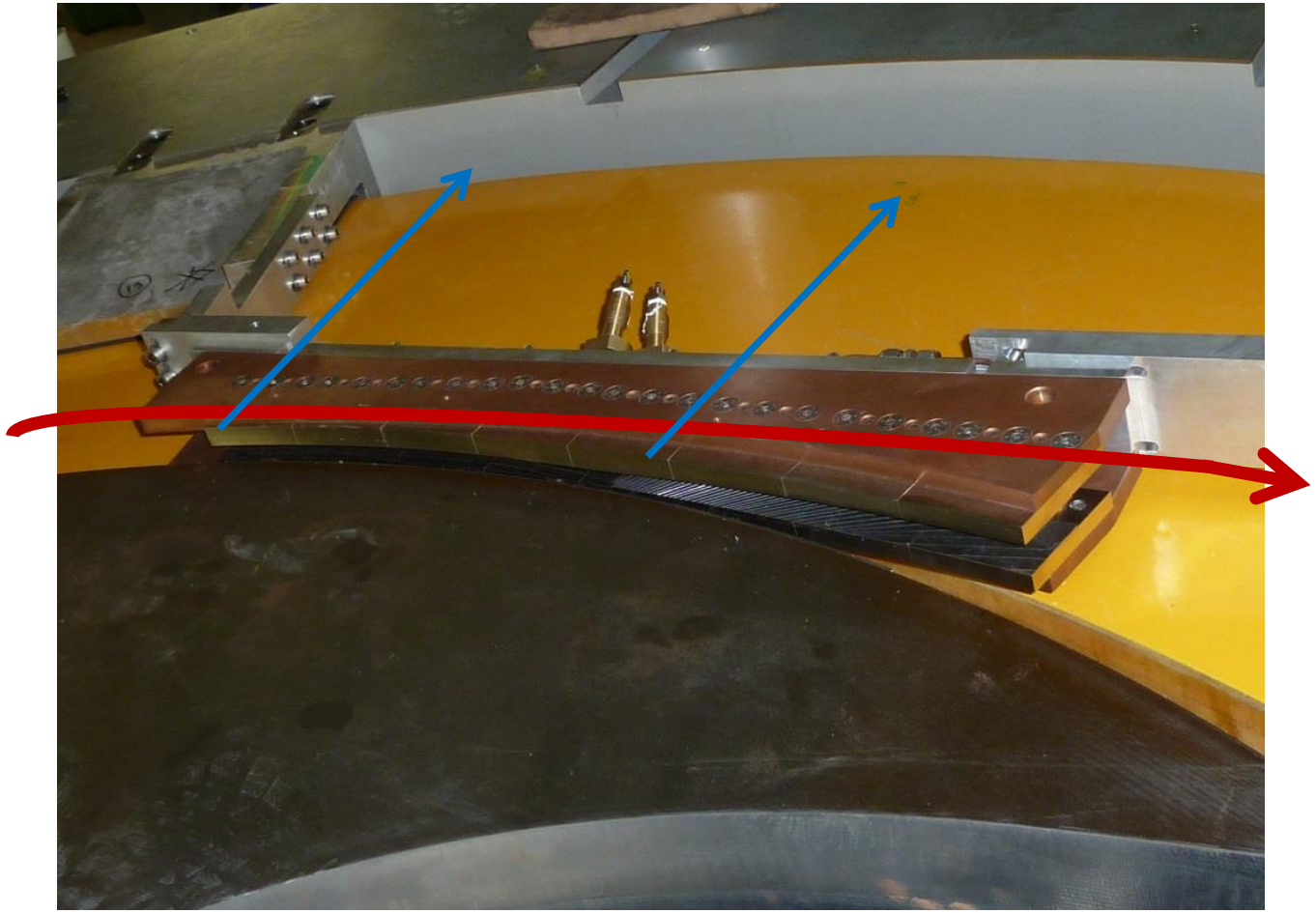
- Different types
  - Passive: soft iron magnetized by the main field
  - Active:
    - Using permanent magnets
    - Using coils

- Designed in such a way as to minimize adverse effects on internal orbits

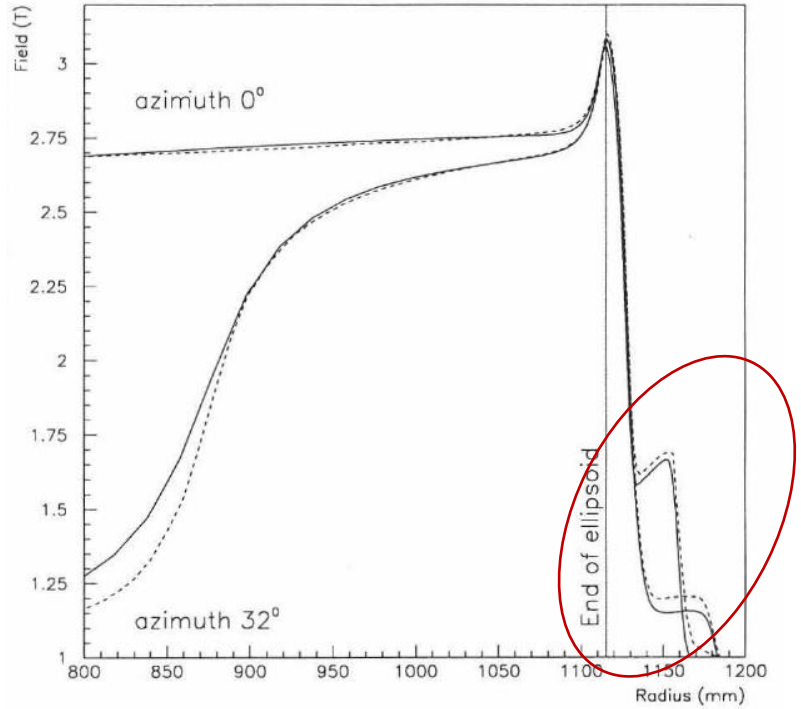


C235 Gradient Corrector

- A passif channel, magnetized by the cyclotron magnetic field
- Placed between the main coils, against (almost touching) the hill sector.



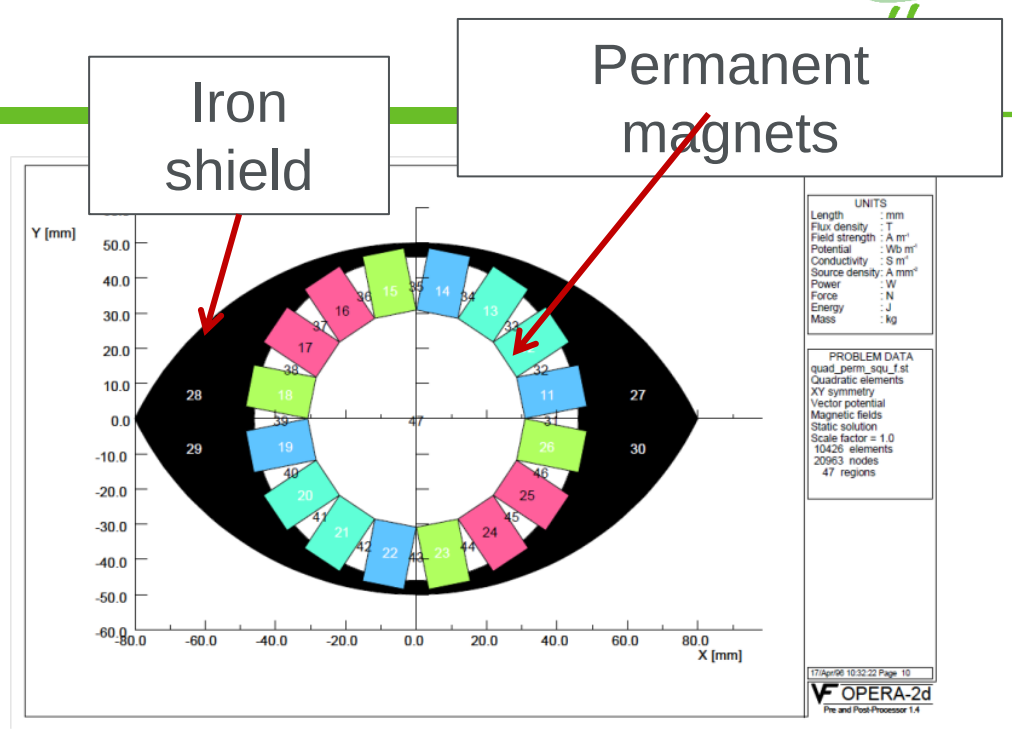
- A descending ‘slider’ of gradually decreasing magnetic field that guides beam gently through fringe field

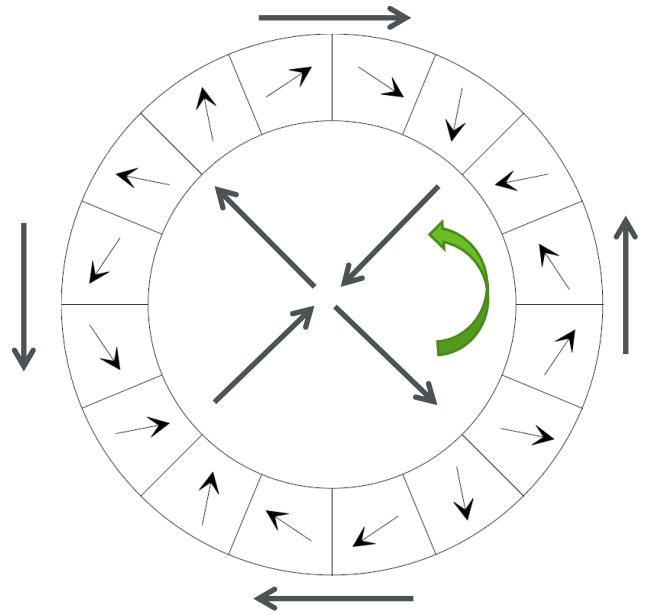


C235 Permanent Magnet Doublet

- Placed in return yoke

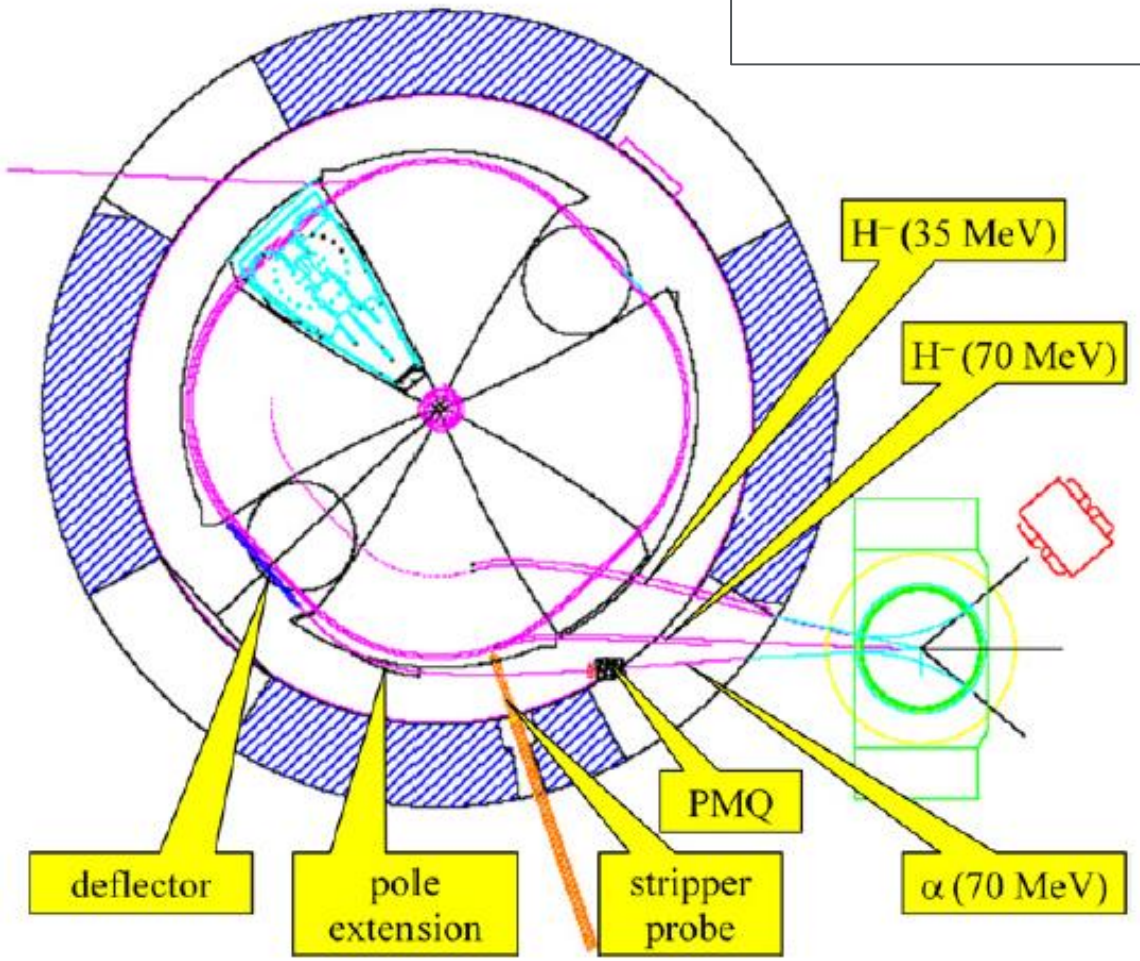






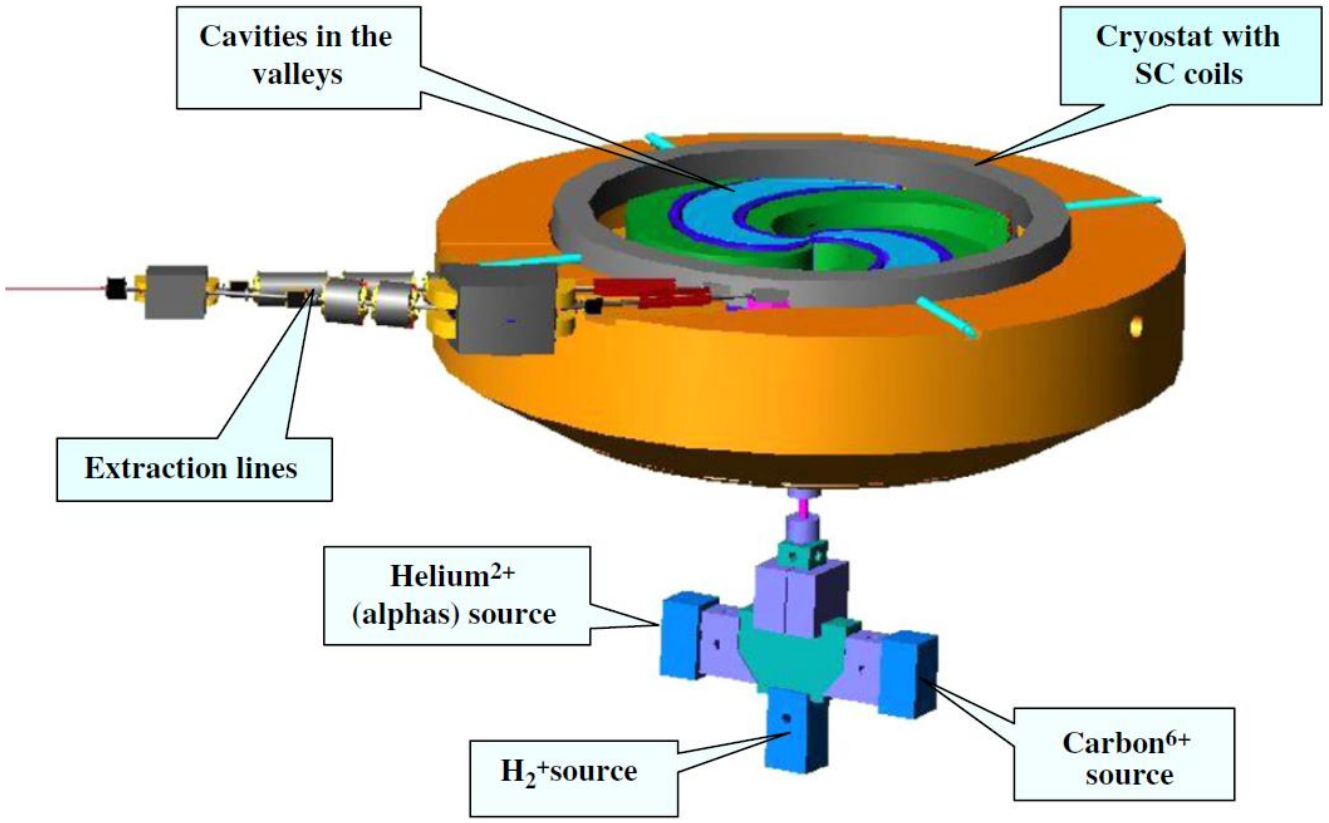
Two extraction systems in one cyclotron

C70 multiple particle cyclotron for Arronax in Nantes
H^-, D^-, α, H_2^+



- Stripping extraction for negative particles
- ESD for α-particle and H_2^+
- Two opposite exit ports
- Simultaneous dual beam capability for H- and D-
- Variable energy for H- and D-
- External switching magnet to direct different energies and particle into the beam lines

The IBA C400 cyclotron



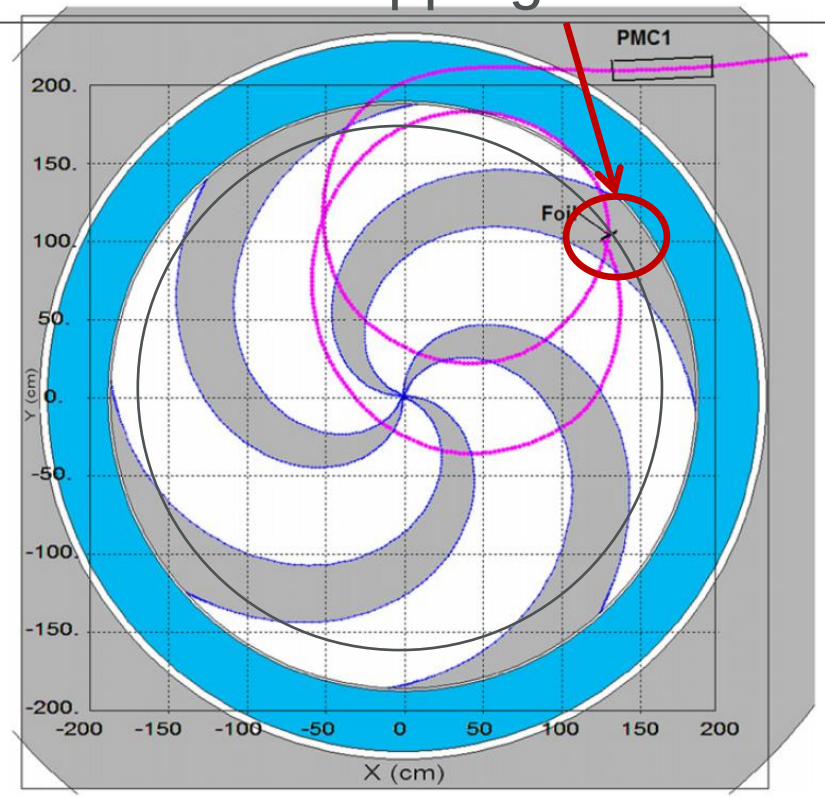
particles	$^{12}C^{6+}$; H_2^+ ; $^4He^{2+}$
Final energy	
ions	400 MeV/A
protons	265 MeV
Bending limit	K=1600
Weight	700 t
Diameter	6.6 m
Hill field	4.5 Tesla
Valley field	2.45 Tesla
Number of cavities	2
RF frquency	75 MHz; h=4
Vdee	80-160 kV
Number of turns	2000
SC coil	NbTi; Helium cooled
Ischronism of H_2^+	Coil in 2 parts

- Full detailed design study was done in collaboration with JINR
- System realized and industrialized by the French company Normandy Hadrontherapy in which IBA is minority shareholder

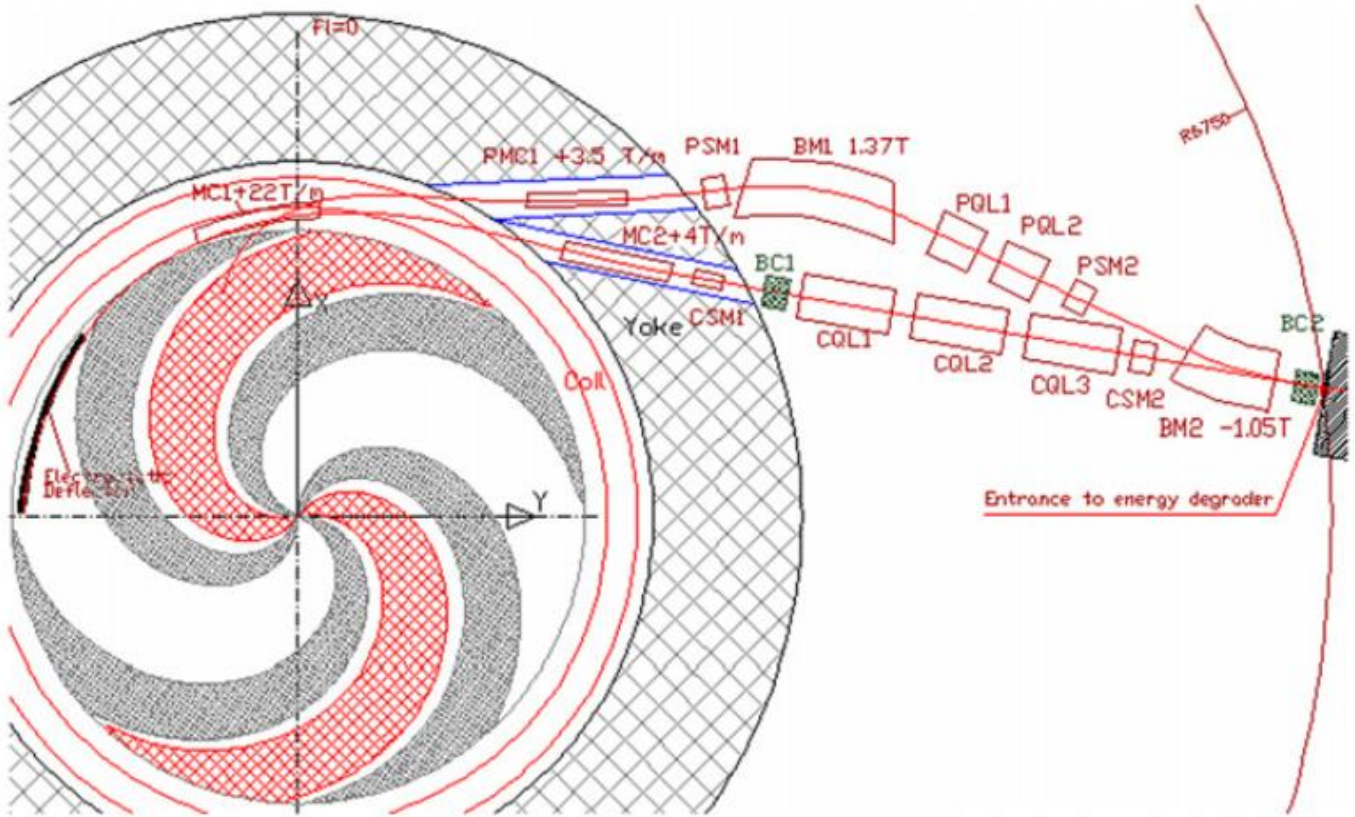
Extraction from the C400

- Protons => stripping of H_2^+ ; $^{12}C^{6+}$ => Electrostatic deflector

H_2^+ two-turn extraction after stripping



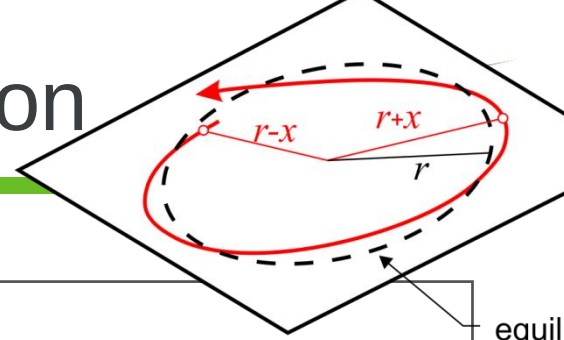
Combining both beams into one beam line



C400 magnet size



Extraction continued: turn-separation in a cyclotron



- A coherent beam oscillation is an oscillation around the equilibrium orbit

$\Rightarrow$

$$r(\theta) = \overbrace{r_0(\theta)}^{\text{EO}} + \overbrace{\underbrace{x(\theta)}_{\text{amplitude}} \sin(\underbrace{\upsilon_r\theta + \theta_0}_{\text{phase}})}^{\text{betatron oscillation}}$$

$\Downarrow$

- There are 3 different mechanisms to create turn separation

$$\Delta r(\theta_i) = \overbrace{\Delta r_0(\theta_i)}^{\text{acceleration}} + \overbrace{\Delta x \sin(2\pi n(\upsilon_r - 1) + \theta_0)}^{\text{resonance}} \overbrace{+2\pi(\upsilon_r - 1)x \cos(2\pi n(\upsilon_r - 1) + \theta_0)}^{\text{precession}}$$

Turn Separation

- Turn separation = radial increase per turn (with $\theta_i = 2\pi n$ where n is the turn number)

$$\Delta r(\theta_i) = \overbrace{\Delta r_0(\theta_i)}^{\text{acceleration}} + \overbrace{\Delta x \sin(2\pi n(\nu_r - 1) + \theta_0)}^{\text{resonance}} + \overbrace{2\pi(\nu_r - 1)x \cos(2\pi n(\nu_r - 1) + \theta_0)}^{\text{precession}}$$

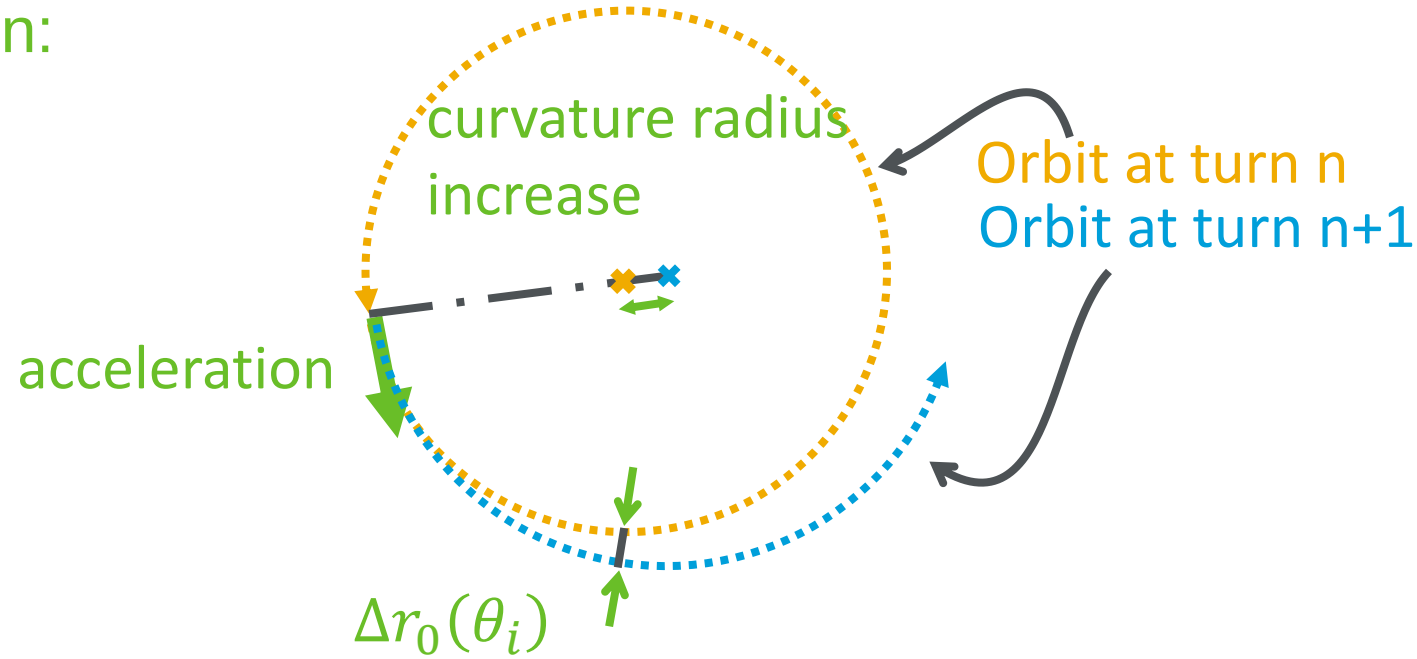
Protect, Enhance and Save Lives

Turn Separation

- Turn separation = radial increase per turn (with $\theta_i = 2\pi n$ where n is the turn number)

$$\Delta r(\theta_i) = \overbrace{\Delta r_0(\theta_i)}^{\text{acceleration}} + \overbrace{\Delta x \sin(2\pi n(\nu_r - 1) + \theta_0)}^{\text{resonance}} + \overbrace{2\pi(\nu_r - 1)x \cos(2\pi n(\nu_r - 1) + \theta_0)}^{\text{precession}}$$

  - Acceleration:



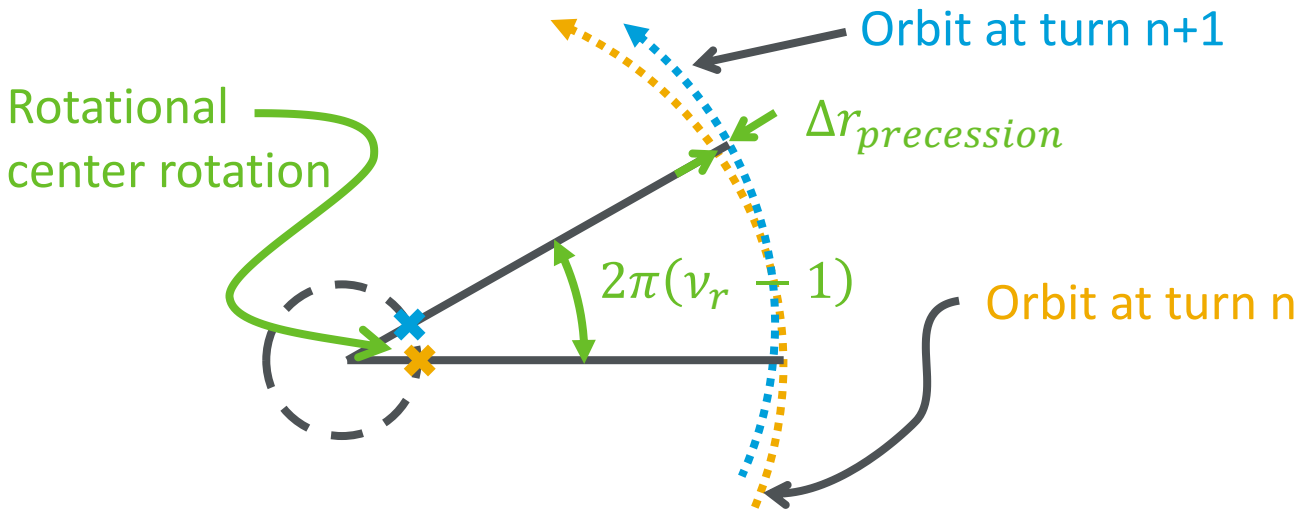
Protect, Enhance and Save Lives

Turn Separation

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  - Precession: change of the betatronic oscillation frequency



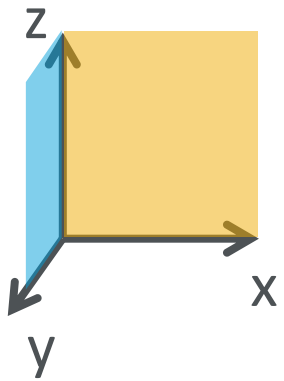
Protect, Enhance and Save Lives

Turn Separation

- Turn separation = radial increase per turn (with $\theta_i = 2\pi n$ where n is the turn number)

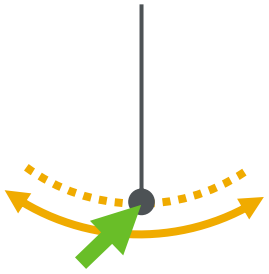
$$\Delta r(\theta_i) = \overbrace{\Delta r_0(\theta_i)}^{\text{acceleration}} + \overbrace{\Delta x \sin(2\pi n(\nu_r - 1) + \theta_0)}^{\text{resonance}} + \overbrace{2\pi(\nu_r - 1)x \cos(2\pi n(\nu_r - 1) + \theta_0)}^{\text{precession}}$$

  - Resonance: exchange of energy between different oscillations mode (axial to radial oscillation in the case extraction case)

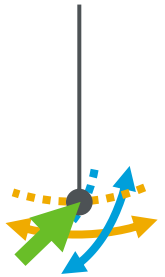


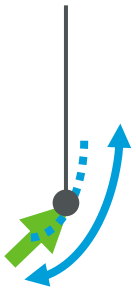


Oscillations in plane zx



Periodic impulse in plan yz





Oscillations in plane yz

Protect, Enhance and Save Lives

How to create turn-separation for extraction

1. By acceleration ⇒ high dee-voltage

⇒ IBA/SHI C235

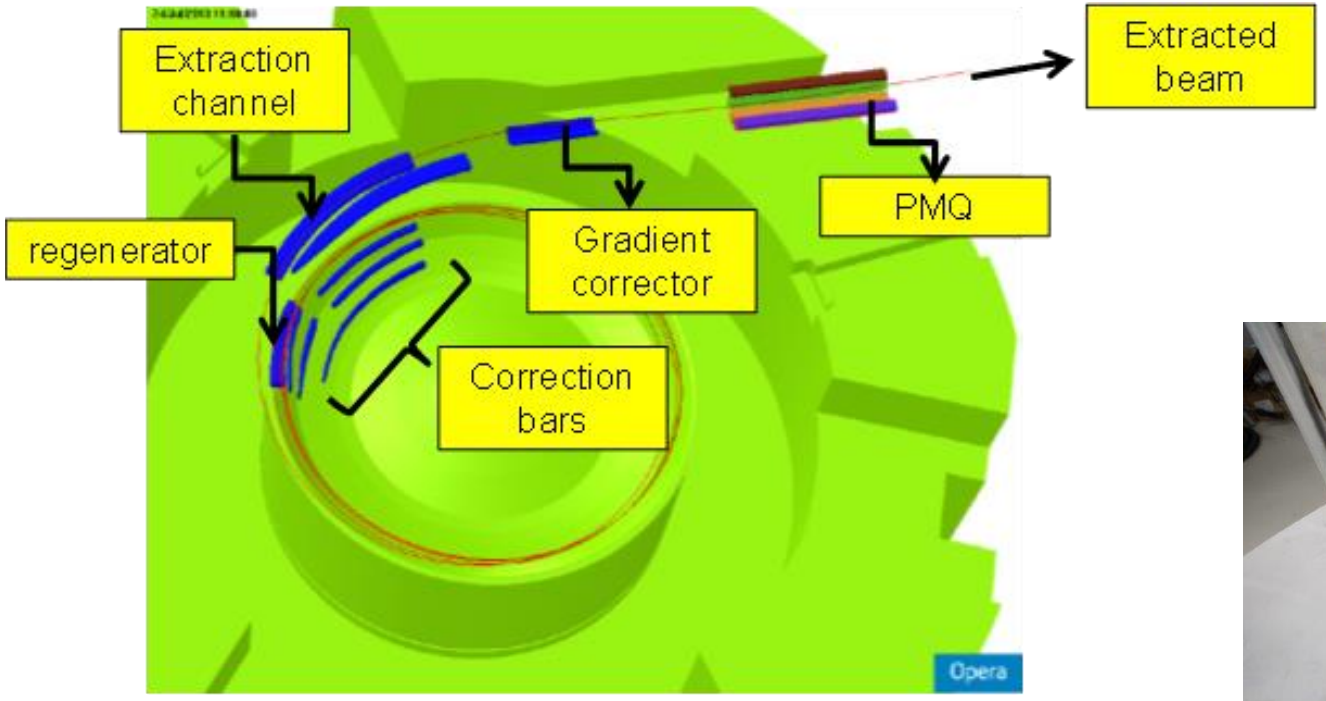
2. Precessional extraction (more subtle)
   - Create oscillation amplitude with 1st harmonic or beam off-centering
   - Accelerate into fringe field where ν_r~0.7
   - Turn separation obtained from orbit-centering move (linked to betatron phase advance)

⇒ Varian SC cyclotron

3. Regenerative extraction (even more subtle)
   - Second harmonic gradient bump: $2\nu_r=2$; ν_r is locked to 1 in the stopband
   - Exponential growth of betatron amplitude

(IBA S2C2, Mevion Monarch)

The IBA S2C2 extraction system



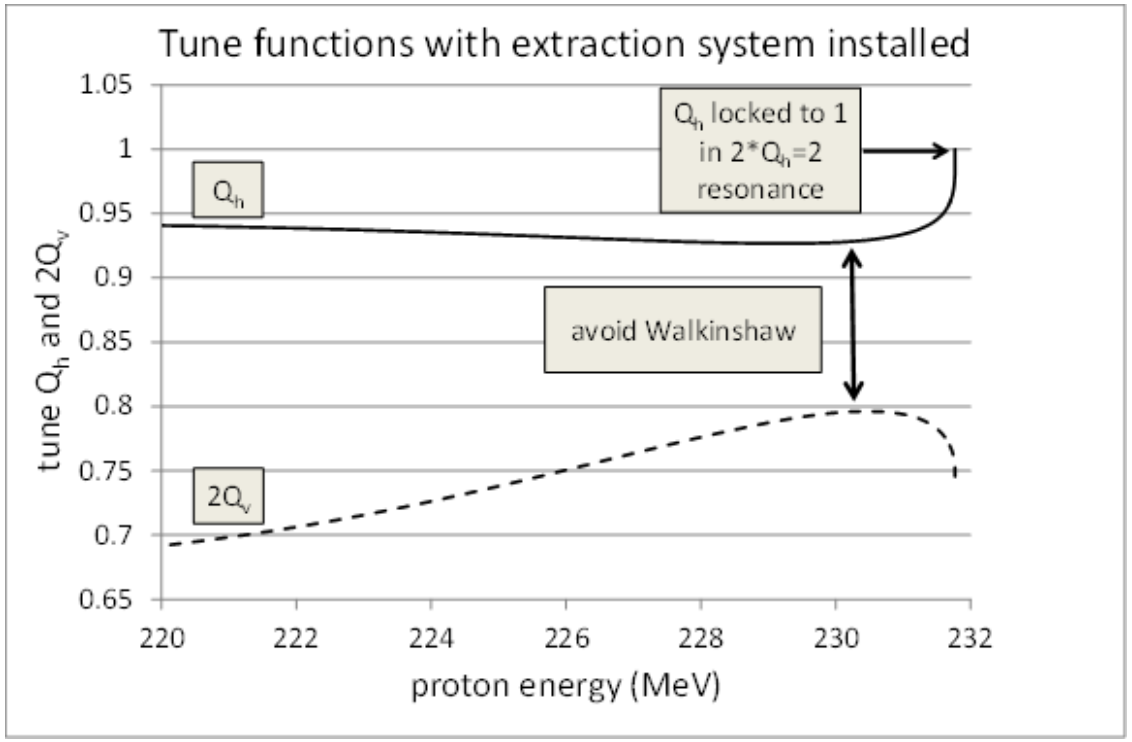
- Fully passive system => only soft iron
- Correction bars are needed to reduce strong first harmonic error during acceleration.



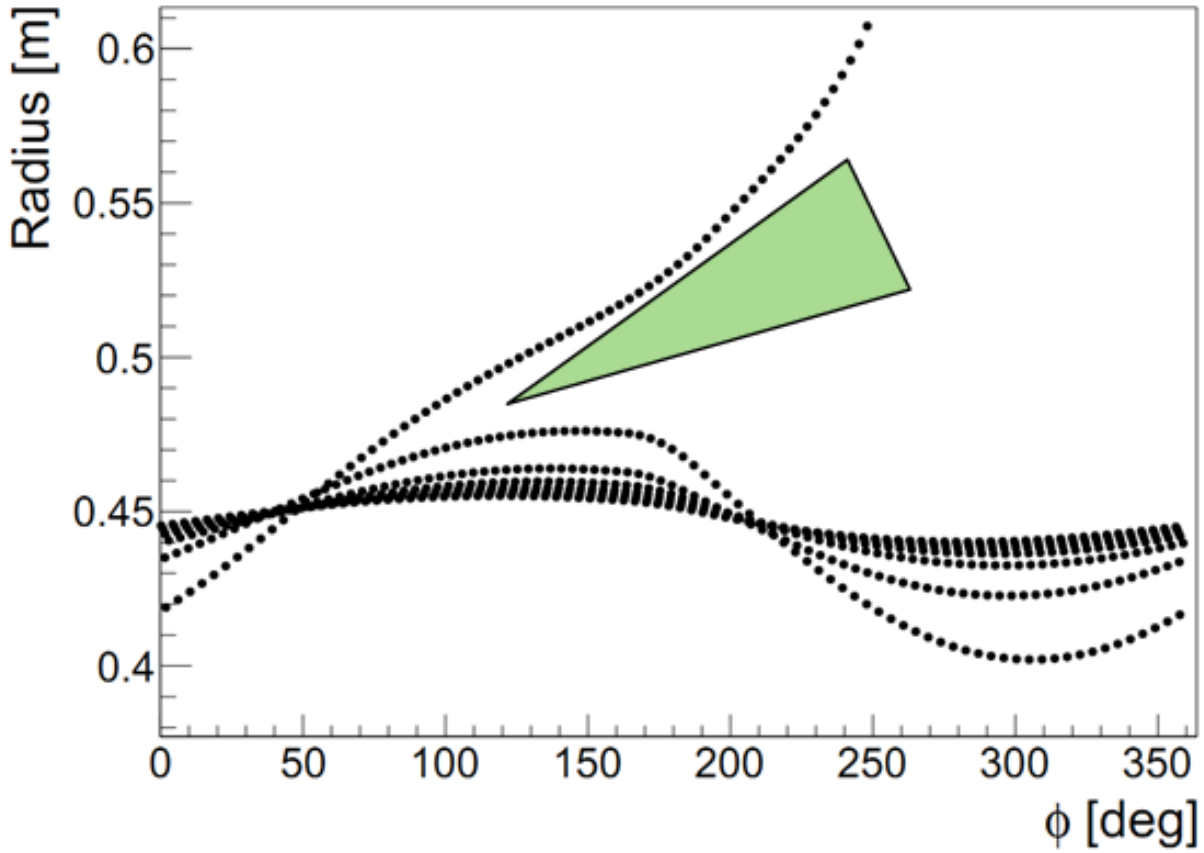
- Strong local field bump due to regenerator increases horizontal betatron frequency and locks it to unity.
- Use resonant extraction based on $2\nu_r=2$ resonance. NB: the “2” implies 2nd harmonic!
- Unstable orbit is pushed towards the extraction channel by the first harmonic

Regenerative extraction based on $2\nu_r=2$ resonance

- A strong regenerator bump increases ν_r and locks it to 1.
- 2nd harmonic drives instability / blow up
- Avoid Walkinshaw resonance ($\nu_r=2\nu_z$)

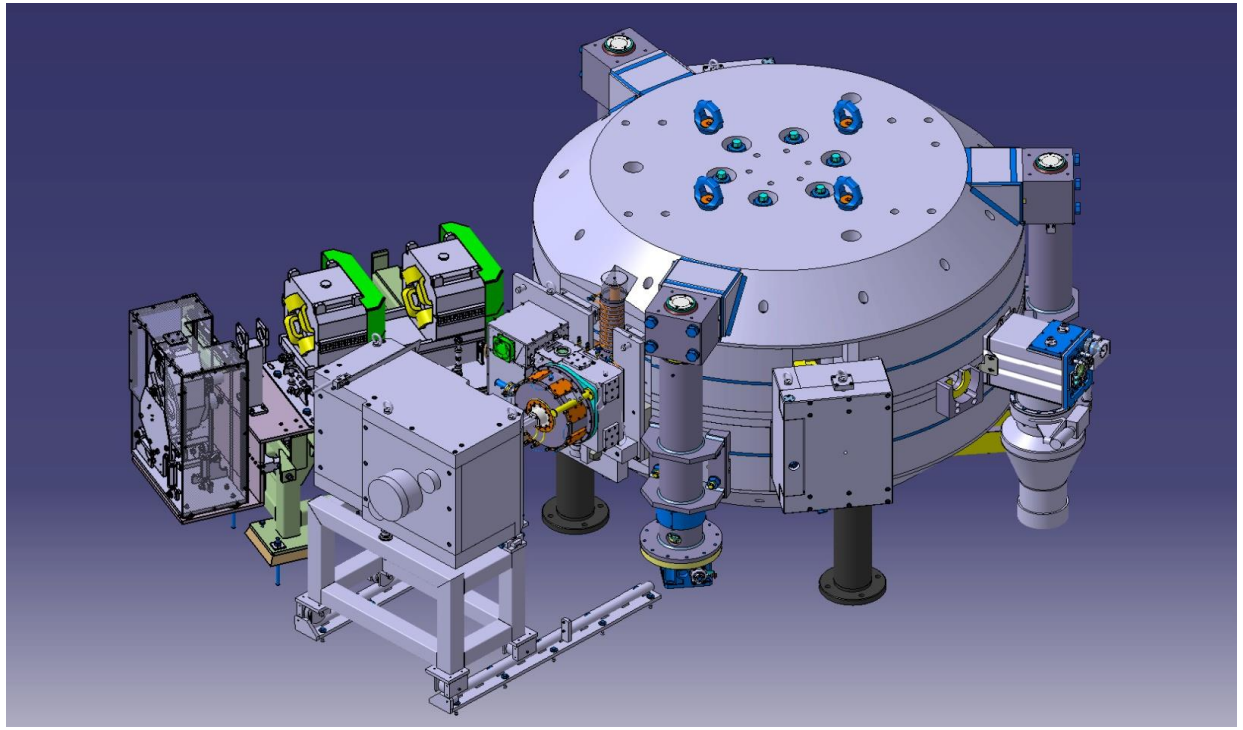


- A steady shift of the beam towards the extraction channel builds up
- Harmonics' phase determines the azi value of the knot:



S2C2

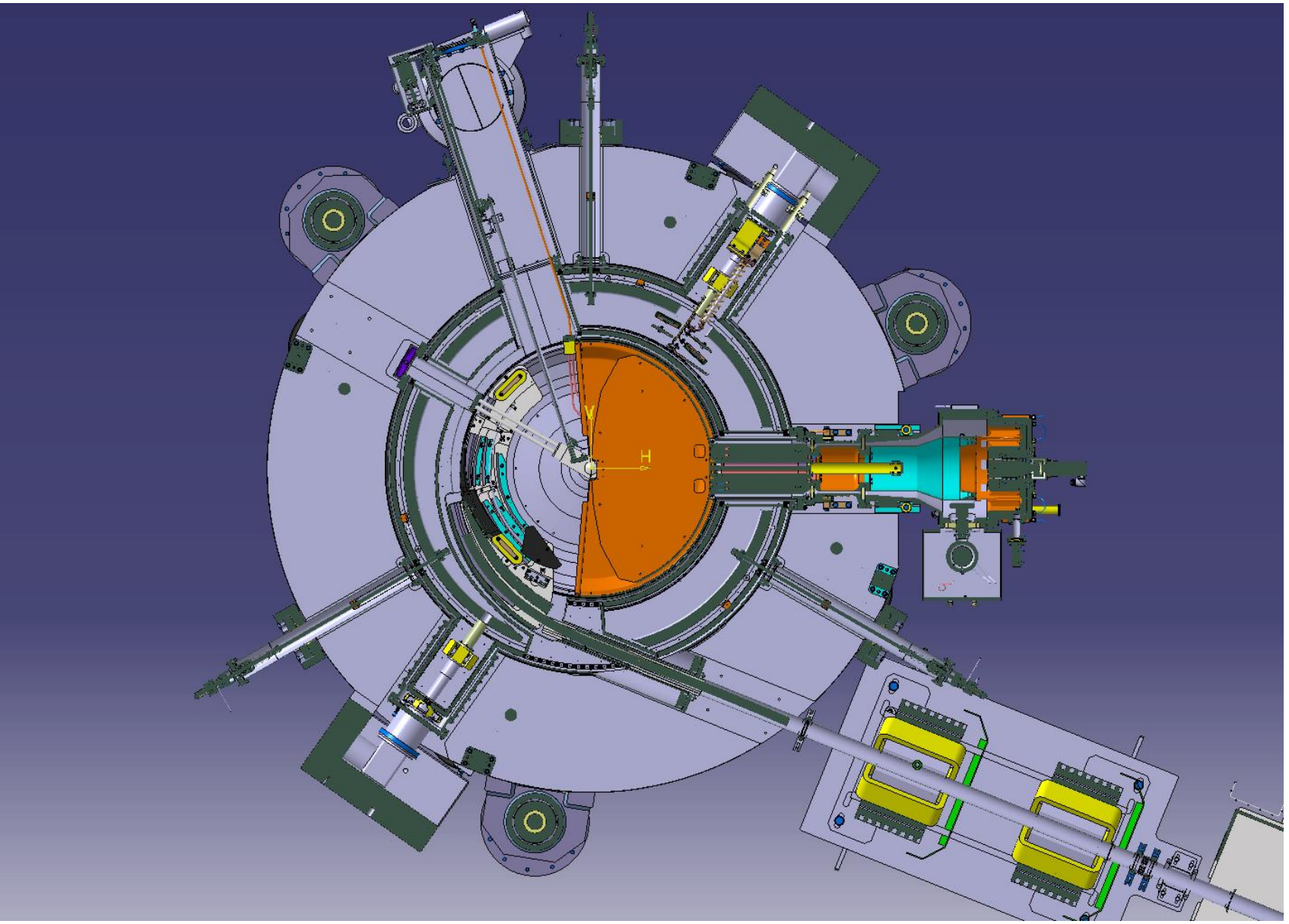
S2C2 – General system layout and parameters



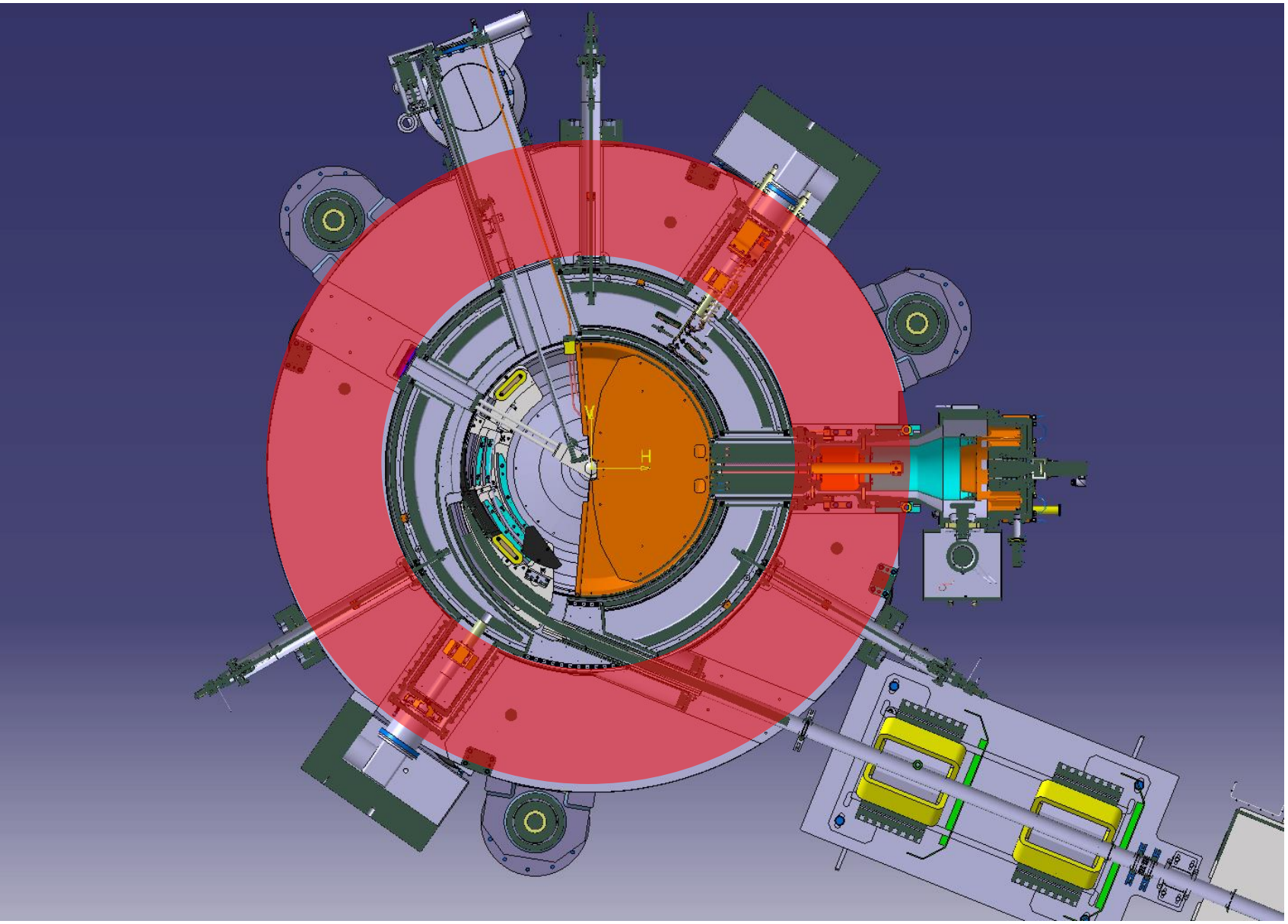
Maximum Energy	230/250 MeV
Size	
yoke/pole radius	1.25 m/0.50 m
weight	50 tons
Coil	NbTi - wire in channel
ramp up rate / time	2-3A/min / 4 hours
windings/coil	3145
stored energy	12 MJ
Magnetic field	
central/extraction	5.7 T/5.0 T
Cryo cooling	conductive
	4 cryocoolers 1.5 W
initial cooldown	12 days
recovery after quench	less than 1 day
Beam pulse	
rate/length	1000 Hz/7 μsec
RF system	self-oscillating
frequency	93-63 MHz
voltage	10 kV
Extraction	Passive regenerative
Ion source	PIG cold cathode
Central region	removable module

- An invited talk on this project was presented at the 2013 cyclotron conference in Vancouver
- Several contributions can be found on the ECPM2012-website

S2C2 overview

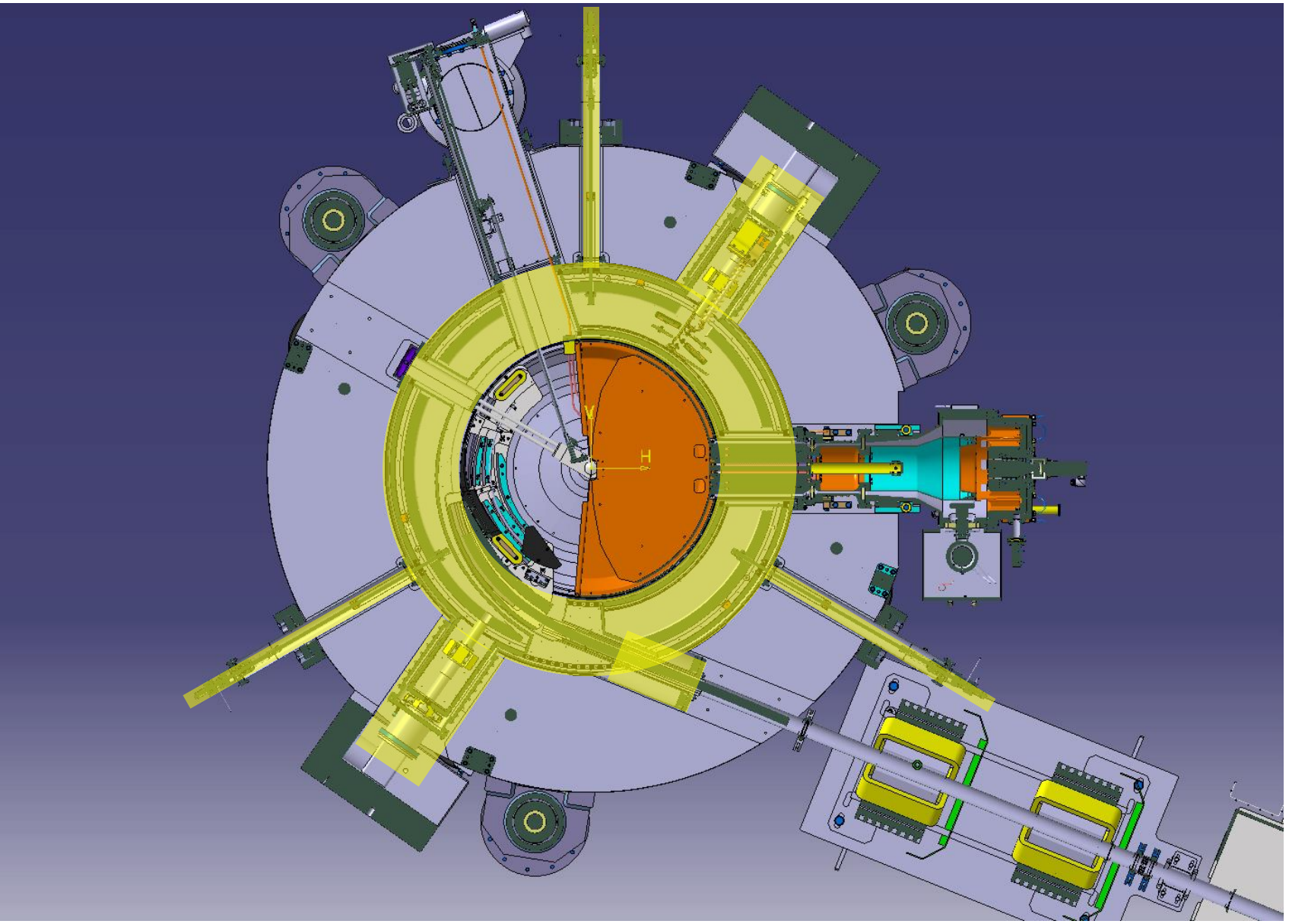


S2C2 overview



Magnet return yoke

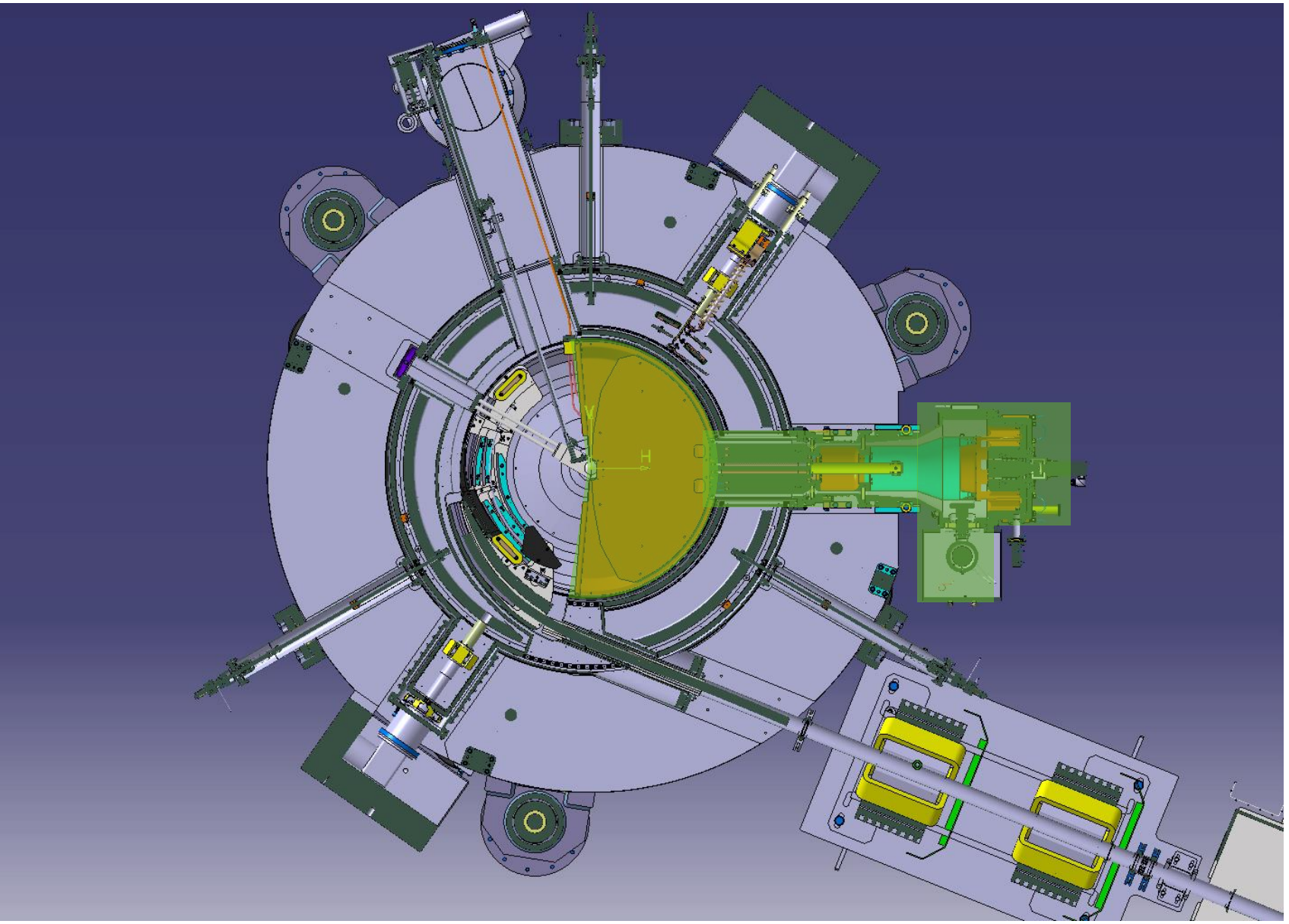
S2C2 overview



Magnet return yoke

Cryostat with coil

S2C2 overview

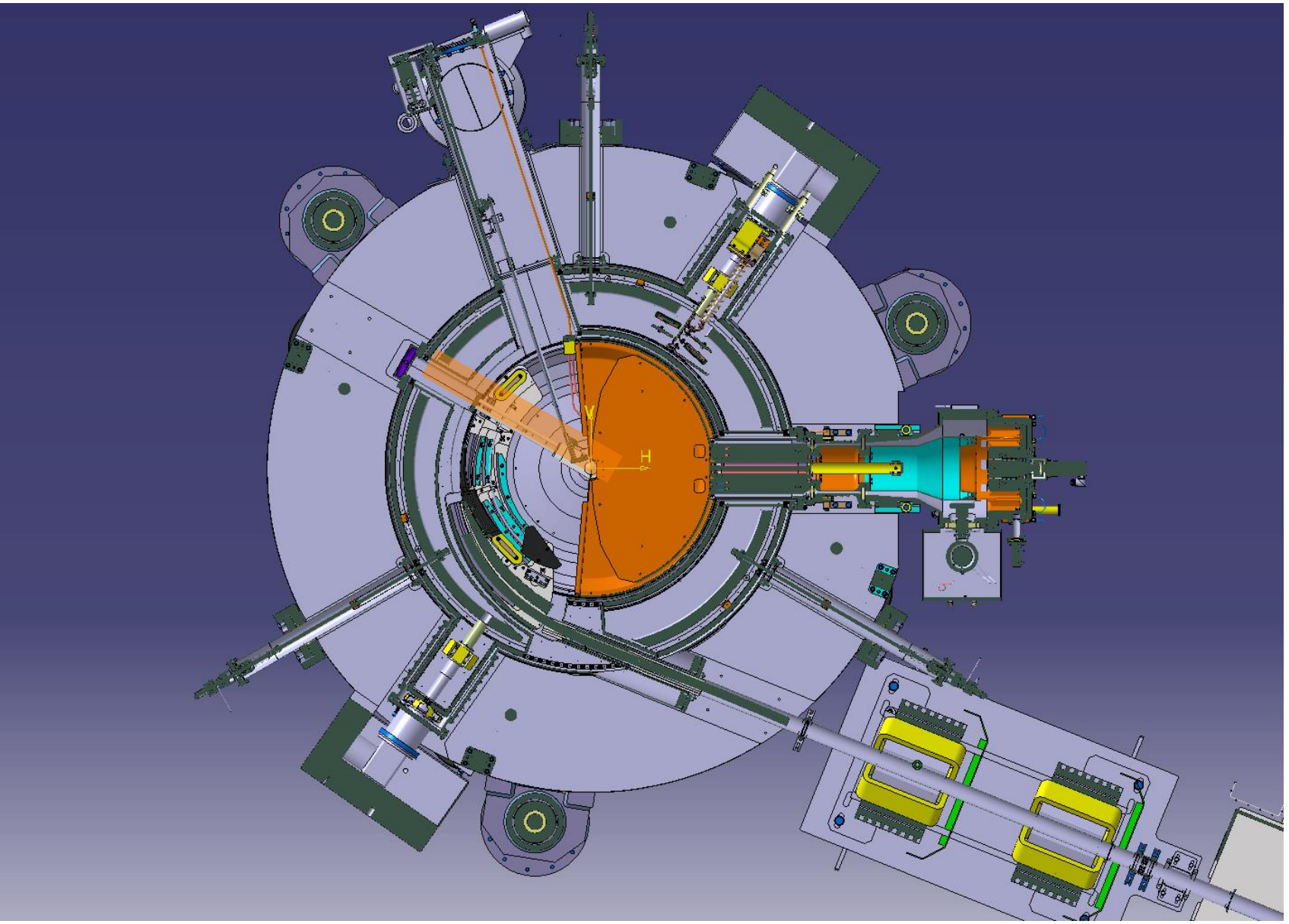


Magnet return yoke

Cryostat with coil

RF system, incl RotCo

S2C2 overview



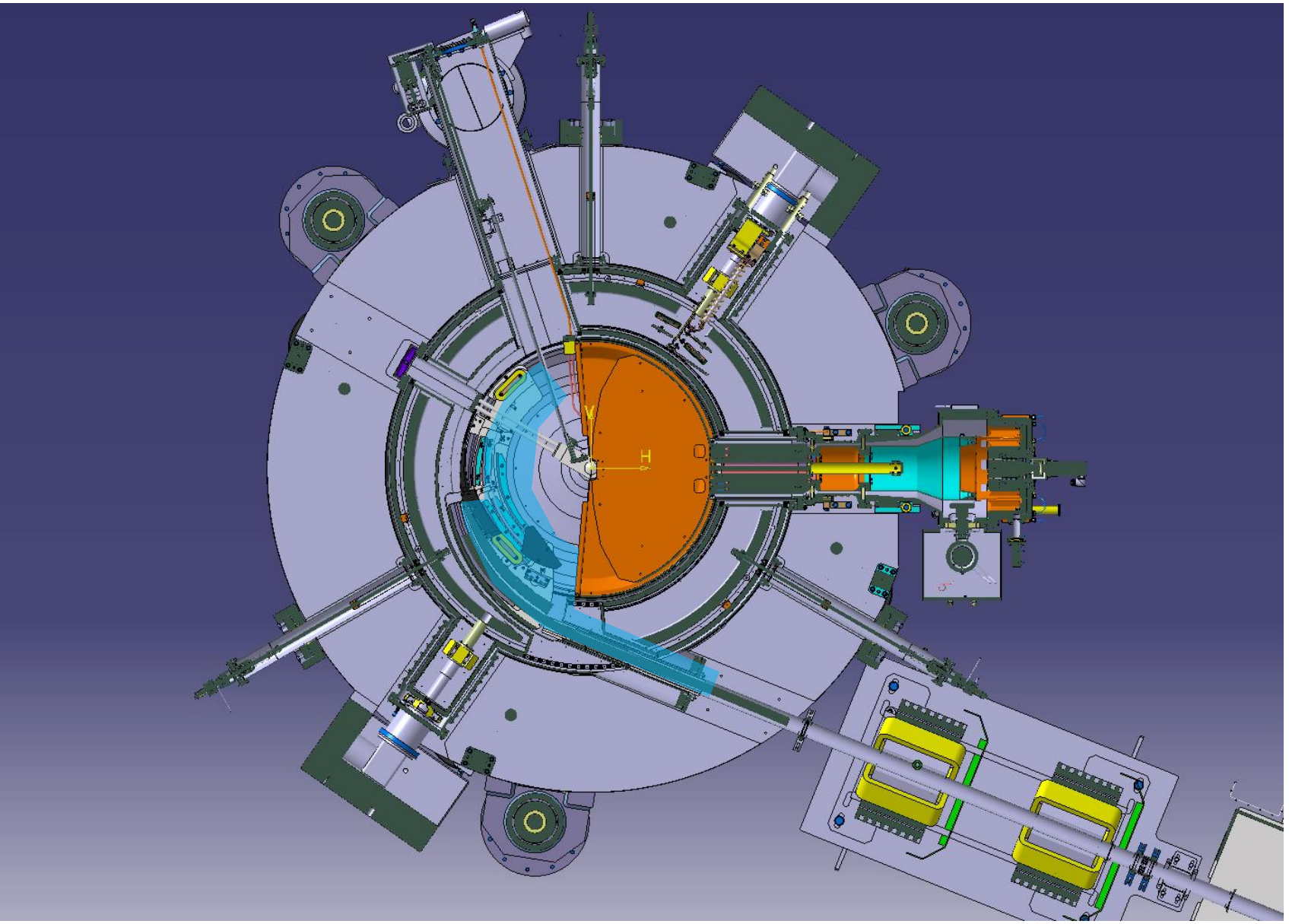
Magnet return yoke

Cryostat with coil

RF system

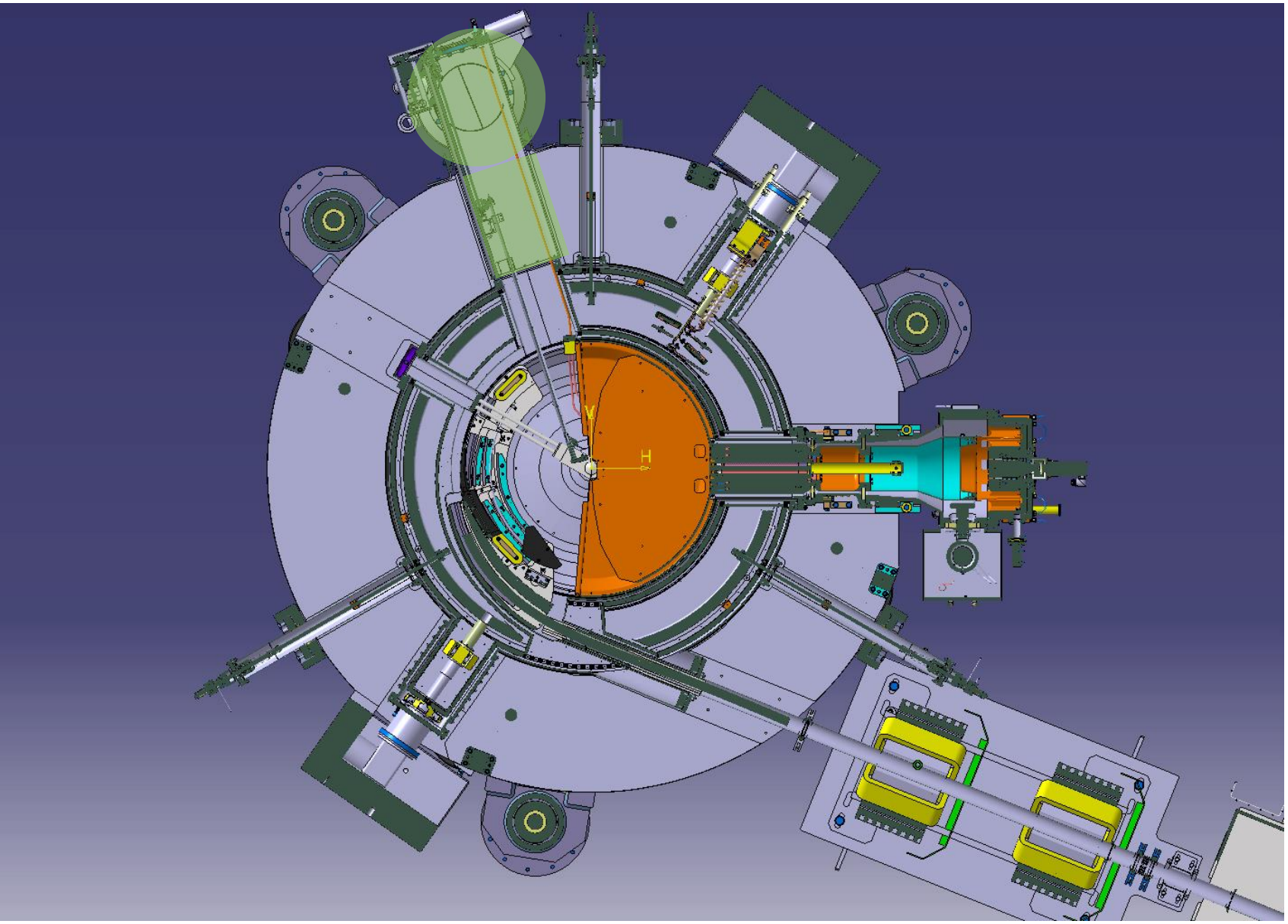
Ion source+central region

S2C2 overview



- Magnet return yoke
- Cryostat with coil
- RF system
- Ion source+central region
- Extraction system

S2C2 overview



Magnet return yoke

Cryostat with coil

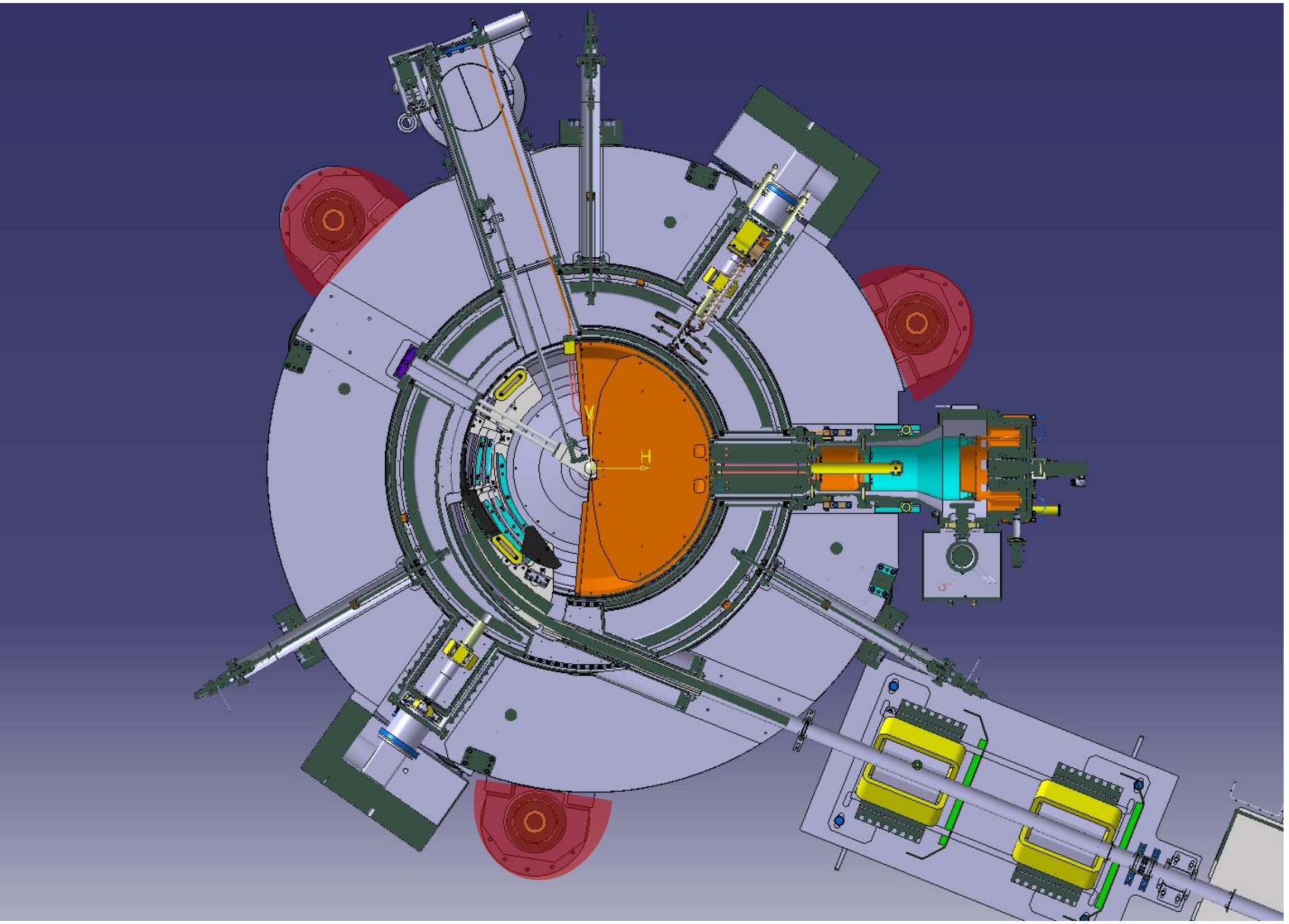
RF system

Ion source+central region

Extraction system

Vacuum system

S2C2 overview



Magnet return yoke

Cryostat with coil

RF system

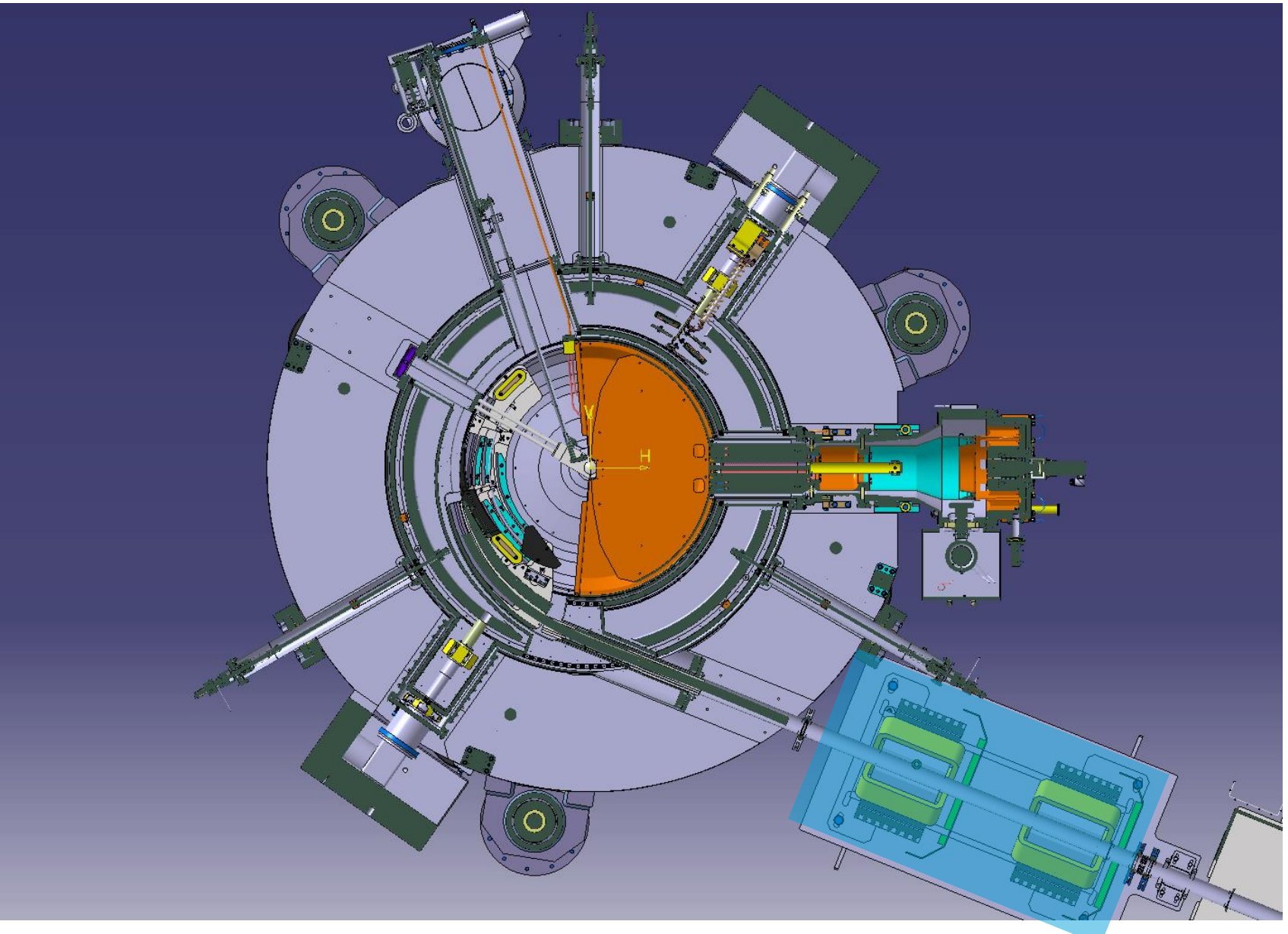
Ion source+central region

Extraction system

Vacuum system

Yoke lifting system

S2C2 overview



Magnet return yoke

Cryostat with coil

RF system

Ion source+central region

Extraction system

Vacuum system

Yoke lifting system

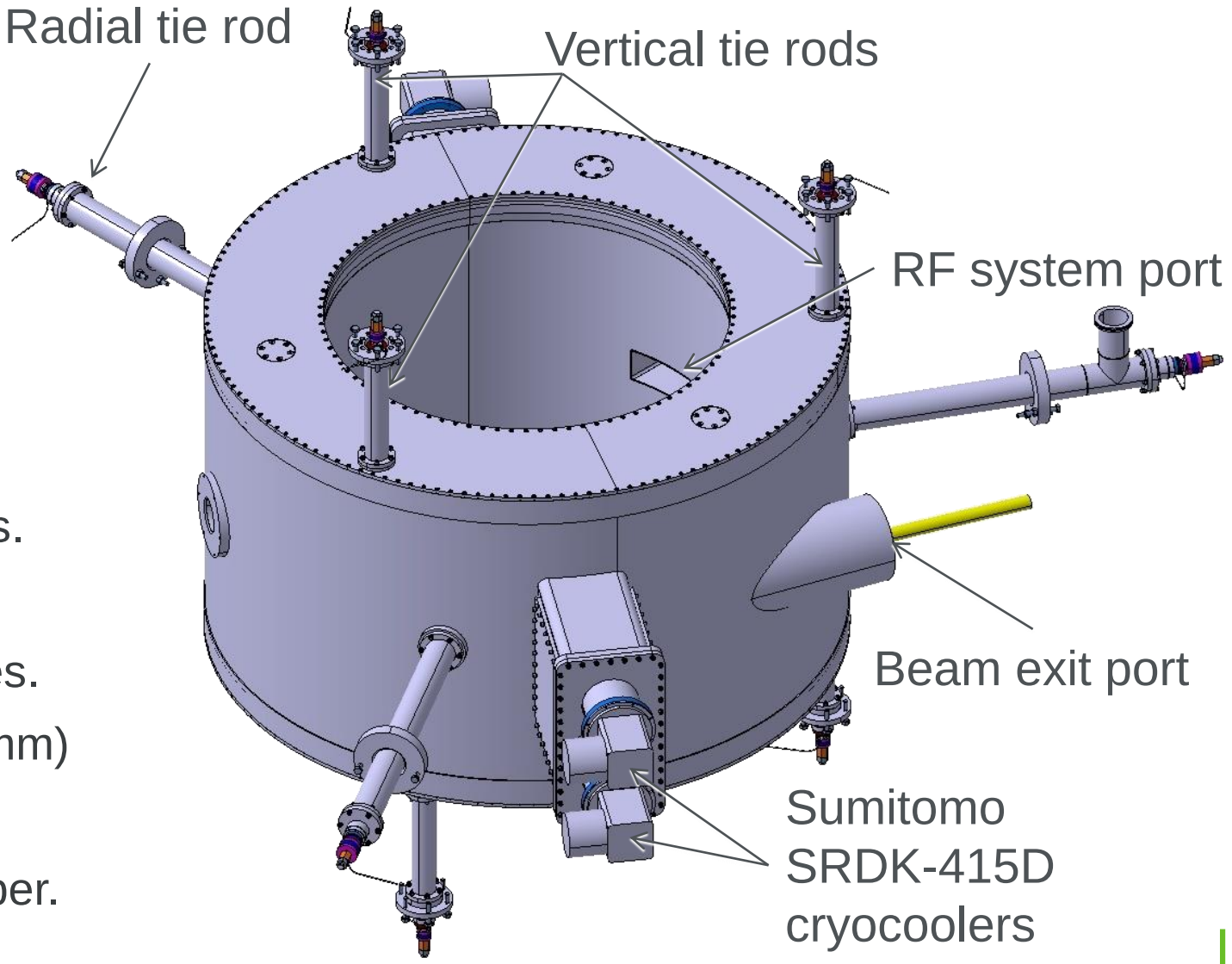
Extracted beam line

Superconducting coil

Designed and manufactured by ASG (Genua, Italy)

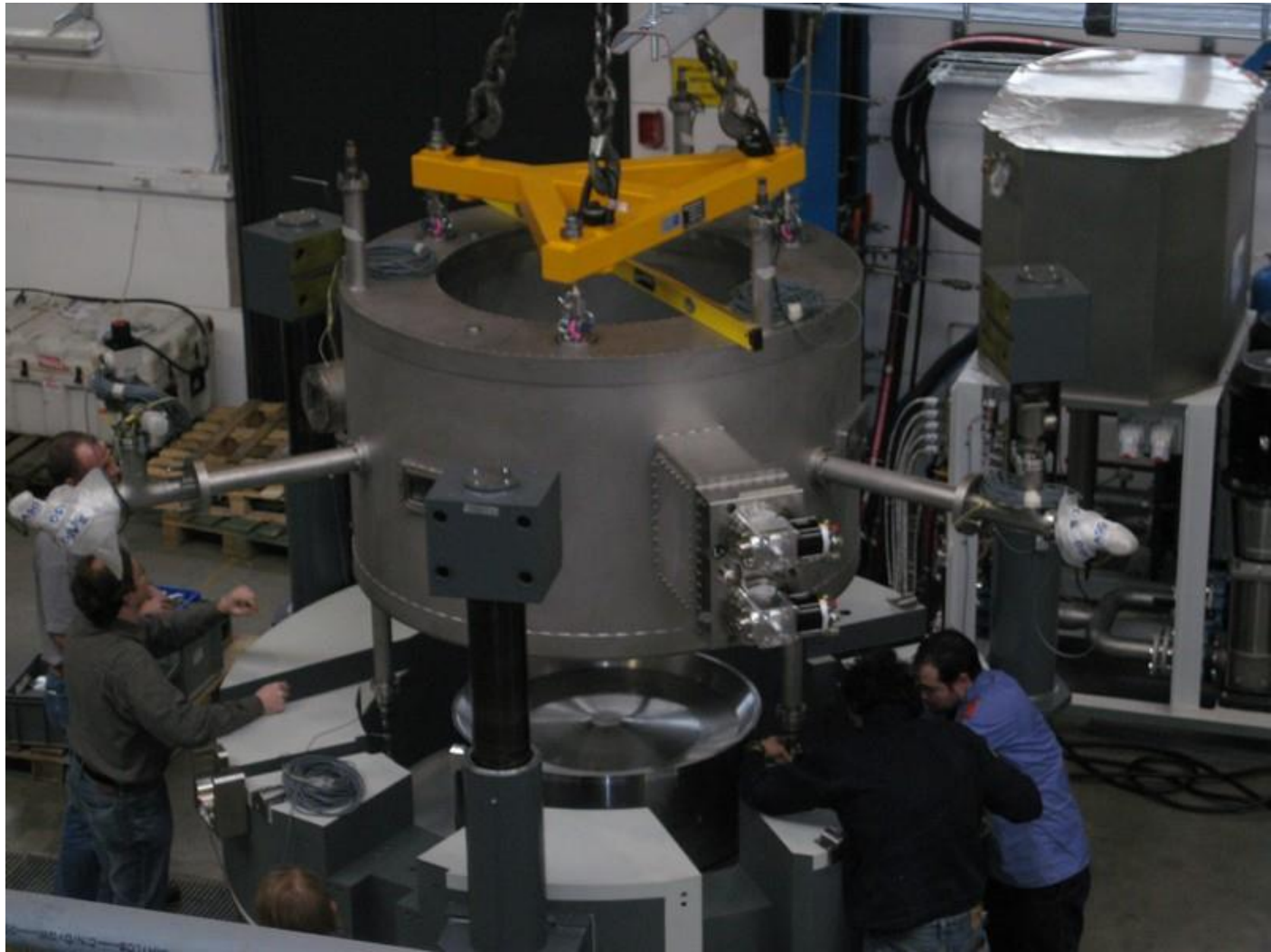
- NbTi wire in channel coil.
- Suspended cold mass: 3tons.
- Nominal current: 650A (56 A/mm²).
- Nominal ampere-turns: $4.3x10^{+6}$At.
- Conduction cooled by 4 SHI cryocoolers.
- Overall weight: 4 tons.
- 9 Inconel tension rods with strain gauges.
  (radial ø14mm; upper ø8mm; lower ø6mm)

- Cryostat is the cyclotron vacuum chamber.



Superconducting coil





Tight fit, 1 mm clearance between yoke and outer wall of cryostat.

QUIZ #5

Which element on the S2C2 gave us the biggest headaches?

1. Superconducting coil
2. RotCo
3. Regenerative extraction



PART II:

Particle therapy of cancer

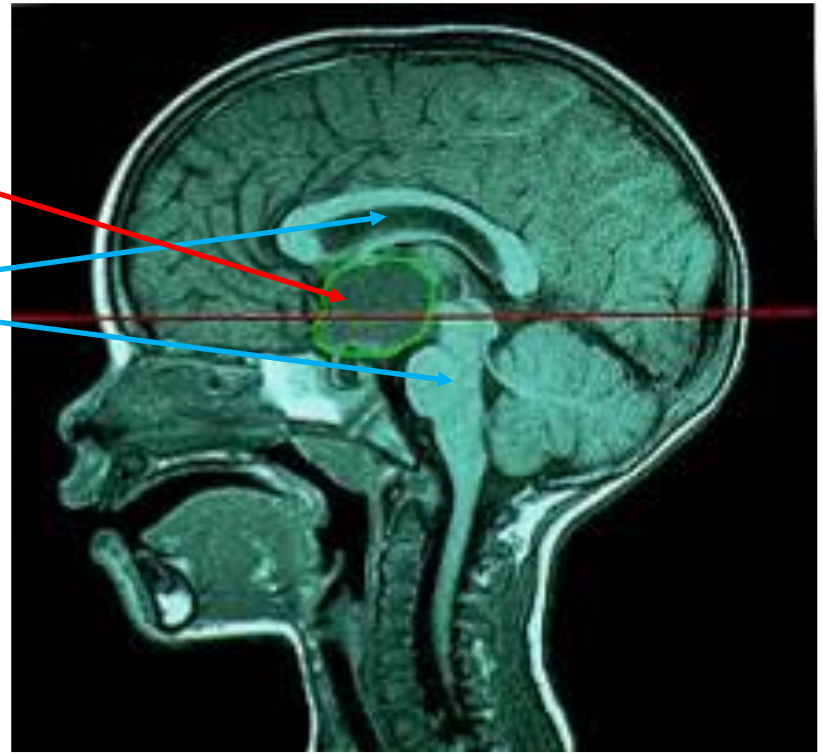
106

HOLY GRAIL of Radiation Therapy

Ideal Situation:

- Provide a lethal dose to the tumor

  and
- Spare perfectly the surrounding healthy tissue



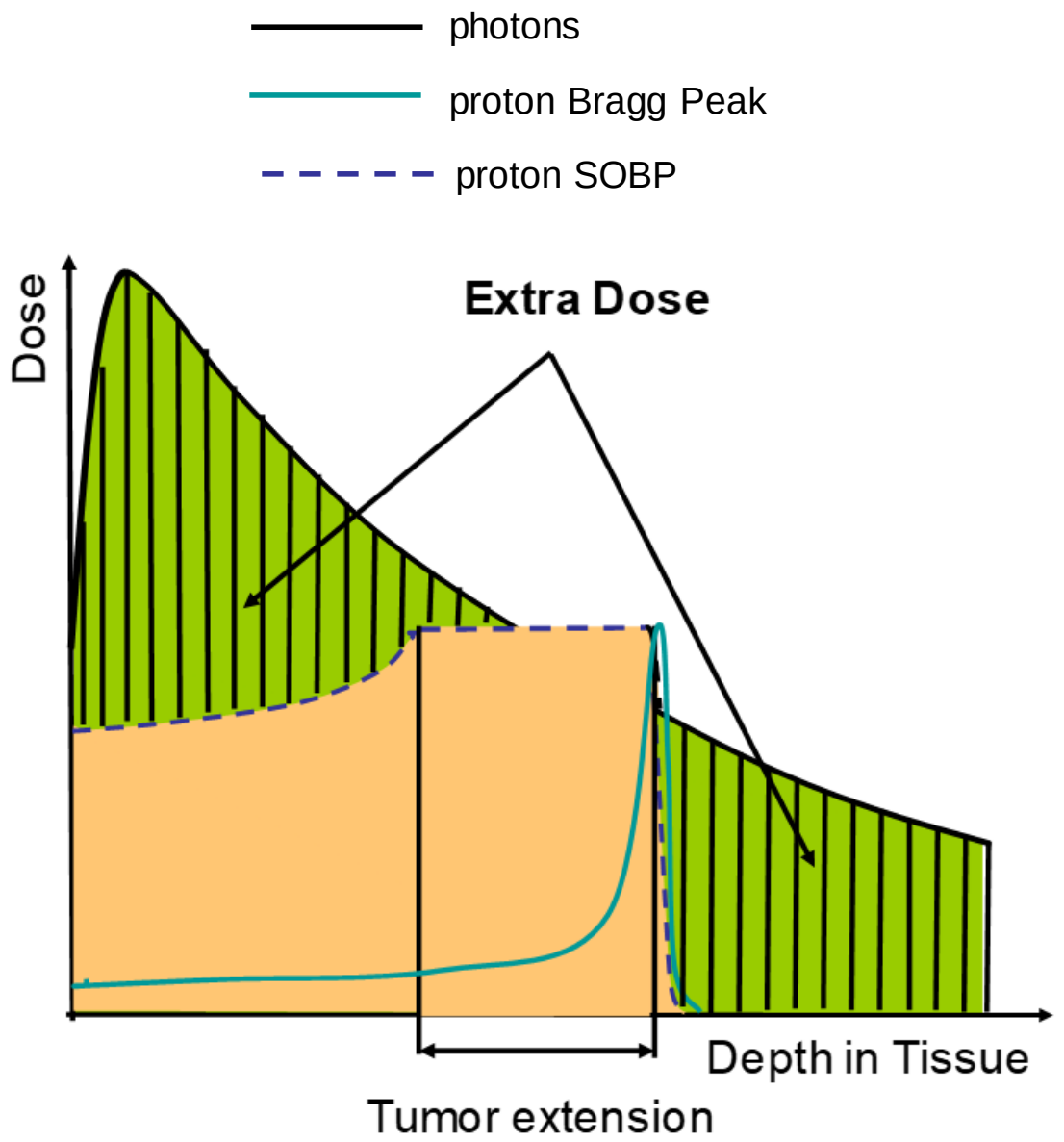
In Practice:

- Deposit the radiation dose more precisely in the target volume with less dose in the surrounding healthy tissues.

Photon-Proton dose distribution comparison

- Position of the Bragg peak depends on beam energy
- Incoming energy modulation ⇒ Spread Out Bragg Peak : SOBP

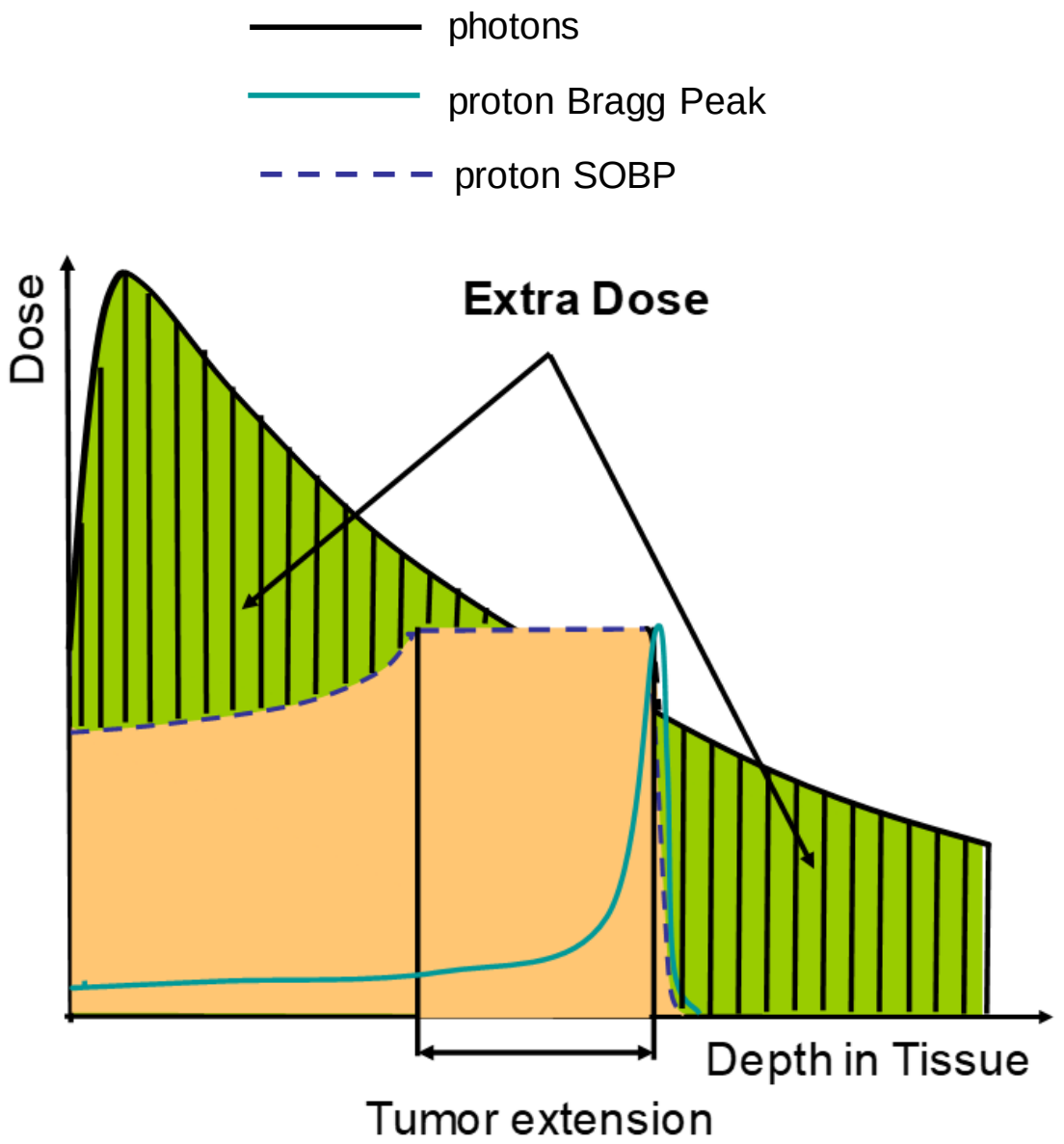
➢ No dose downsteam and low dose upstream of tumor



Photon-Proton dose distribution comparison

- Position of the Bragg peak depends on beam energy
- Incoming energy modulation ⇒ Spread Out Bragg Peak : SOBP

- ➢ No dose downsteam and low dose upstream of tumor
- ➢ Up to 50% reduced risk of radiation induced secondary cancer
- ➢ Drastically lower risk of adverse effects (treatment toxicity, side effects, growth abnormality) – better quality of life



Main requirements for a proton therapy system

1. Ability to reach the tumor
   - Range in patient: up to 32 g/cm²
   - Range modulation: up to full range, with steps of 0.5 g/cm²
   - Field size: up to 30 x 40 cm

2. Ability to reach the tumor from any selected direction
   - Isocentric Gantry
   - Precise, robotic patient positioner

3. Ability to verify and control the dose deposition using IC’s

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   - Precise, robotic patient positioner

3. Ability to verify and control the dose deposition using IC's

4. Ability to reach the tumor accurately
   - Distal dose falloff: maximum 1 mm above physical limit
   - Penumbra: maximum 2 mm at skin
   - Patient positioner accuracy and reproducibility: 0.5 mm for small displacements
   - Gantry accuracy and reproducibility: 1 mm radius circle of confusion
   - Patient alignment methods: lasers, light fields, X-rays

Accelerator parameters driving technology choice

Energy:

- defines the range in the patient (230 MeV enough)

Energy definition:

- defines range accuracy and distal falloff

Beam current:

- defines dose rate (10^{11} p/sec enough (10 nA))

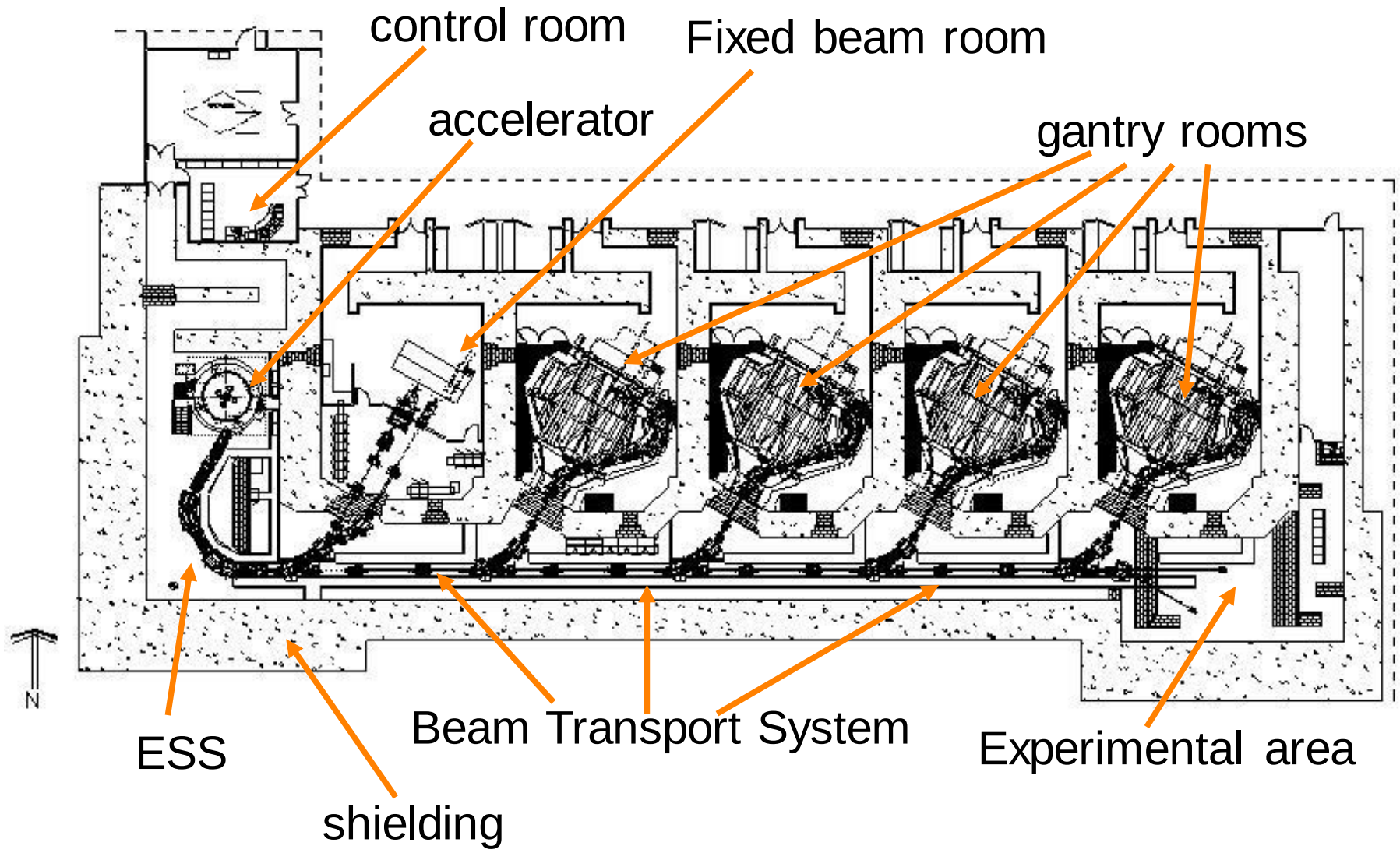
Beam current stability and noise:

- defines ability to use scanning

Accurate and fast beam intensity control:

- needed for conformal therapy

Typical Proton Therapy Facility Layout



Main Sub-systems of a cyclotron based PT facility

- 230 MeV isochronous cyclotron
- Energy Selection System (ESS)

- Beam Transport and Switching System
- Isocentric Gantries (typically 3) and one Fixed Beam Line

- Nozzles for matching the beam wrt the required treatment (scattering, scanning, diagnostics)
- Robotic Patient Positioners
- Software Control and Safety System

The 230 Mev Cyclotron at MGH/NPTC in Boston



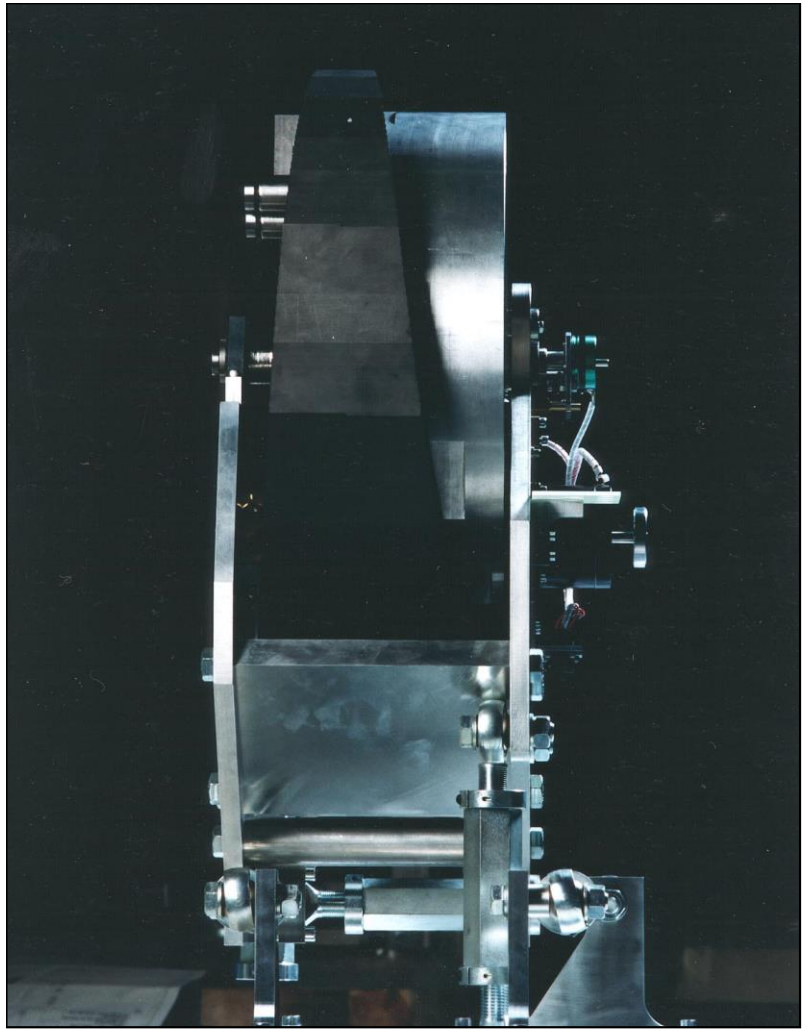
Protons only

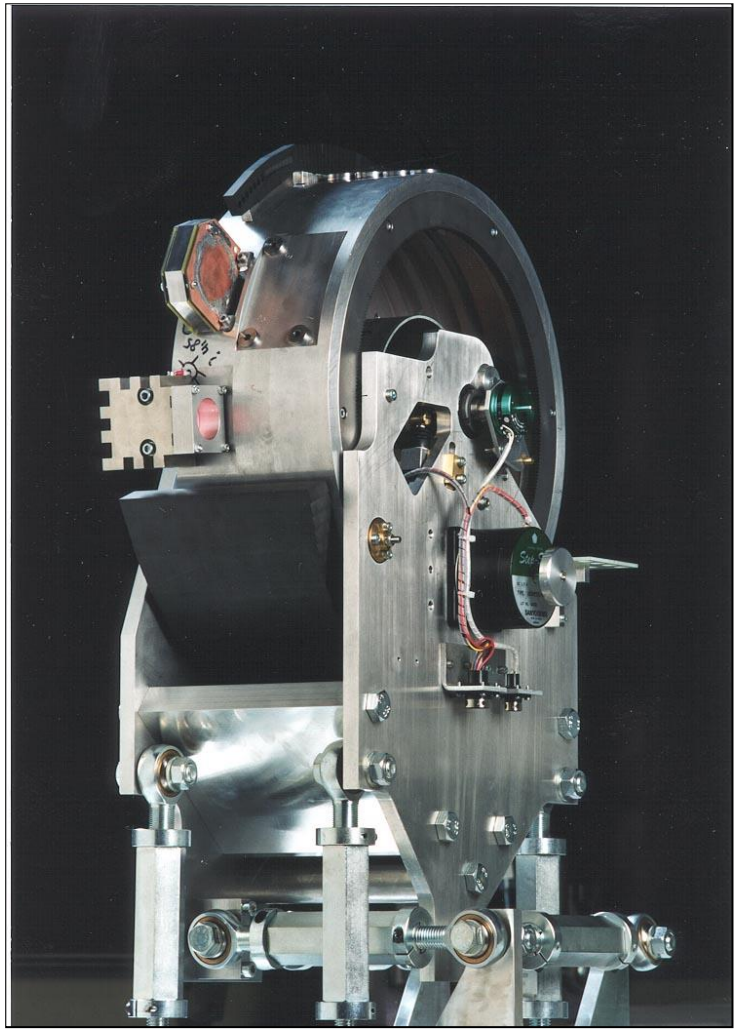
Fixed energy

200 tons

∅ = 4.7 m

The carbon wedge degrader





IBA PT subsystems : the beam transport lines.

The energy selection system. WPE, Essen, 2010.



The Energy Selection System

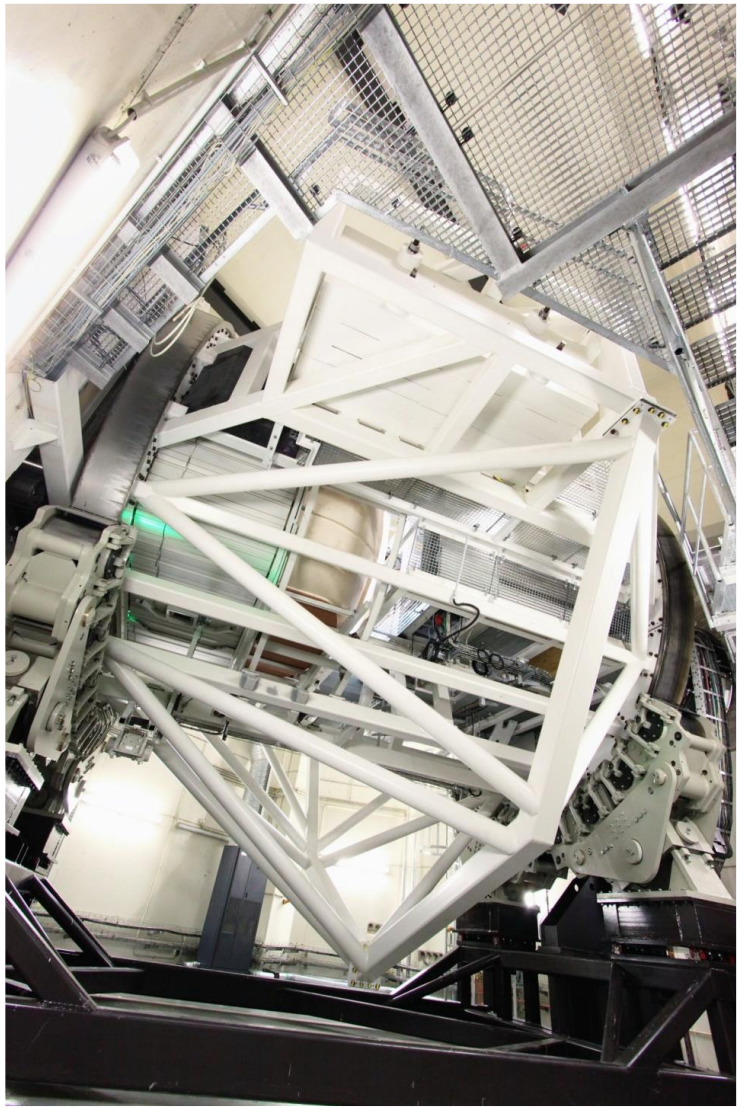
- Graphite or Beryllium wedge is used for coarse energy definition
- Emittance slits are used to define the emittance of the transmitted beam
- Analyzing magnet system defines accurately the range at nozzle entrance
- Laminated magnets and quads allow 10% energy change in 2 seconds

The Beam Transport and Switching System



The isocentric gantry => about 10 m high

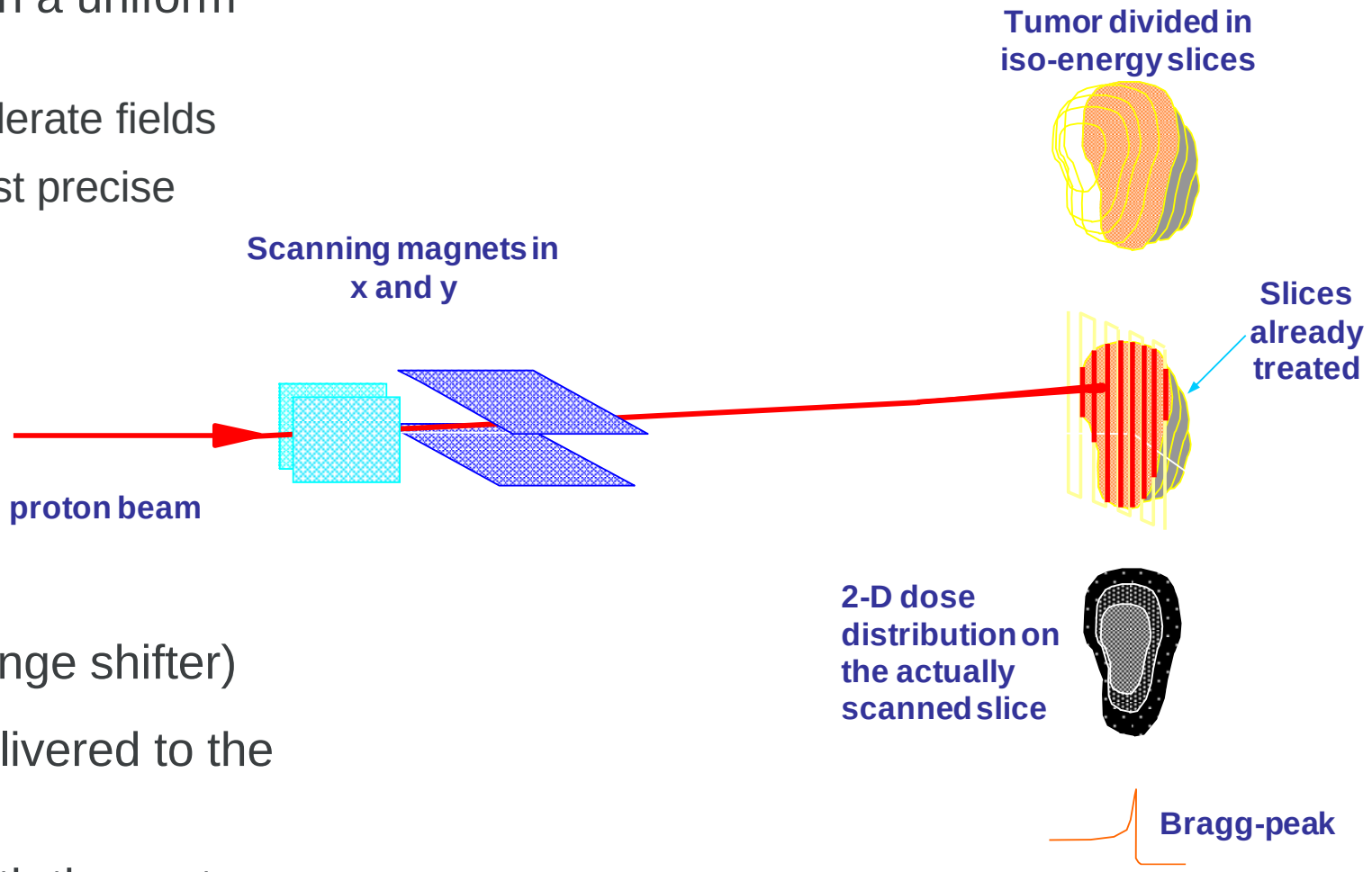




Good alignment is of crucial importance

Purpose of the nozzle

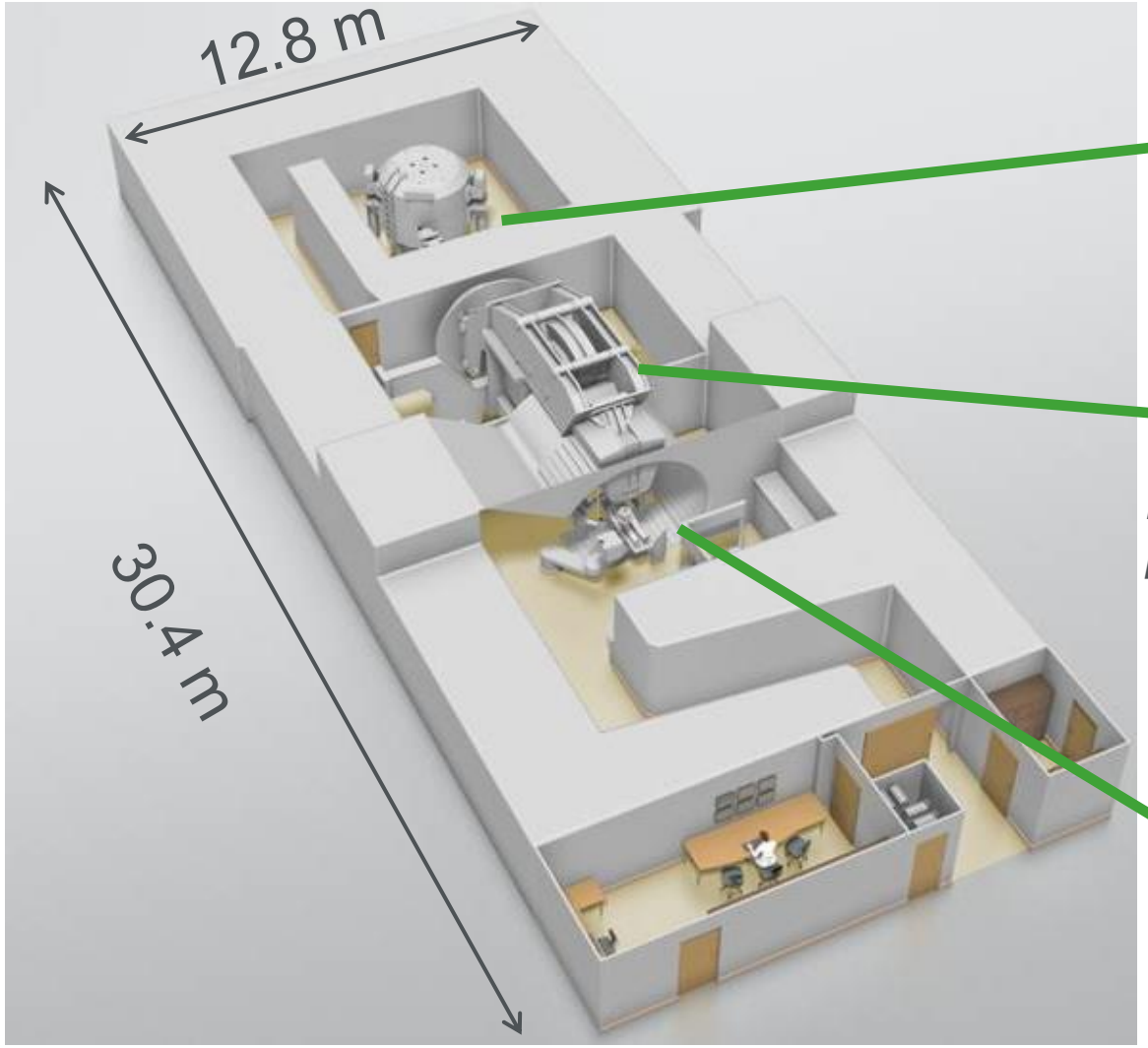
- to spread the proton beam to obtain a uniform dose distribution in a large volume
  - Double scattering for small to moderate fields
  - Pencil Beam Scanning for the most precise conformal mapping, see →

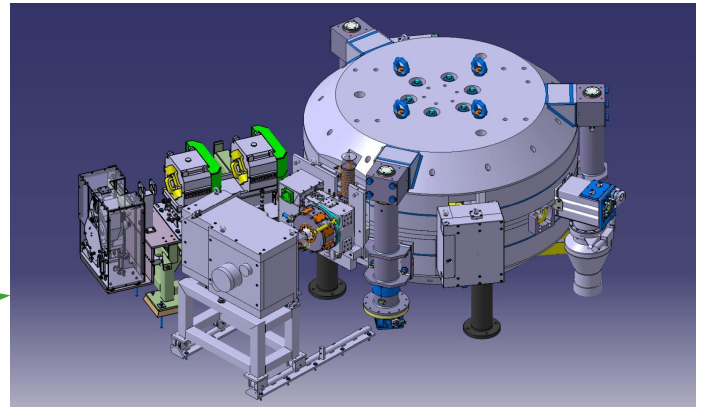


- Modulate the proton energy (via range shifter)
- to measure accurately the dose delivered to the patient
- Provide alignment of the patient with the proton field

IBA Single Room PT Solution: ProteusONE®

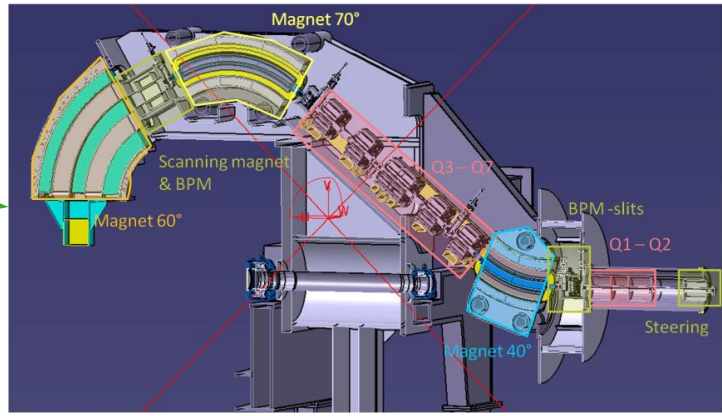
- Compact and affordable





Synchrocyclotron with superconducting coil: S2C2

New Compact Gantry for pencil beam scanning





Patient treatment room

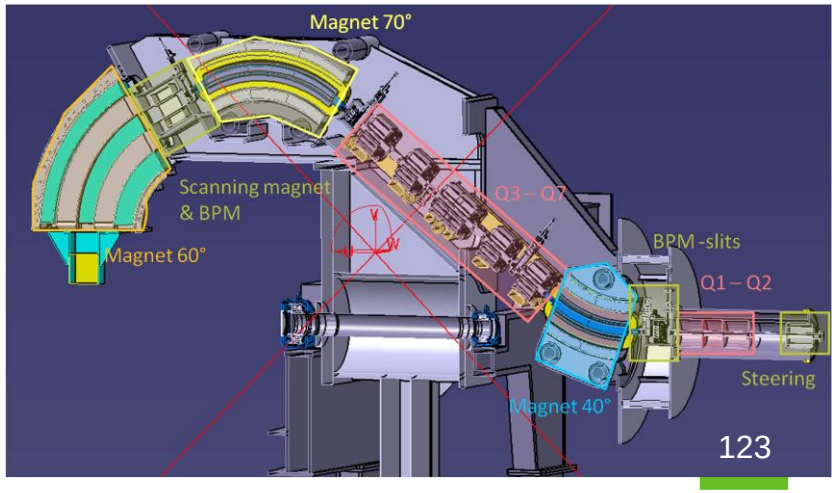
The new compact gantry for pencil beam scanning

Design aimed at reducing footprint and cost

- Scanning magnets are placed upstream of the last bending magnet
- ESS integrated in the 45 deg inclined part
- Rotation angle 220°=> more compact treatment room
- Transport and installation in one part

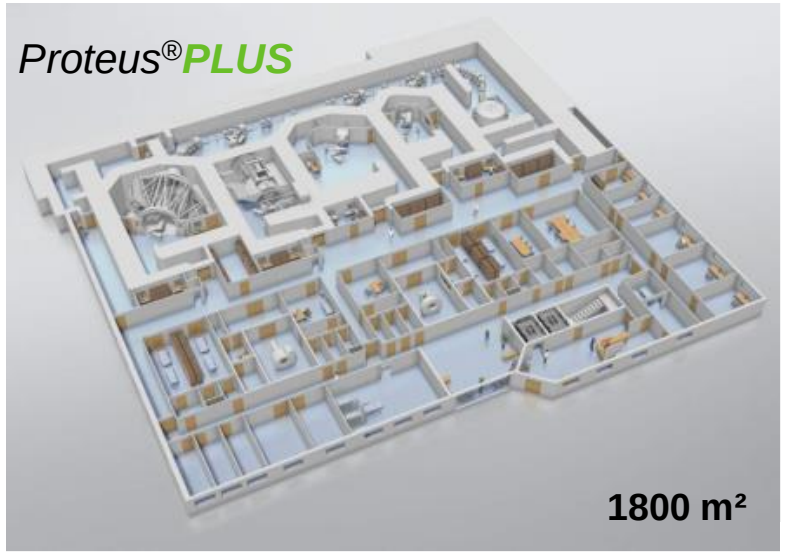
- Prototype installed in 2014

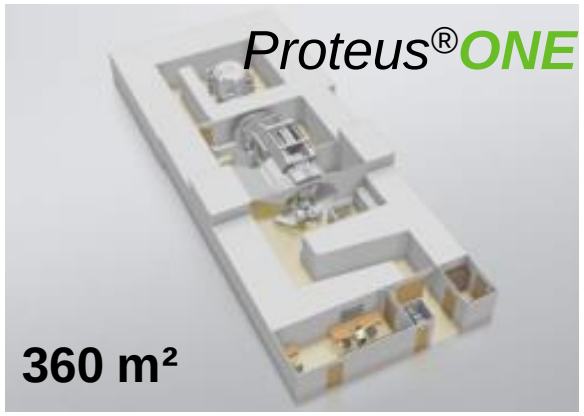




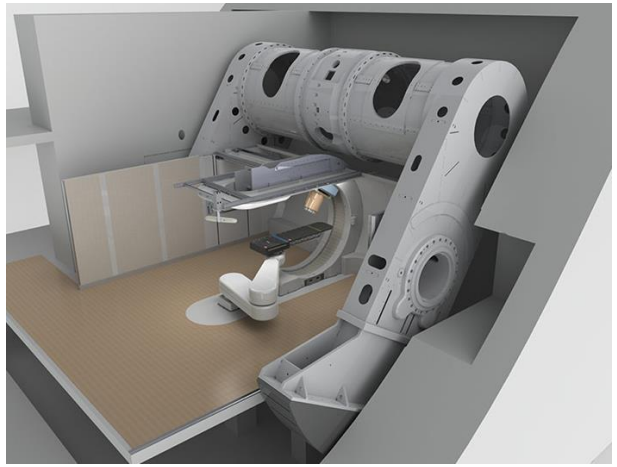
Typical proton therapy systems

IBA

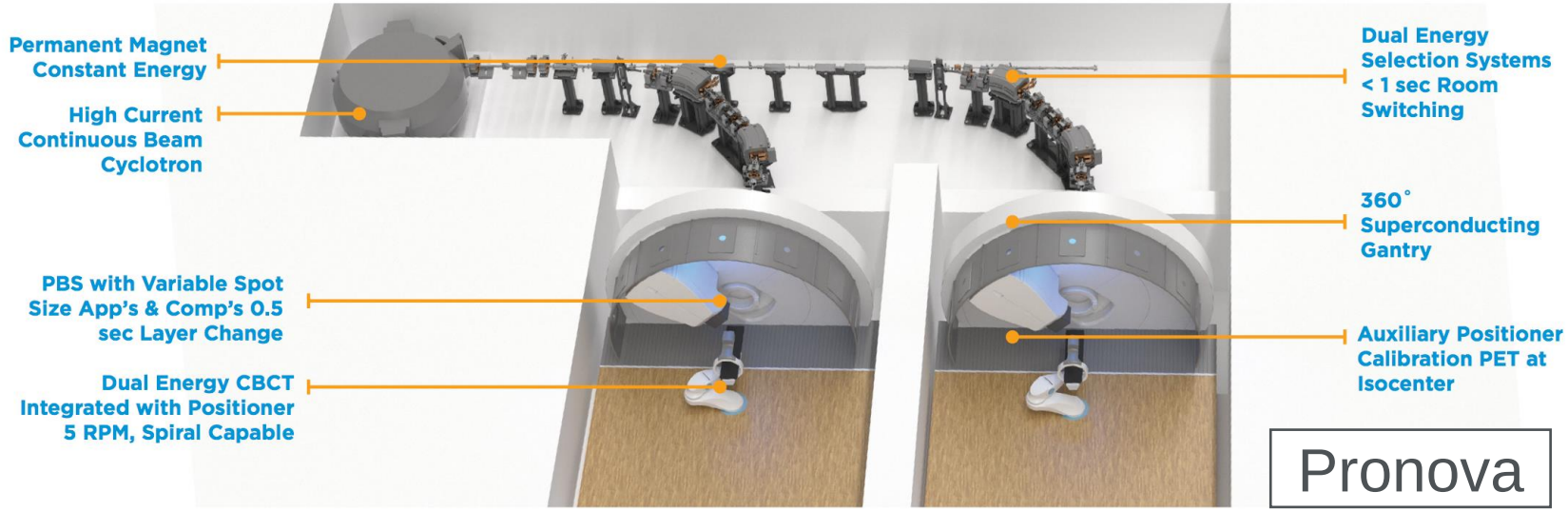




Mevion



Pronova



Cyclotrons for Proton & Carbon therapy?

- In 1991, when IBA entered in PT, consensus was that best accelerator for PT was a synchrotron
- IBA introduced a very effective cyclotron design, and today majority of PT centers use cyclotron technology (IBA, Varian, Mevion, SHI)

- Over the last 25 years, users came to appreciate the advantages of cyclotrons:
  - Simplicity & reliability
  - Intense, continuous (non pulsed) beam current
  - Lowest cost and size
  - Most importantly, ability to modulate rapidly and accurately the proton beam current

Why is fast current modulation important?

- A big issue with scanned beam is the motion of the target during irradiation
- If you cannot control accurately and rapidly the current ,or if the beam is slowly pulsed, your only choice is step-and-shoot (spot scanning)

- Assuming a 10 mm (FWHM) beam spot size, a 50% overlap and a 20 Hz pulse rate, the maximum scanning speed will be 0.2 m/sec
- With this speed, for a large size tumor, repainting many time each layer is not really an option
- In contrast, with a cyclotron you can scan at 20 m/sec and rescan many times each layer

- Note: FLASH therapy goes for extreme high doses!

Change of energy?

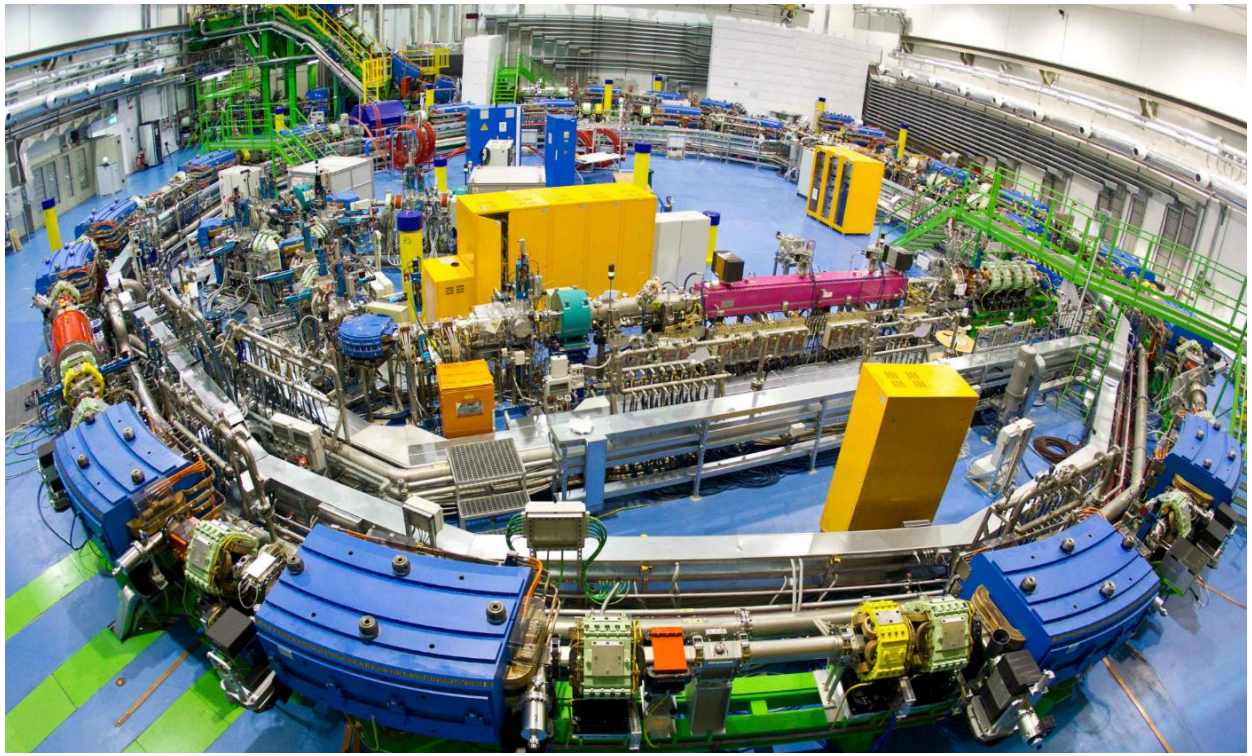
- Cyclotrons are simpler at fixed energy
  - Variable energy from the cyclotron itself is still under research
- Energy change by graphite degrader at waist after cyclotron exit, followed by divergence slits and energy analyzer
  - This very effectively decouples the accelerator from the patient
- Fragmentation products are effectively eliminated in slits and ESS
  - Yes, neutrons are produced, but ESS is well shielded, and the average beam currents are very low > little activation
- How fast? 5 mm step in energy in 100 msec. Respiration cycle is 2…4 seconds => 100 msec is fine

Synchrotrons



Hitachi

- 70-250 MeV protons
- Slow cycle
- 7 m Diameter



"PIMMS" (CERN) design

- Up to Carbon
- 25 m Diameter
- Rep. rate: 5 Hz
- Installed @CNAO, Medaustron

http://www.protominternational.com/about/about-radiance-330/

Rossi, EurPhysJPlus2011-126-78.pdf

Trend: more compact

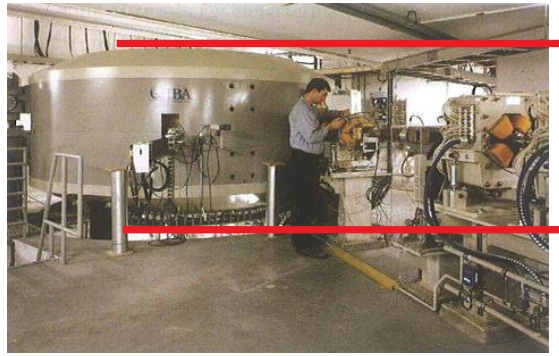


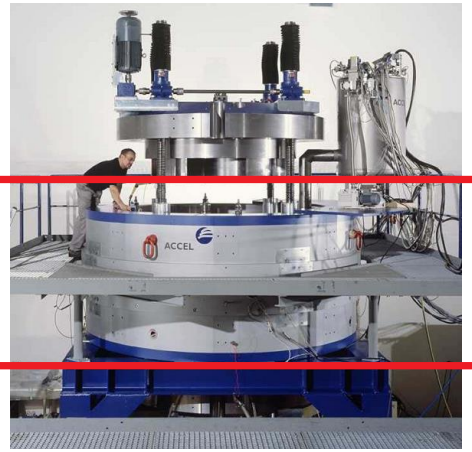
Protom

- Up to 330 MeV protons
- 5 m Diameter, ~16 tons
- Installed @MGH; recently first patient

http://www.protominternational.com/about/about-radiance-330/

Cyclotrons in PT – commercial models









IBA C230
- 230 MeV protons
- 4.3 m Diameter
- CW beam
- Normal conducting
- Magnet: 200 kW
- RF: 60 kW

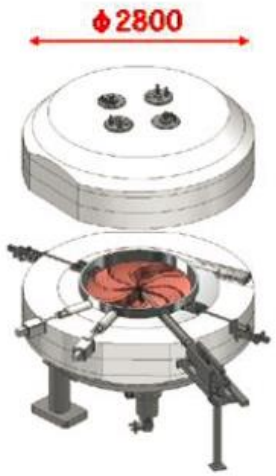
Varian-Accel Probeam
- 250MeV protons
- 3.1 m Diameter
- CW beam
- Superconducting (NbTi)
- Magnet: 40 kW
- RF: 115 kW

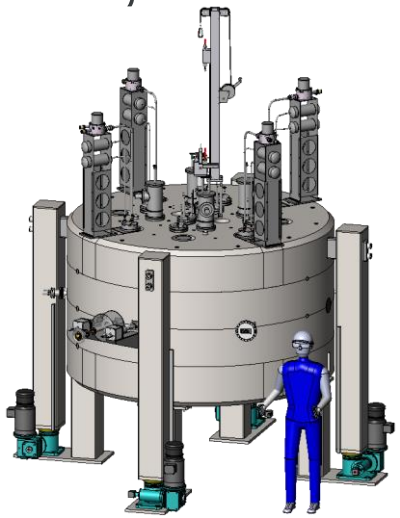
Mevion SC250
- 250 MeV protons
- ~1.5 m Diameter (shield)
- Superconducting (Nb_3Sn)

IBA S2C2
- MeV protons
- 2.2 m Diameter
- Rep. rate: 1 kHz
- Superconducting (NbTi)
- RF: 11 kW

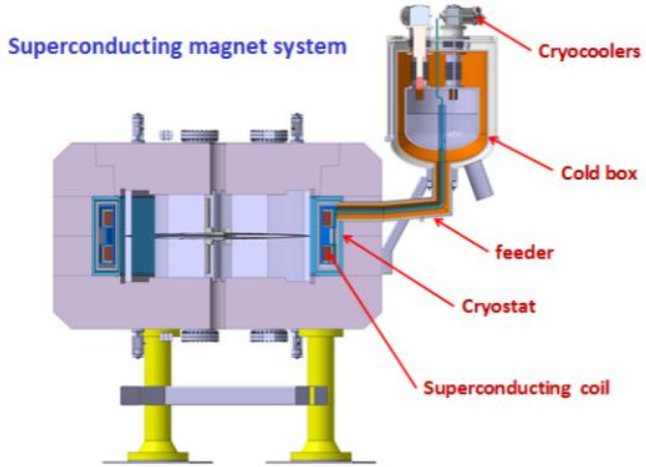
Proton cyclotrons - Ongoing developments

- 1. Isochronous: SHI, Varian/Antaya, Pronova/Ionetix, Heifei/JINR

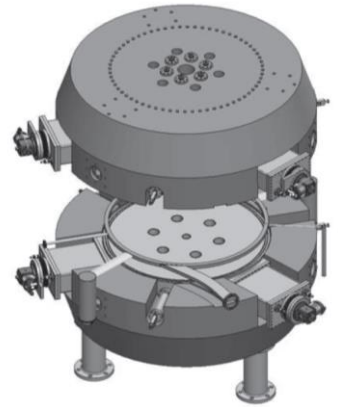




Derenchuck - NAPAC 2016



Karamysheva - THP20 cyclotrons 2016



Antaya – CAS 2015

SHI
- 230 MeV protons
- 2.8 m Diameter
- CW beam
- Superconducting (NbTi)
- 55 tons
- 4 T (extr.)

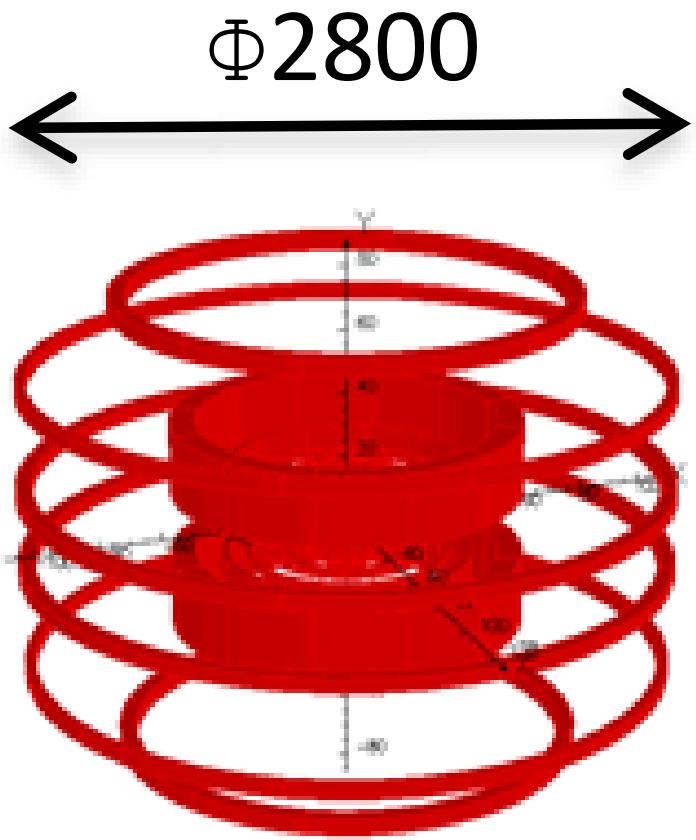
Pronova/Ionetix
- 250 MeV protons
- 2.8 m Diameter
- CW beam
- Superconducting (Nb_3Sn)
- 60 tons
- 3.7 T (extr)

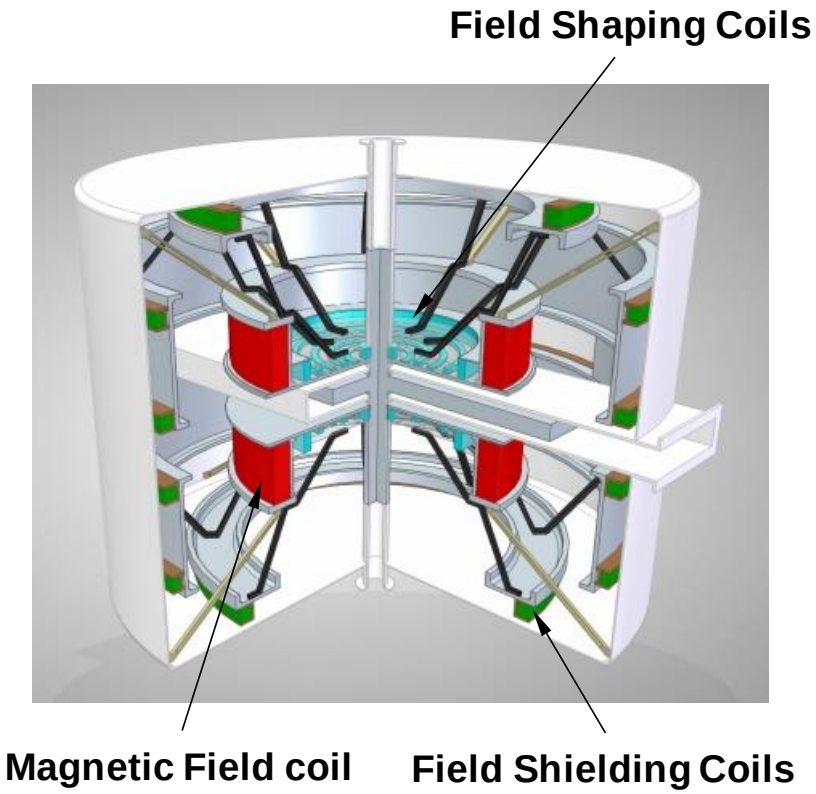
Heifei/JINR
- 200 MeV protons
- 2.2 m Diameter
- CW beam
- Superconducting
- 30 tons
- 3.6 T (extr.)

Varian/Antaya
- 230 MeV protons
- 2.2 m Diameter
- CW beam
- Superconducting (Nb_3Sn)
- 30 tons+
- 5.5 T (extr.)
- “Flutter” coils

Proton cyclotrons - Ongoing developments

- 2. Synchrocyclotrons: MIT ironless





- 250 MeV protons
- (2.4-)2.8 m Diameter
- Pulsed beam
- Superconducting (Nb_3Sn)
- 4 tons
- B_extr = 5T
- Variable-energy possible

QUIZ #6

Next biggest breakthroughs we are looking for in cyclotron application for PT ?

1. Beam current increase by 10
2. Decrease of beam emittance
3. Variable energy
4. Increase of magnetic field to reduce footprint



Part III - Industrial applications

Brief overview / Some examples

More extensive course by prof. W. Mondelaers tomorrow

Sterilization Processing Comparison

Same technologies from a radiation point of view

Ethylene Oxide





E-beam





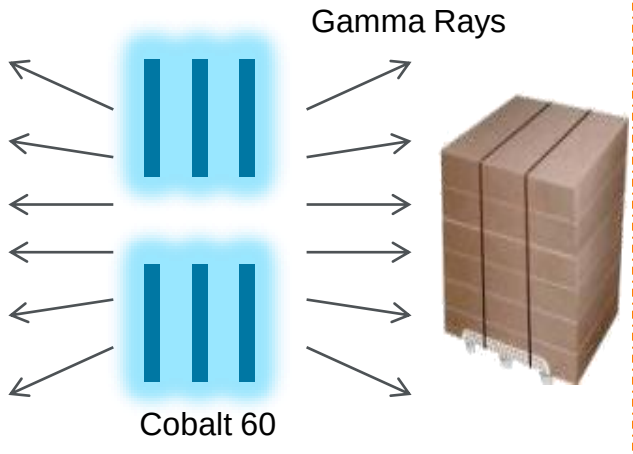
X-ray

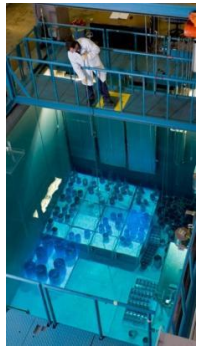




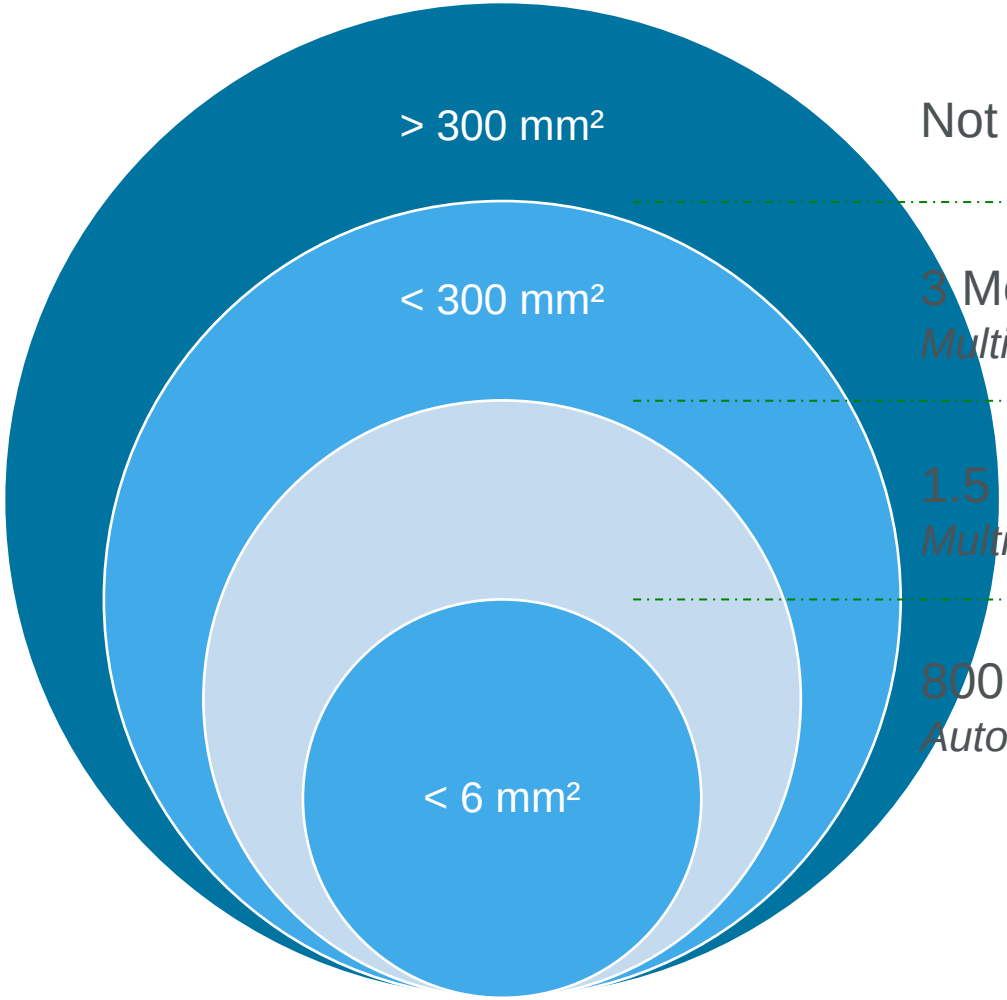
Gamma







Cables treatment with e^- : Dynamitron => crosslinking



Not crosslinked with radiation

3 MeV
Multiple applications



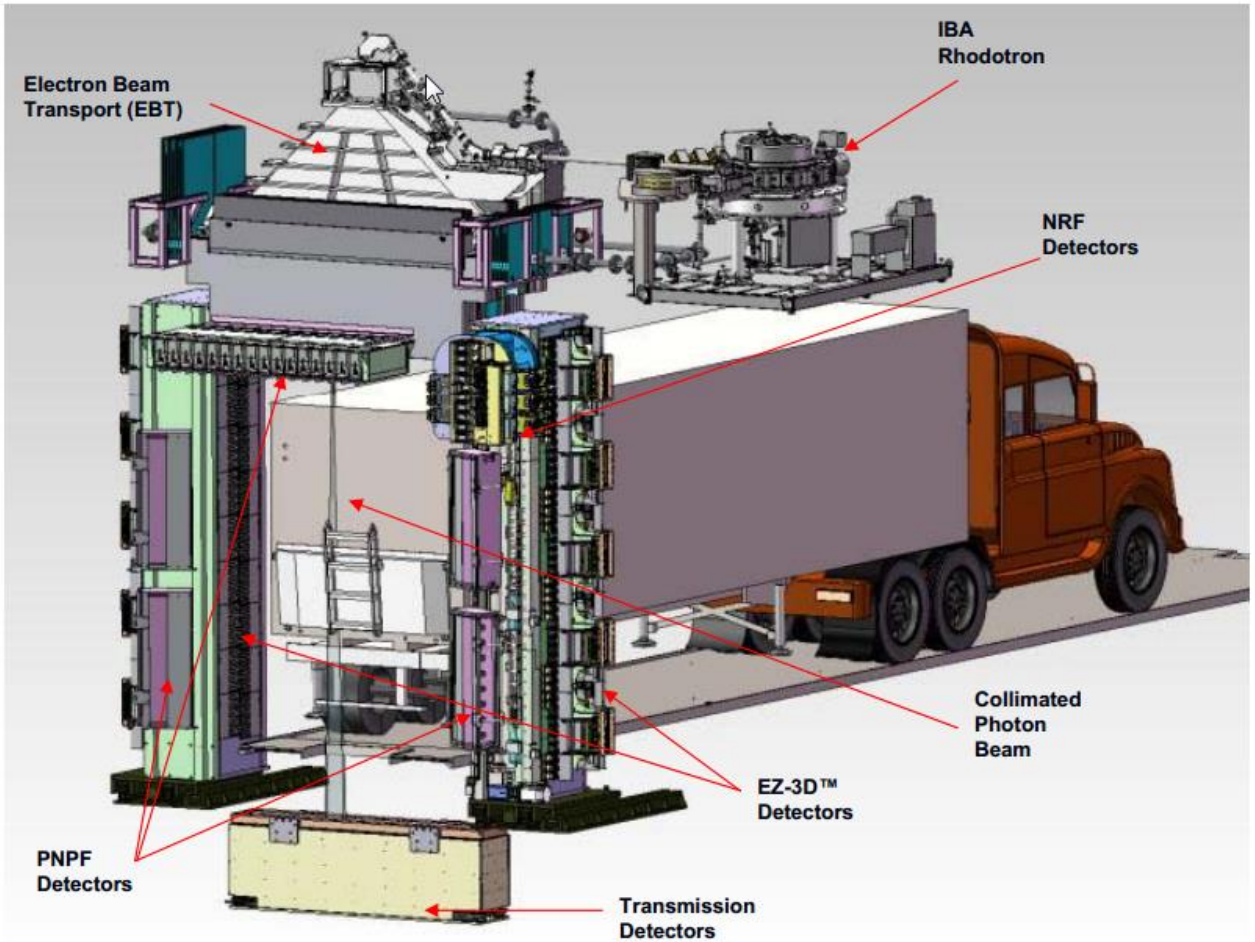
1.5 MeV
Multiple applications



800 keV
Automotive



Cargo screening => TT50



This full configuration identifies anomalies and resolves potential threats

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Report Constitutes Sales Technology Exempt from U.S. Export Control Pursuant to 15 CFR 740.13(b).

Passport Systems Inc.

IBA Industrial's Product Portfolio

Dynamitron

0.5 -> 5 MeV | 160 mA

Electron beam



Main application

E-beam Crosslinking

Rhodotron

3 -> 10 MeV | 42 mA | 420 kW

Electron beam and X-rays



Main application

E-beam box sterilization

eXelis

5 – 7 MeV | 80 mA | 560kW

X-rays

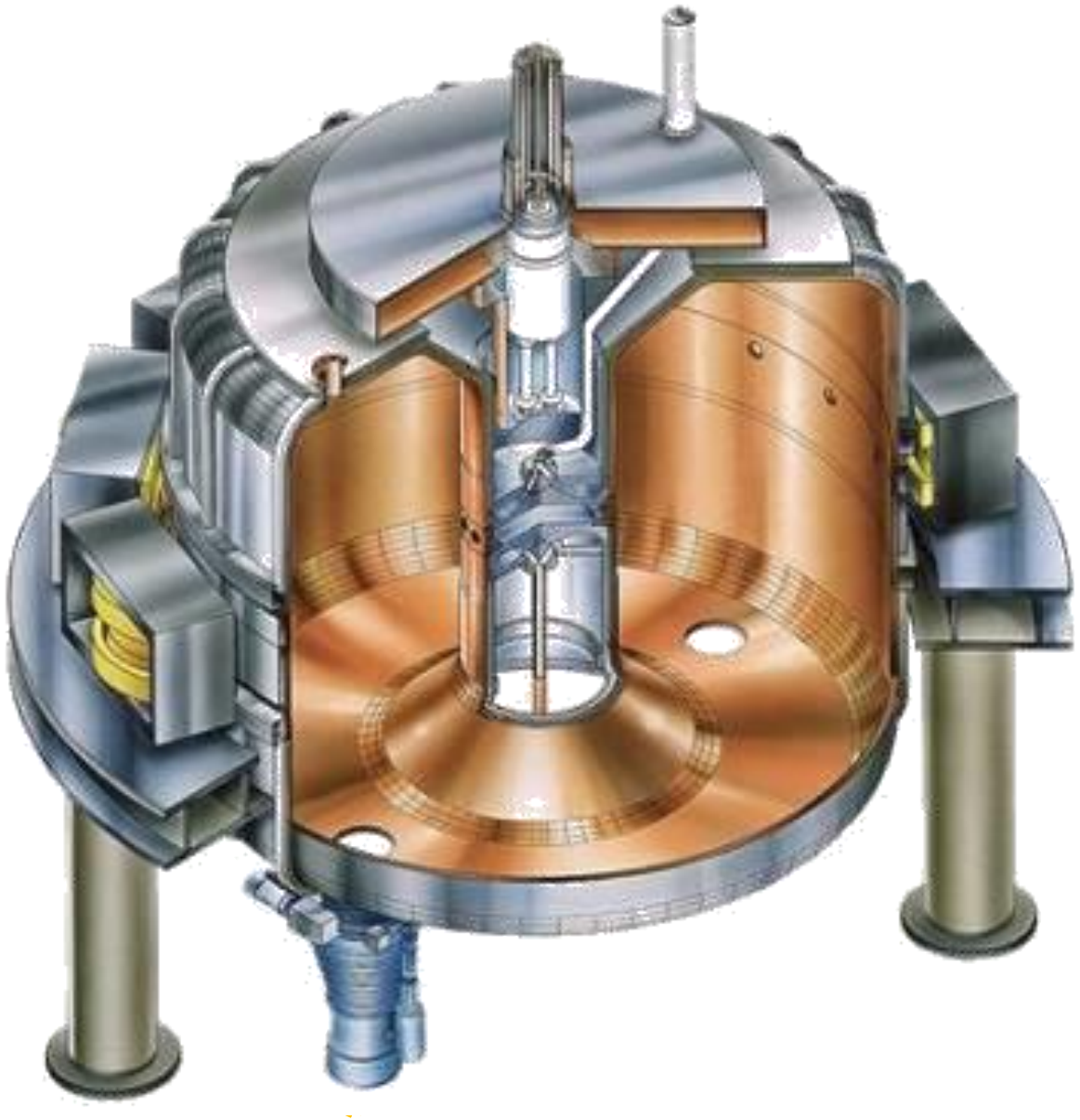


Main application

X-ray pallet sterilization

Linac's reach about 40-60 kW

Brief explanation of the Rhodotron:

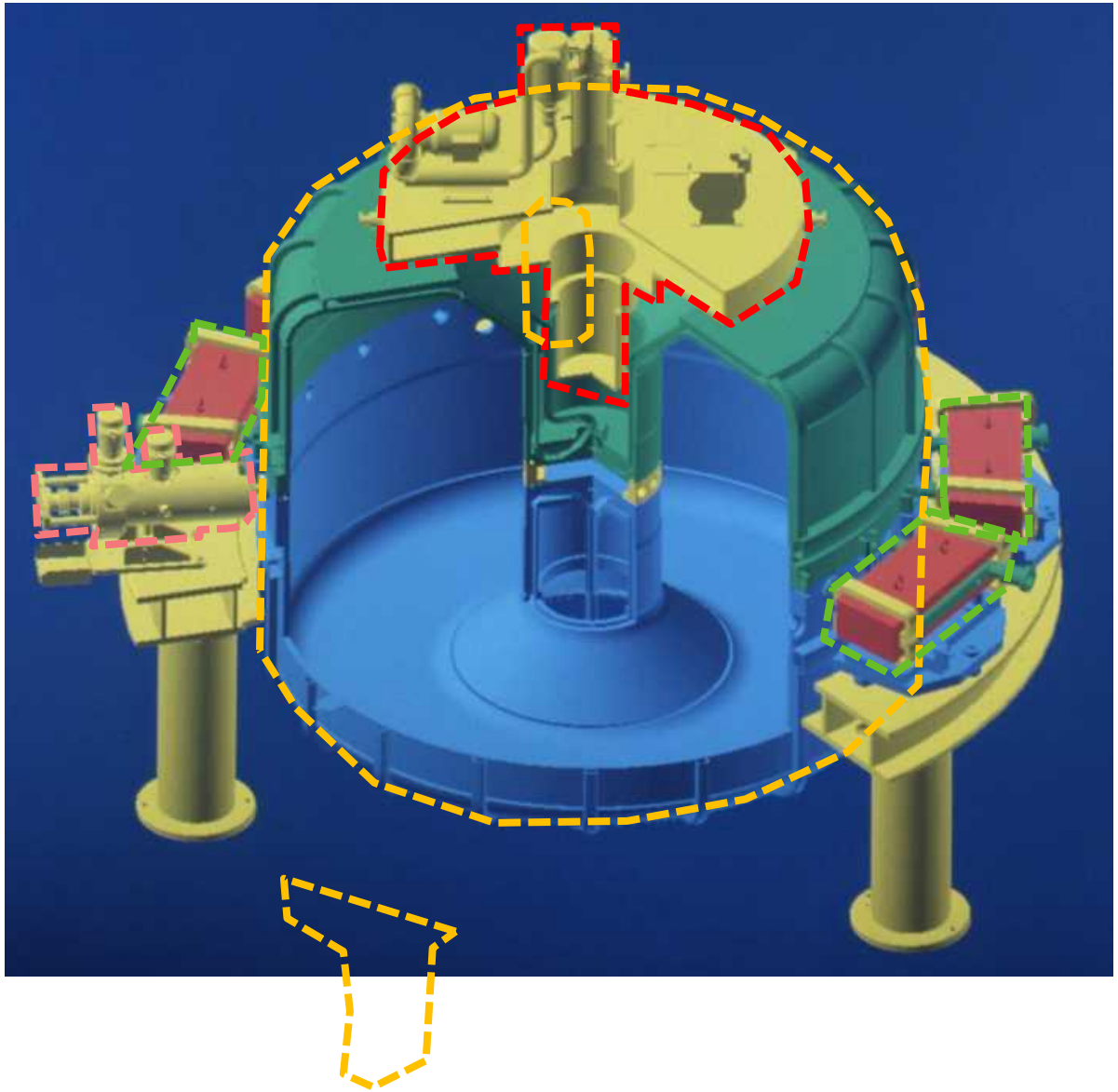


Main components:

- **RF Cavity**
- **E-Gun**
- **Magnets**
- **Final Power Amplifier**

- **RF tube (Tetrode)**
- **Vacuum system**

Brief explanation of the Rhodotron:



Main components:

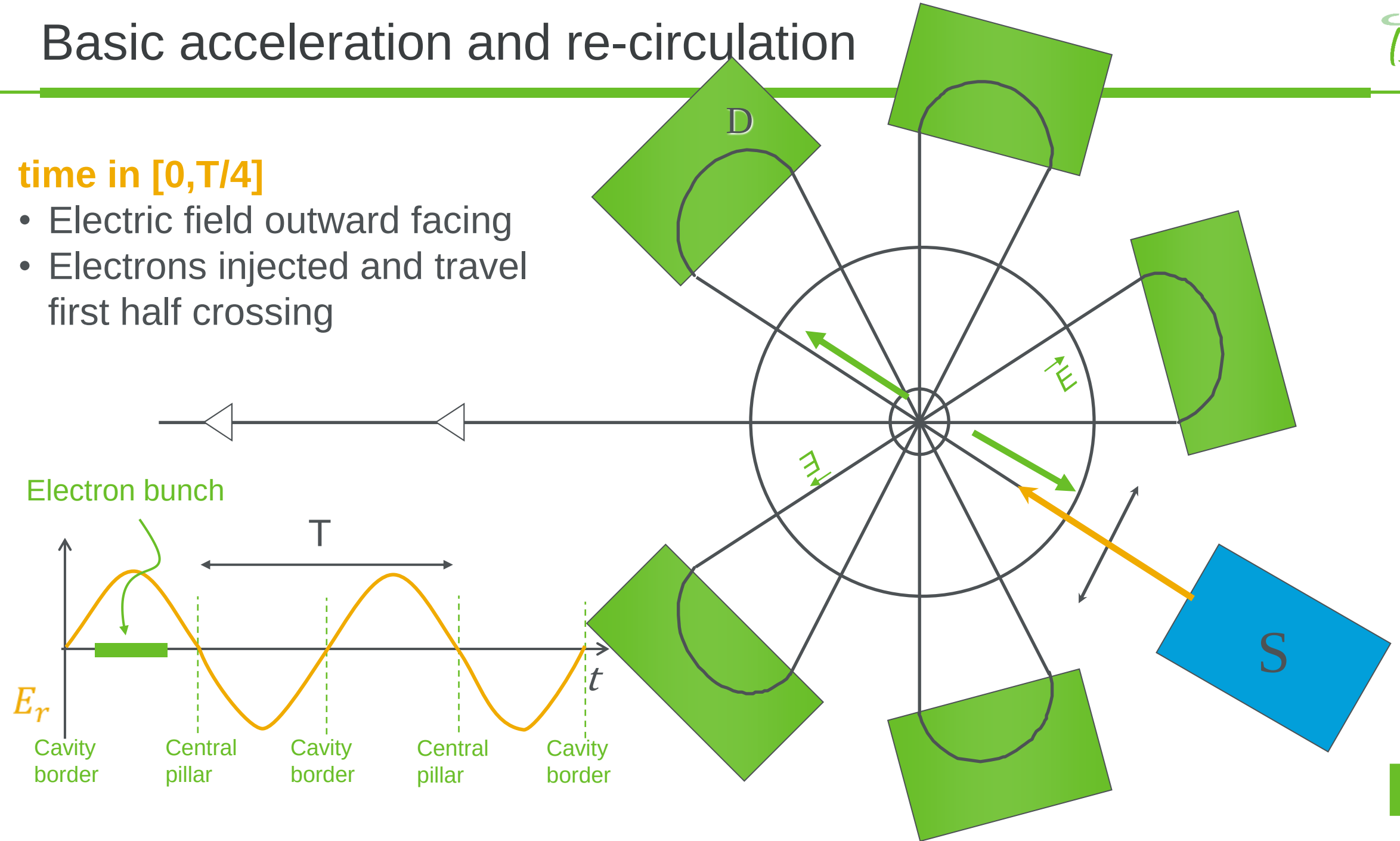
- **RF Cavity**
- **E-Gun**
- **Magnets**
- **Final Power Amplifier**

- **RF tube (Tetrode)**
- **Vacuum system**

Basic acceleration and re-circulation

time in [0,T/4]

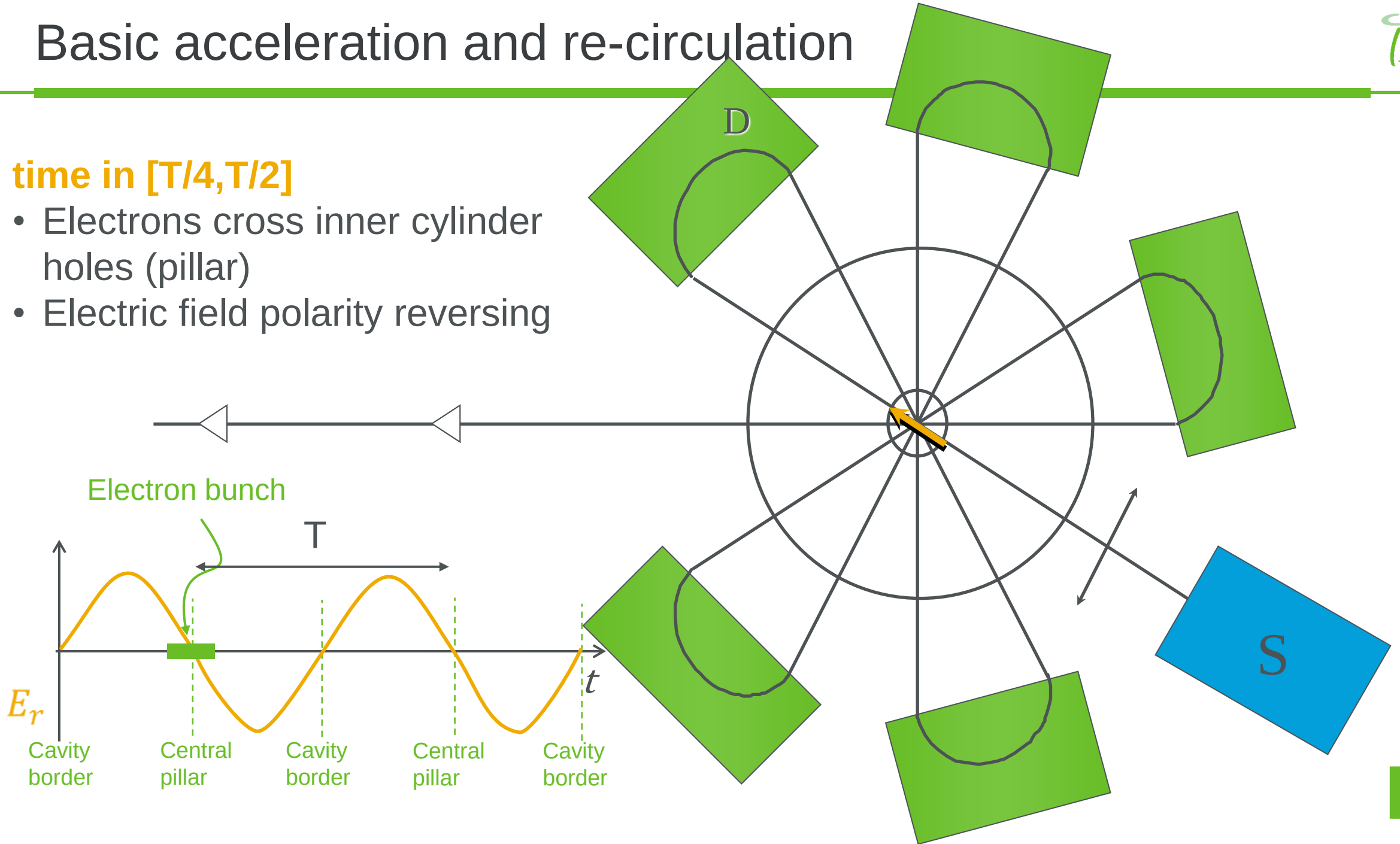
- Electric field outward facing
- Electrons injected and travel first half crossing



Basic acceleration and re-circulation

time in [T/4,T/2]

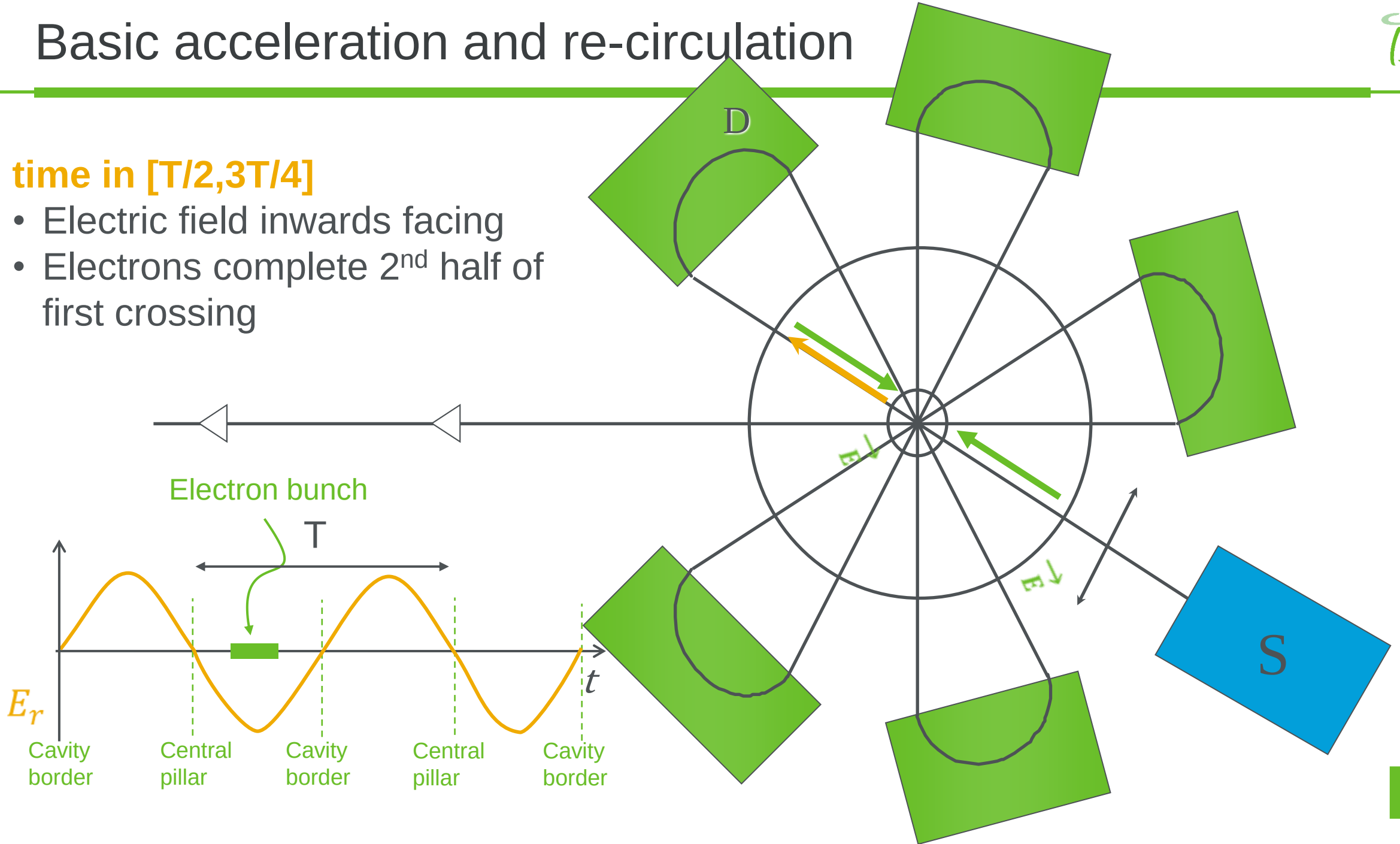
- Electrons cross inner cylinder holes (pillar)
- Electric field polarity reversing



Basic acceleration and re-circulation

time in [T/2,3T/4]

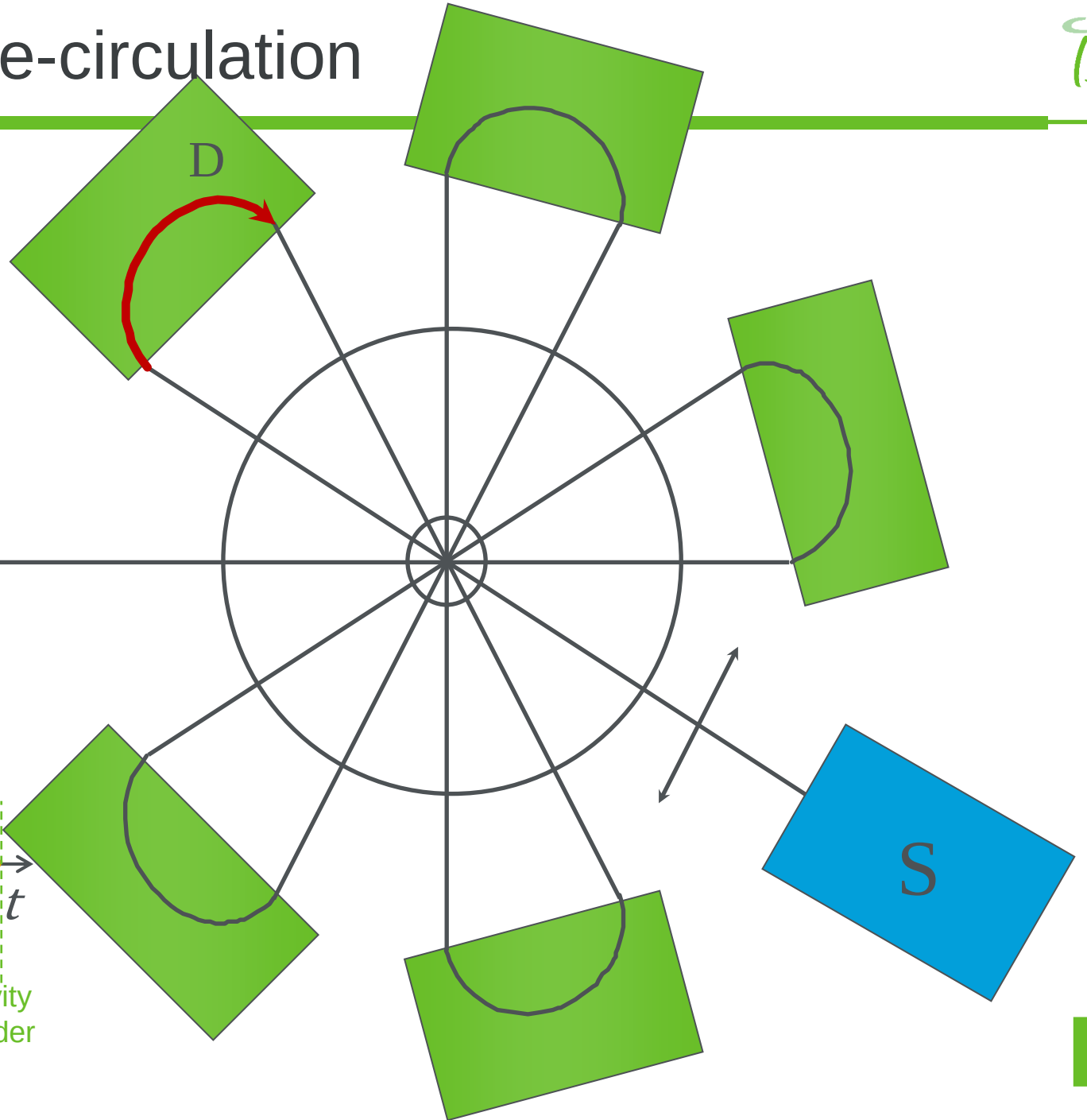
- Electric field inwards facing
- Electrons complete 2nd half of first crossing

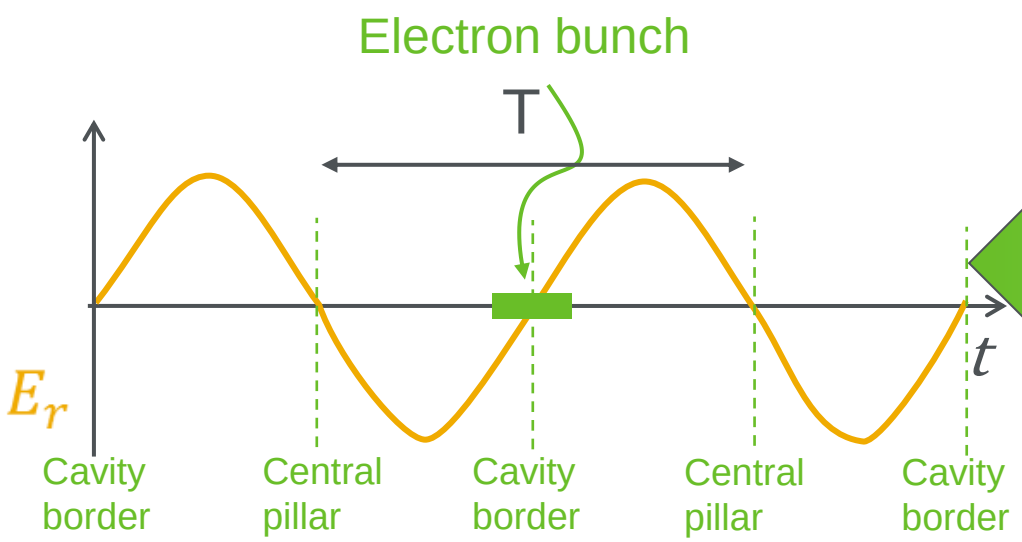


Basic acceleration and re-circulation

time in [3T/4,T]

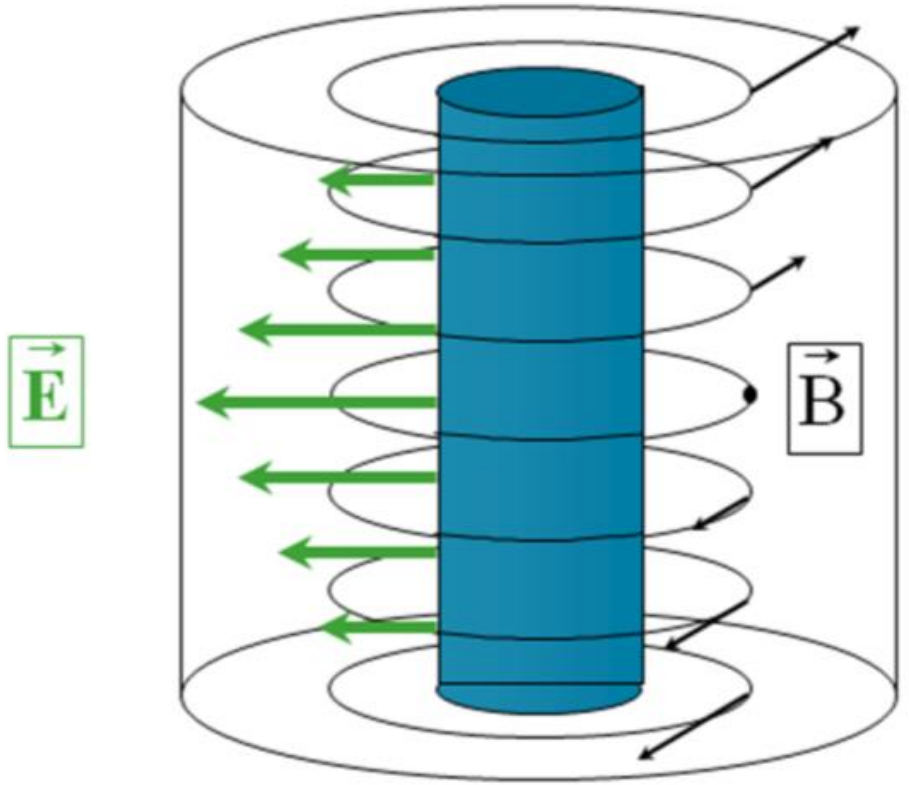
- Electrons deflected back into the cavity
- Electric field polarity reversing





Rhodotron Cavity Design

- RF sinusoidal electrical field ➔ coaxial cavity !
- Frequency is 107 MHz (215 MHz for TT100). Depends on tube availability ➔ best is FM band
  - The size of the cavity is fixed by f:
    - Height = 0.5 λ
    - Radius ideal is 0.35 λ to allow transit in magnets
  - Fundamental mode (TEM 1):
    - Radial E-field and azimuthal B-field
    - E-field varies as cos (z) / r
  - Electrical losses increase with $f^{1/2}$
  - Cost increase with size, small is complicated for beam optics: phase acceptance & transmission
  - ➢ **Maximize energy gain vs losses & cost !**



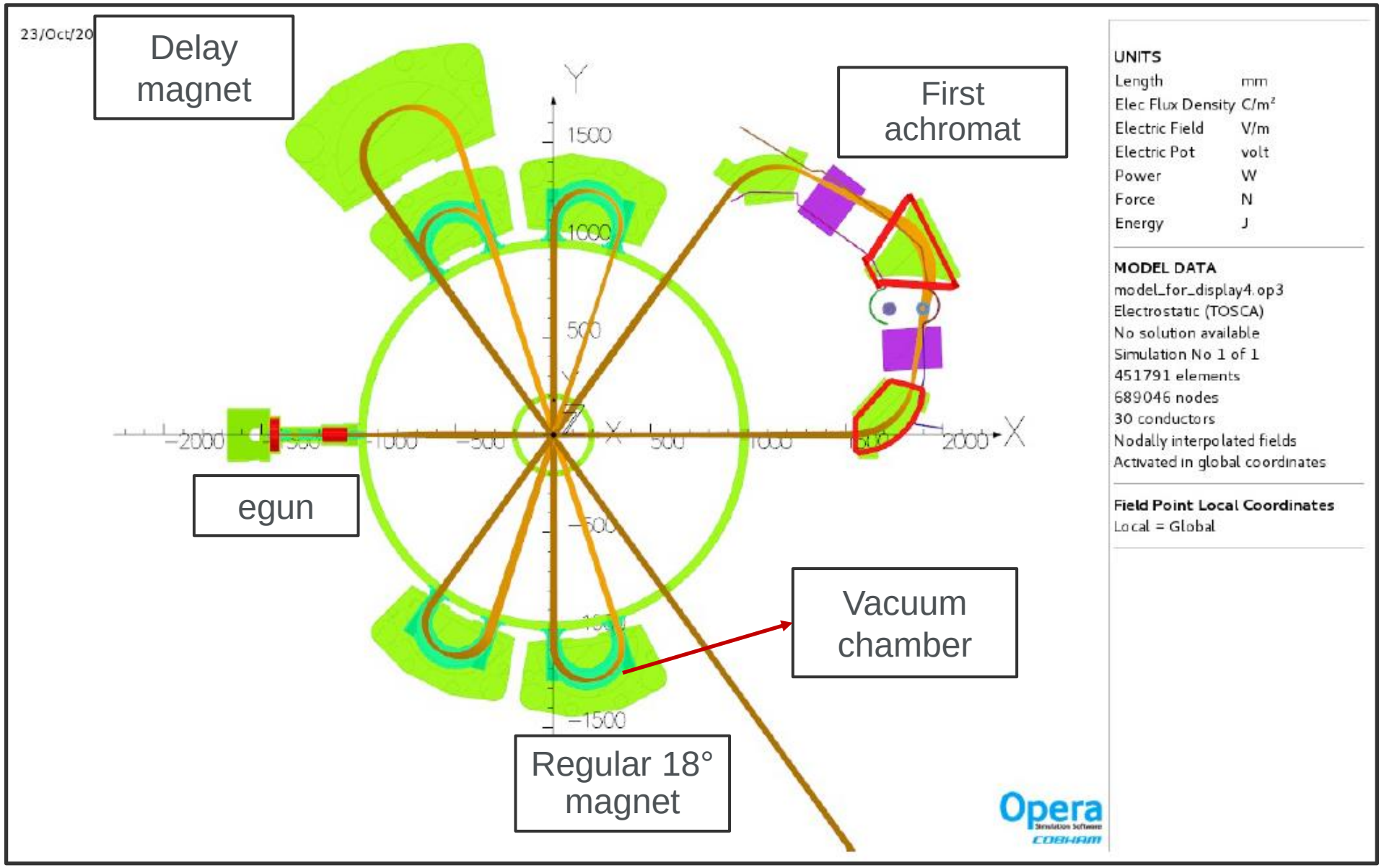
Electric ($\vec{E}$) and magnetic ($\vec{B}$) fields in Rhodotron coaxial cavity

QUIZ #7

What is the purpose of the Delay Magnet:

1. Bunching
2. Boosting
3. Decelerating

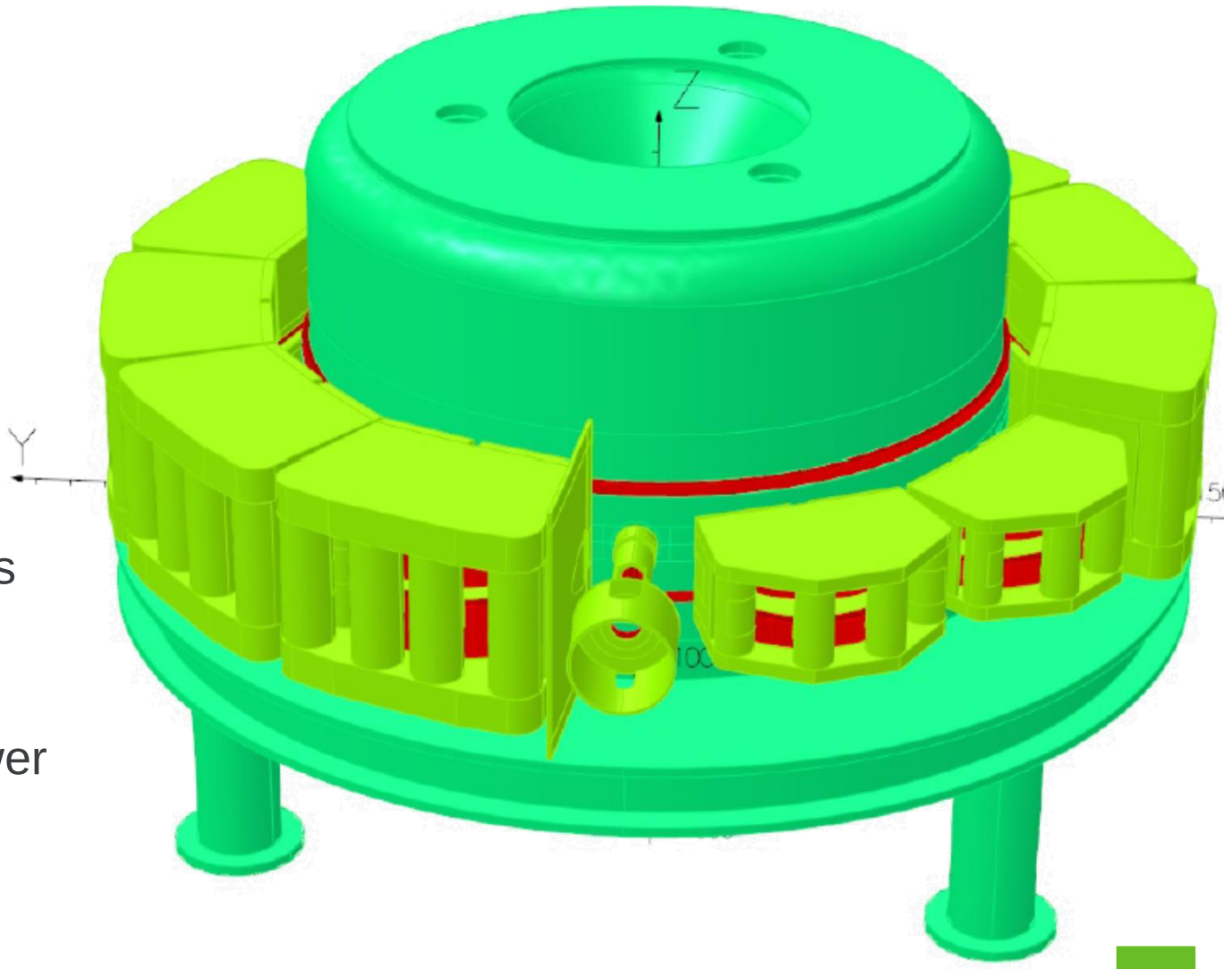
Specific layout or TT1000



TT300-40MeV for Tc-99 production

- 12 passes with 3.3 MeV/pass
- RF pulsed to obtain higher cavity voltage with reasonable cooling

- Full magnetic model includes
  - All dipoles
  - Two egun solenoids with return yokes
  - Global tuning solenoid
  - Cavity structure in iron
  - Magnets support ring and feet (in lower quality) iron
  - Egun shielding plate



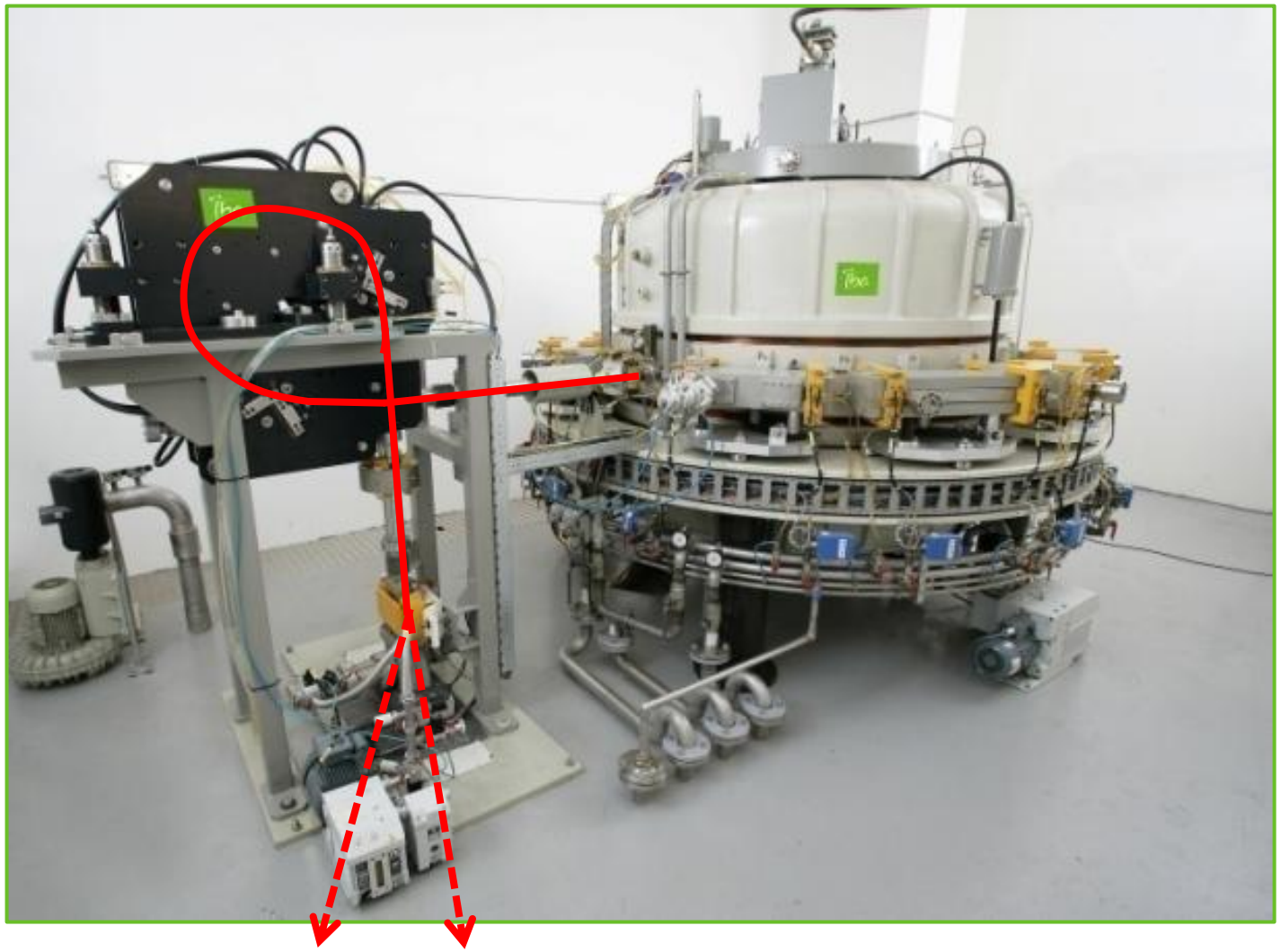
Rhodotron Beam transport line, Scanning, Horn



Goal of the BTL and horn ?

- Transport beam from Rhodo vault to target without any losses
- Scan beam across material to be treated according to conveyor speed
- Convert electrons to X-ray when needed

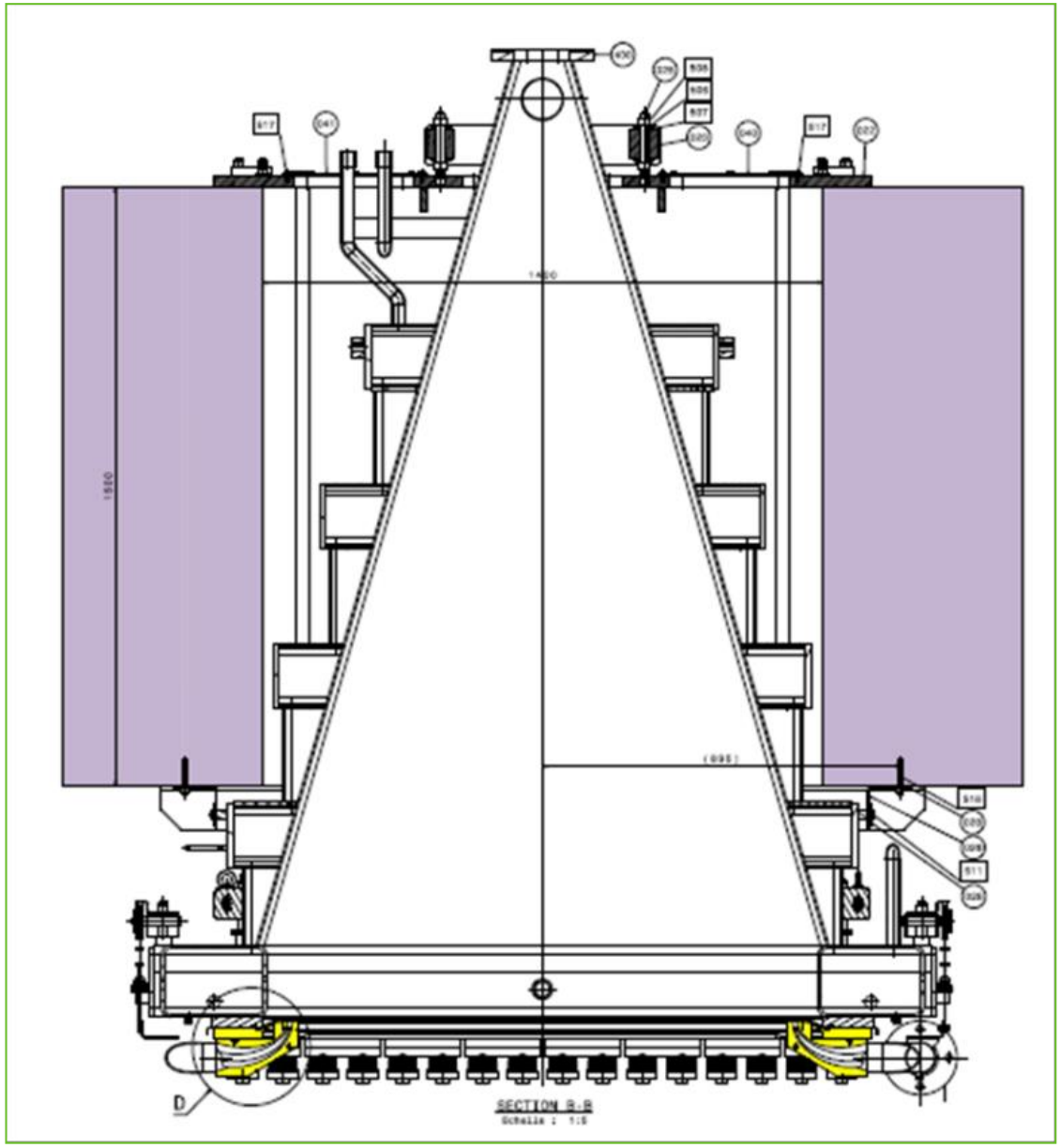
Rhodotron Beam transport line, Scanning, Horn



Goal of the BTL and horn ?

- Transport beam from Rhodo vault to target without any losses
- Scan beam across material to be treated according to conveyor speed
- Convert electrons to X-ray when needed

Rhodotron Beam transport line, Scanning, Horn



Goal of the BTL and horn ?

- Transport beam from Rhodo vault to target without any losses
- Scan beam across material to be treated according to conveyor speed
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Rhodotron Beam transport line, Scanning, Horn



Goal of the BTL and horn ?

- Transport beam from Rhodo vault to target without any losses
- Scan beam across material to be treated according to conveyor speed
- Convert electrons to X-ray when needed

Addendum
Part A: radio-isotopes for medical diagnosis



SPECT: Single Photon Emission Computed Tomography

PET: Positron Emission Tomography

How is imaging done with radio-tracers ?

SPECT

PET

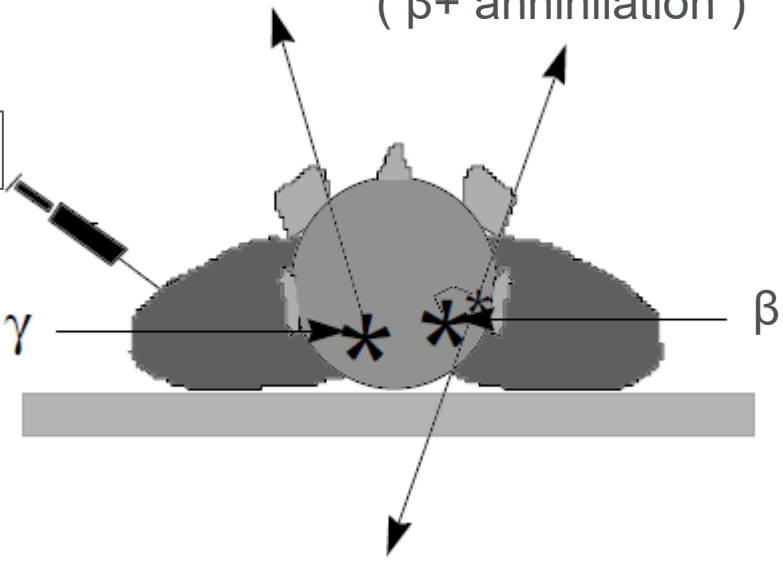
Single photon

2 photons of 511keV
(β+ annihilation)

Radio-isotope

γ emitter γ

β+ emitter



A camera (Gamma or PET) detects the photons emitted from the body and computes 3D distributions of the radio-activity

Full production & application cycle of PET radio-isotopes

SYNTHESIS



CYCLOTRON





PET-SCAN

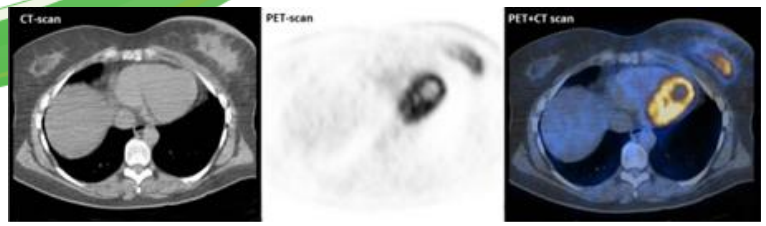




TREATMENT

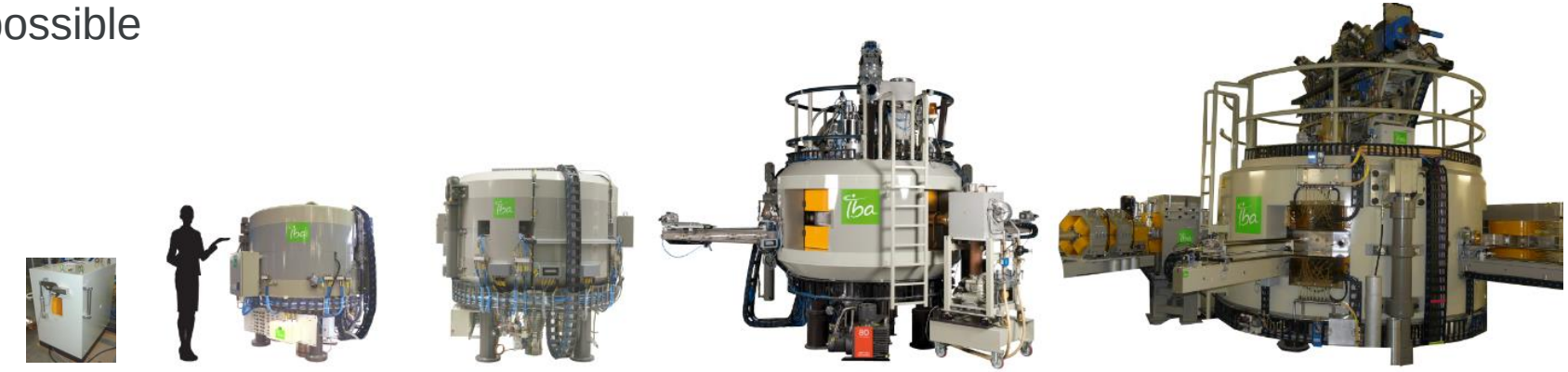
DIAGNOSIS

IMAGING



A family of cyclotrons for isotope producion

- Cyclone 3D — 3 MeV D+ (production of Oxygen-15)
- Cyclone 11: — 11 MeV H-, 120 µA (~1300 W)
- Cyclone 18/9: — 18 MeV H-, 150 µA (~2700 W)
- Cyclone Kiube — 18 MeV H-, 300 µA (~5400 W)
- Cyclone 30(xp): — 30 MeV, 1.2 mA (36 kW) — alpha/deuteron possible (xp)
- Cyclone 70(xp): — 70 MeV, 750 µA (53 kW) — alpha/deuteron possible

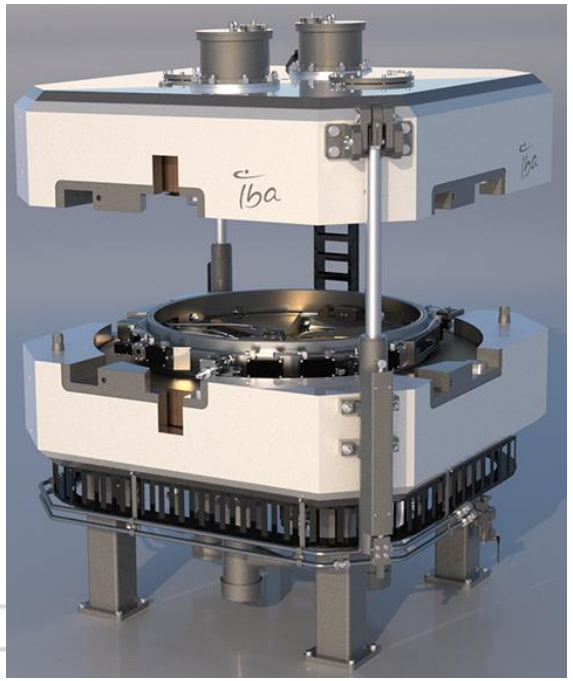


Cyclone Kiube: main features

- More compact, more production capacity

Cyclone®	18/9	18P
Shape	Circle	Square
Height (m)	2.2	1.85
Dimensions (m) (overall)	Ø 2.3	1.95 x 1.95
Median plan height (m)	1.25	1.12
Minimum vault dimensions (WxLxH)	4 x 4 x 3	3.7 x 3.7 x 2.5
Accelerated particles	Proton/ deuteron	Proton ONLY
Rigging hole	2.5 x 2.5	2.2 x 2.2
Production capacity (Ci)	16	Up to 30
Weight (tons)	25	17



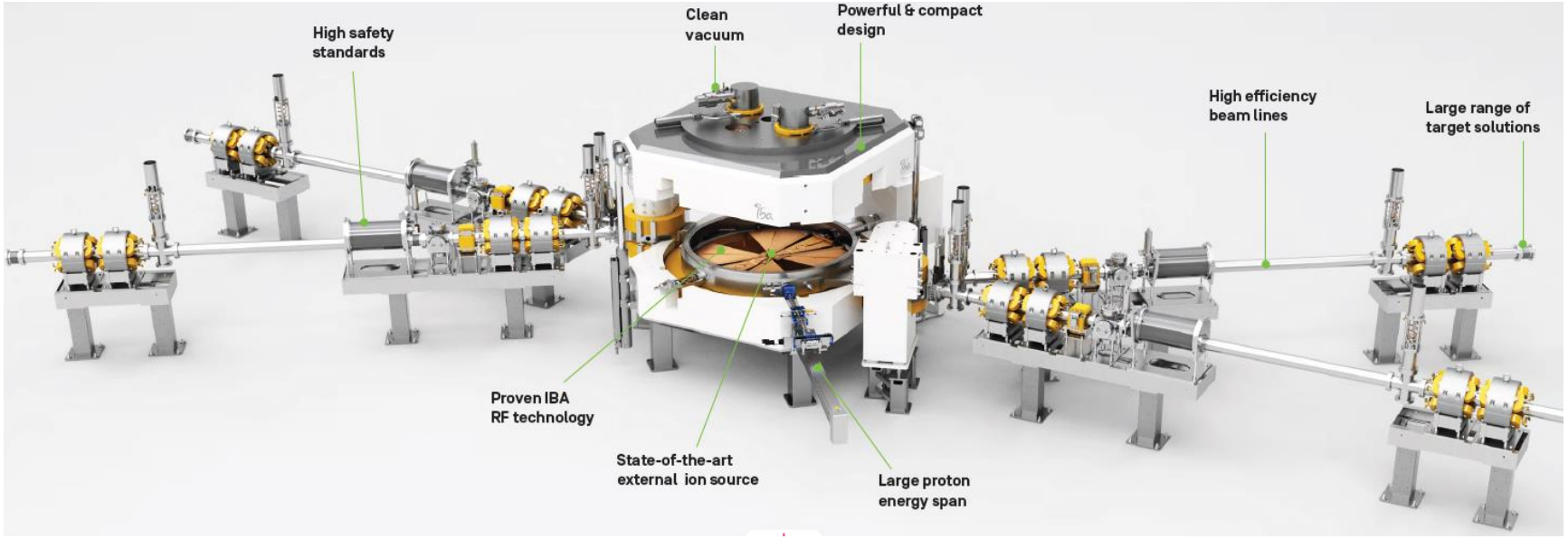


C30

Old vs new:

- 15 to 30 Mev, 12 to 30MeV
- 55 ton vs 30 ton
- R*h = 2.8m * 2m, vs 2.2*2.2*1.46m
- Current is equal :
  1.5mA extraction, 1.2mA target





The use of Radio Isotopes for medical imaging

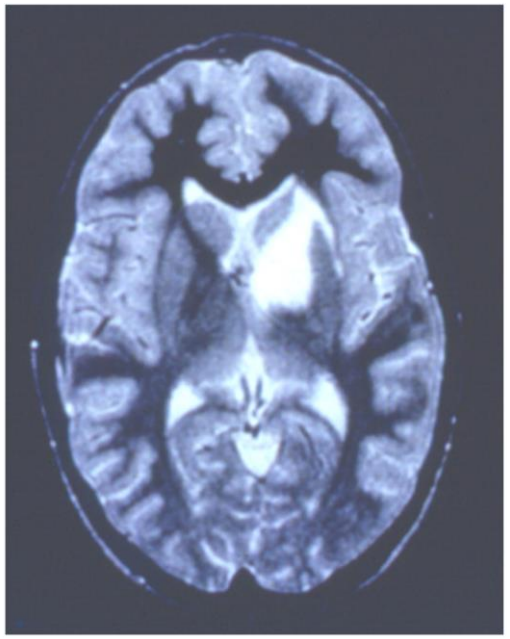
- Radio tracers can be used to label a specific bio-chemical molecule.

- They allow to see metabolism
  - X-ray (CT-) scan or MRI are better to see the anatomy (structure)

- Nuclear medicine (imaging of metabolism using molecules labeled with an appropriate radioisotope) is therefore not in competition, but in complement of imaging techniques such as X-ray, X-ray CT-scan or MRI.

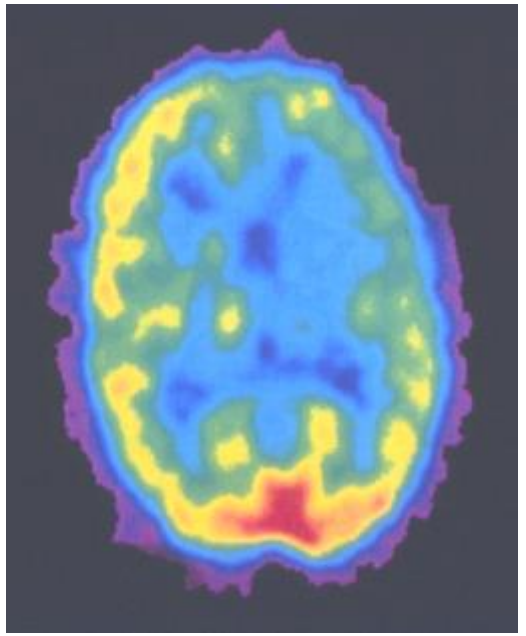
Metabolic versus anatomic imaging

MRI



Anatomic View
(Tissue-structure)

PET



Metabolic imaging
(Biological-function)

How is imaging done with radio-tracers ?

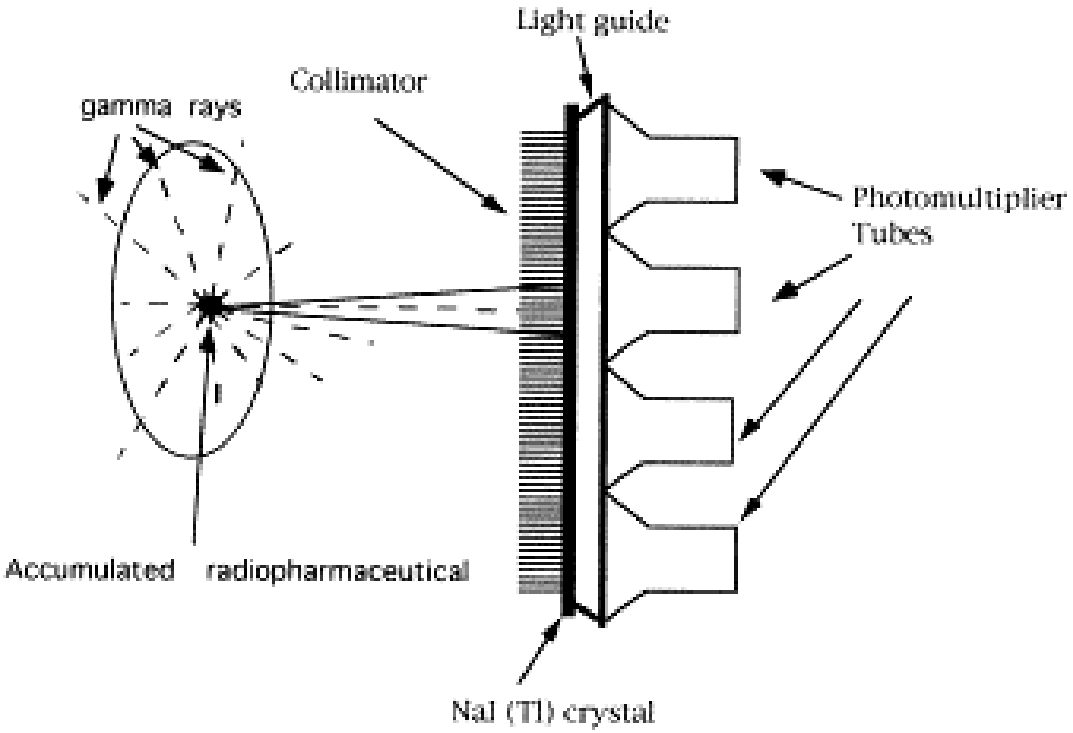
Single photon isotopes (SPECT)

- The imaging of single photons emitters requires:
  - a collimator (causes a loss of efficiency !).
  - a position-sensitive detector (with good detection efficiency): the Gamma (or Anger) camera.

- The image obtained is a projection.
- Multiple (perpendicular) projections can be mathematically correlated to produce a 3D representation.
- SPECT (Single Photon Emission Computed Tomography).

The SPECT gamma camera (Anger camera)

Anger camera



The collimator prevents photons that are not approximately perpendicular to the collimator holes from interacting with the detector.

The spatial resolution degrades as the distance to the object is increased.

So the detector should be placed as close as possible to the patient.

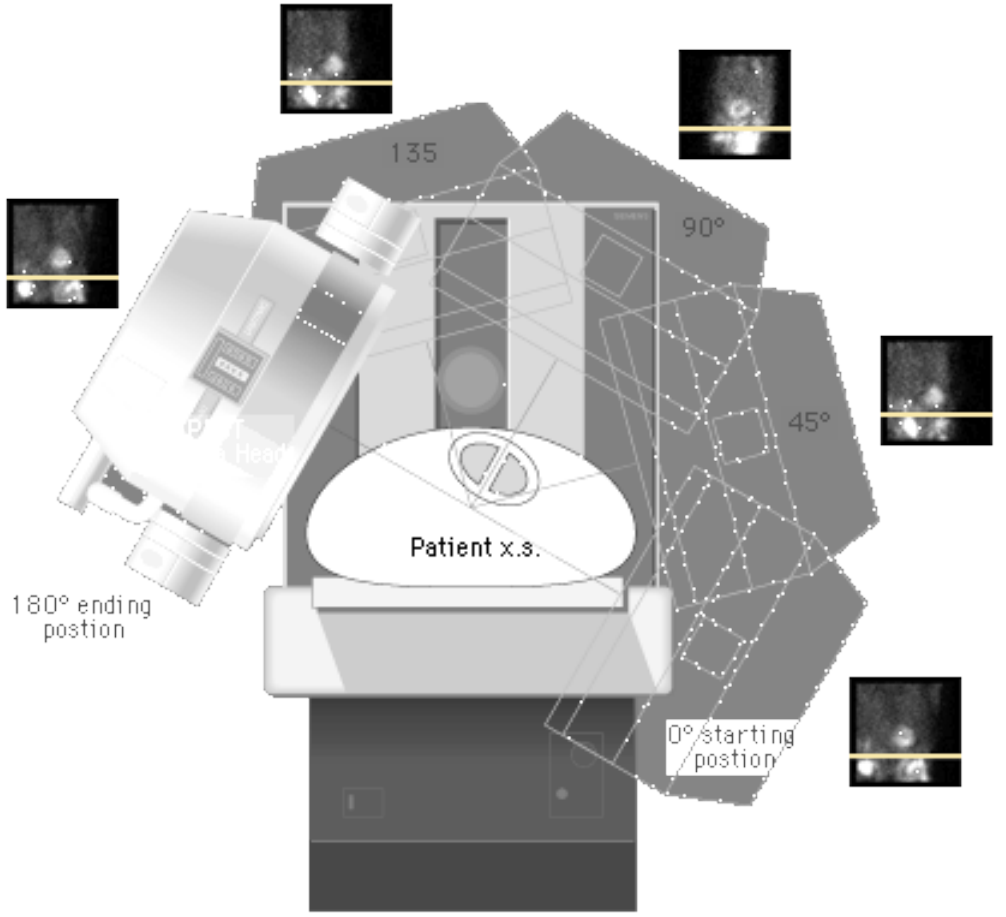
Collimators are usually made of lead.

Typical dimensions: holes 3mm, walls 1mm, depth 40mm.

The SPECT gamma camera



Projections from different angles are taken by rotating the camera around the patient



“Traditional” nuclear medicine

- Technetium 99m, the most commonly used radio-isotopes in nuclear medicine is produced in reactors.
  - 90% of diagnostic studies in hospitals is done with ^{99m}Tc !
- But a number of other, very important nuclear medicine radio-isotopes are produced with cyclotrons of higher energy.
  - ^{201}Tl (Cardiac studies).
  - ^{123}I (Thyroid, Various examinations).
- For these longer life isotopes, international distribution is possible.
- Large, very powerful cyclotrons are owned by radiopharmaceutical companies.

How to select a good single-photon radio-tracer?

1. The energy of the emitted photon
   - Low enough for good detector efficiency and efficient collimation
   - High enough to leave the body: 100 keV $\leq$ E $\leq$ 300 keV is optimum
2. The half-life:
   - Short enough to minimize the patient's exposure
   - Long enough to allow distribution to the hospitals
   - Practically 10h $\leq T_{1/2} \leq$ 100h
   - Generators are great too !

^{99}Mo (66 hours) = $^{99}Tc_m$ (6 hours)

^{81}Rb (4.6 hours) => ^{81}Kr (13sec)

3. The chemistry
   - Essential bio-chemical behavior of the labeled molecule should remain intact:
     - Halogens (Fluor, Iodine), Technetium $\Rightarrow$good;

Nuclear reactions for SPECT radio-Isotopes

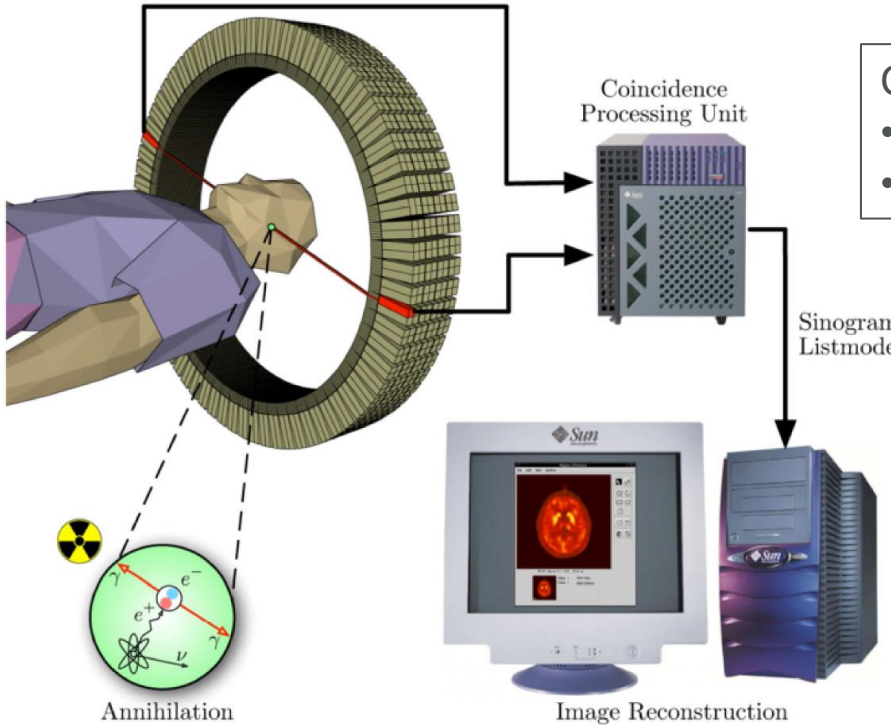
Radioisotope	Half-life	Reaction	Energy (MeV)
^{201}Tl	73.1 h	^{203}Tl (p,3n) => ^{201}Pb => ^{201}Tl	17~28
^{67}Ga	78.3 h	^{68}Zn (p,2n) => ^{67}Ga	12~28
^{111}In	67.4 h	^{112}Cd (p,2n) => ^{111}In	12~28
^{123}I	13.2 h	^{124}Te (p,2n) => ^{123}I	20~25
		^{124}Xe (p,2n) => ^{123}Cs => ^{123}I	20~30
		^{124}Xe (p,pn) => ^{123}I	
		^{127}I (p,5n) => ^{123}Xe => ^{123}I	45~68

A 30 MeV cyclotron can often do the job

Positron emitting radio-isotopes (PET)

- The emitted positron travels a few millimeters, then meets an electron and annihilates, emitting two anti-parallel photons of 511keV.
- These two photons can be detected in coincidence by a ring of detectors surrounding the region of interest.
- One knows then that the origin of the photons is on the line connecting the two detectors
- no collimator needed=> better sensitivity and resolution

The PET scanner



Currently available are
- PET scanners integrated with CT: PET-CT
- PET scanners integrated with MRI: PET-MRI

PET scanner in a hospital



- Coincidents events are grouped into projected images (sinograms) and sorted by the angle of view
- 3D image re-construction is similar to that of CT scanners

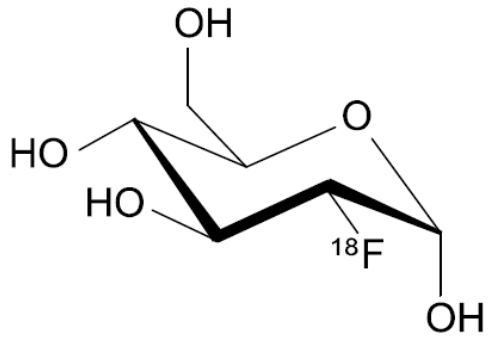
Common positron emitting radioisotopes for PET

Radioisotope	Half-life (min)	Positron energy (MeV)	Reaction	Energy (MeV)
^{11}C	20.4	1.0	^{14}N (p,α)=> ^{11}C	5=>16
^{13}N	9.96	1.2	^{16}O (p,α)=> ^{13}N	8=>16
			^{12}C (d,n)=> ^{13}N	3=>8
^{15}O	2.07	1.7	^{15}N (p,n)=> ^{15}O	5=>14
			^{14}N (d,n)=> ^{15}O	3=>8
^{18}F	109.8	0.6	^{18}O (p,n)=> ^{18}F	5=>14

Cyclotrons 10-18 MeV

FDG = Fluoro-Deoxy-Glucose

- Most commonly made PET scan (90% of cases) is done with 18F-FDG (Fluoro-Deoxy-Glucose)
- Metabolic activity by virtue of glucose uptake in tissue
- This tracer is mainly used to explore the possibility of cancer metastasis and the response to treatment



- In glucose one OH-group is replaced by a 18-F atom
- Both atoms have about the same size =>
- Bio-chemical behaviour almost not altered

Addendum
Part B: radio-isotopes for cancer therapy

Part II-B: radio-isotopes for cancer therapy

In a radioisotope, the nucleus decays spontaneously, giving off particles with some kinetic energy.

β^+ γ

diagnosis

PET - $T_{1/2}$:
- F-18- 2h
- Ga-68- 1h
- Zr-89-3 d
- I-124-4 d
- O-15- 2min

SPECT- $T_{1/2}$:
- I-123-13d
- Tc-99m-6h
- Ga-67
- In-111
- Tl-201

α β^-

therapy

- I-131
- Y-90
- Re-188
- Lu-177
- At-211*

Brachy therapy

- Dose is delivered by placing the radiation source directly inside the area requiring treatment
- Commonly used for cervical (uterus), prostate, breast and skin cancer

- Irradiation affects only a very localized area => healthy tissues are spared
- Much higher doses can be delivered. For comparison:
  - Proton therapy: about 40 Gray
  - Prostate brachytherapy: about 100 to 150 Gray

- Brachytherapy can often be completed in less time
  - Reduce the possibility of recovery of cancer cells between treatment intervals

Prostate brachytherapy with Pd-103 or I-125

Seeds placed with 3D precission verified with ultrasound probe
Seeds are not harmfull and can stay in place after treatment

